

Science for Little Thinkers

Michael Gannotti



Figure 1: Cover

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This illustrated edition is the teaching-manual of 15 September 2026. Endnotes run in one series. Chapter-opening images are original still-lives, not portraits of persons or children. Documents still constrain the facts.

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Welcome

You do not need to be a scientist to teach a five- to ten-year-old real science looking.

You need a lookable on the table before the explanation, and one good question you are willing to wait for.

That is what this book is for.

Why this book exists

Many adults remember science as a list of vocabulary words and a laminated scientific-method poster. That memory can leave you feeling rusty when a child asks why the ice is smaller, or why the plant leans. Feeling rusty is normal. It is not a character flaw.

At ages 5–10, science does not begin with a kingdom chart or a five-step method. It begins with looking and trying. A child can notice ice changing before anyone says *melt*. A child can draw what happened. A child can hang a weather card and say what is the same as yesterday. A child can push a ramp harder and say what they saw. A child can sort living from not-living by a need they can see.

Those moves are not a waiting room for middle school. They are the work.

This book teaches you how to run a short **Science Hour** at home — roughly twenty to forty minutes — with the lookable down before the explanation, and with **observe before explain** as the spine. Fair-test light lives beside it: one deliberate change, and the question *can we tell for sure?* Inside the chapters you will see two planning bands: **5–7** and **8–10**. Place your child by skill, not by birthday. A fluent eight-year-old and a careful six-year-old may sit in the same looking move on different days.

The focus lock for the whole book is simple:

Observe before explain. Fair test light.

Hold that, and the rest of the manual becomes practice.

What a good Science Hour looks like

Picture a kitchen table. Not a museum field trip every day. Not a silent “read the chapter” hour. Not an app chase.

You put ice in a cup, or a leaf, or a flashlight and a card. You cover any vocabulary worksheet title if you need to. You ask what the child notices — before you tell them what this is called. You wait. They look. They try one small thing. They draw what happened (at 5–7) or add a sentence under the picture (at 8–10). You ask one authentic question about what they saw, and you wait again. Then you name the idea only as far as the looking can carry. Then you stop talking.

That is the hour. Library visits, museum days, and nature walks motivate. They do not replace the looking move on the table.

You will meet the full session shape in **The Science Hour**. Later chapters point back to it. You do not rebuild the spine every week.

Two readers at once

This is a teaching manual for the parent or co-teacher at the table, with short student sections a child can actually use.

When you read a parent section, you are learning today’s looking move well enough to hear a wrong turn — a name-first rush, a force-as-fuel jab, a confounded “proof,” an adult-filled observation sheet, a generated ice-melt paragraph filed as the child’s looking, a demo with no child’s record, a coverage lecture with no lookable on the table. You learn what to say next without grabbing the answer or filling the silence with the explanation.

When your child reads (or hears) a student section, they get respect. No baby-talk. No lecture about you. They get a tiny worked example, two tries, and a chance to explain it back.

What this book will not do

It will not hand you a 180-day worksheet dump.

It will not sell you a curriculum, crown one program, or promise a score.

It will not reprint *Science for Homeschooling* — that book serves grades 1–12, with later survey, Biology, Chemistry, Physics, Earth, and transcript edges. For that map, see that book once. This book teaches the 5–10 Science Hour.

It is not a sequel to *History for Little Thinkers*, *Critical Thinking Skills for Little Thinkers*, *English for Little Thinkers*, or *Math for Little Thinkers*. One pointer each is enough. Then we teach science as looking and trying.

It will not title your subject **Little Thinkers Science I**. A stranger-readable title is **Science**. Life Science, Physical Science, or Earth Science only when the work is actually that.

It will not turn the hour into a culture-war pamphlet. Safe science objects only: a seed, ice in a cup, a leaf, a shadow, a thermometer, a magnifier, a plant on the windowsill, a sink-and-float tray, a board-as-ramp, a flashlight and card, a weather window card, a dated notebook.

It will not treat kitchen vinegar-and-soda as Chemistry credit. A fizz is a picture of change. Curiosity is not a permit for unauthorized mixing.

It will not ask you to wait until middle school to begin. Children this age can look. They do not become scientists because someone bought a kit.

How the book is built

Front matter tells you how to use the week, five things to remember, and the Science Hour once — including the safety spine and the AI box for ages 5–10, said carefully and only once.

Eight teaching chapters follow: what science looks like at 5–10; looking and drawing what you saw; try it, one change at a time; living or not, needs and life cycles light; weather window and sky patterns; light, path, push and pull; talking and writing science; short assessments and progress checks.

Then records and resources. Records keep a dated notebook, a draw-and-retell,

a one-thing-changed check, and a living-or-not sort under **Science**. Resources name common programs by fit — parent load, style, placement, cost shape — with library, museum, and nature first. Fit, not rank.

Every teaching chapter uses the same eight headings, a talk box with an exact opening question, named try-its inside the home and outside it, and age-band moves for 5–7 and 8–10.

Encouragement that is specific

This week you can learn one looking move well enough to hear a name-first rush and turn it back to the ice, the leaf, or the ramp.

Today your child can point at one thing that changed and say what they notice.

You do not need to be a scientist. You do need to put the lookable on the table before the explanation, and to ask what they noticed.

That is enough to begin.

— Michael Gannotti

How to use this book this week

Pick one looking move. Run the Science Hour a few times. Leave one crumb on Friday.

That is the honest “lesson plan.” Not a 180-day calendar. Not a unit binder that never returns to the ice.

This week’s shape

1. **Choose one chapter focus.** Examples: notice before name on ice or shadows; drawing as the record; one-thing-changed on a ramp; living-or-not by a need you can see; a weather card vs yesterday; a flashlight path; a caption that could only come from looking.
2. **Run three to five Science Hours** of the named shape in **The Science Hour** (~20–40 minutes each). Same focus. Fresh lookable if you can; same ice cup or leaf revisited if that is what the week holds.
3. **Do one in-home try-it** from the chapter’s practice section (ice plate, leaf before name, sink/float, ramp, flashlight, weather card, plant one-thing, magnet bag).
4. **Do one out-of-home try-it** once in the week when you can (park bark rub, puddle edge, grocery produce sort, museum one exhibit, night sky sit, garden insects soft, library science shelf). Ordinary safety. Adult present outdoors.
5. **Friday: one diagnostic wrong + done-enough.** Hear one wrong reason without crushing the look. Check whether the child can show the move two ways, and whether it holds after a night. Details live in each chapter’s “done enough” and in the assessments chapter.

If a day only holds twenty minutes: lookable + notice + exit crumb. One named

move done well beats a rushed medley. There is no nationally representative study of science minutes per day in U.S. homes that this book can quote — so there is no honest “required” dose to post on the fridge — and no need to shame a week that held three short sits.

How to read a teaching chapter

Read **Why this matters** and **For the parent** first — alone, if you can. Do the five-minute parent warm-up with tomorrow’s lookable before the child sits. Practice the sentence “What do you notice?” aloud once. Check that the materials are actually on the table, not still in a cupboard.

Then use **How to teach it this week** as your script for the table. Exact wording is there on purpose. Use it. Age-band moves for **5–7** and **8–10** live inside that heading.

Practice that actually builds learning names try-its with time, materials, safety, the fun, and the skill — at least one inside the home and at least one outside. The talk box gives the locked opening question, follow-ups, how to wait, and what a stuck silence usually means.

Read **For the student** with your child, or let them hear it. Warm, direct, short. Never a lecture about research.

Use **If it isn’t clicking** when something stalls. Three diagnostics. No shame. Correct a wrong reason without crushing the look.

Tools, including AI stays short and points back to the Science Hour AI box. Do not rebuild policy every chapter.

What “done enough” looks like is your leave-this-chapter checklist. Placement by skill, not birthday. A publisher’s “grade 2 science workbook” is a scope, not a legal grade, and not a transcript line.

Sibling and mixed-age homes

Share the lookable. Split the try.

The younger child may draw, notice, or say one thing that changed. The older child may name what stayed the same, write a caption, or catch an obvious confound

with help.

Give each child a turn to speak. After one answers, wait so the next can add — not so the older becomes unpaid tutor as their only role. They need their own looking move too.

Multilingual homes

Looking works in the family's talk language.

Notice, predict, and draw work before English labels arrive. When a book page is in another script, you read the name aloud after the look; the child still records what they saw. You may scribe a caption the child said. Do not delay Science Hour until decoding is “ready.” Looking is not decoding.

Activity banks are a menu

Later chapters pull named try-its from banks (observe, draw, trial, fair test, living things, weather, light, force, talk, write, outings, assessment). A chapter picks a few. It does not assign all eighty. Coverage itch is the enemy of looking.

What to skip for now

Skip chapters you are not ready to teach this month. The map is a path, not a race.

Skip vendor shopping until you have run a few Hours with free lookables: ice, a leaf, a window, a flashlight, a library science shelf, a park tree.

Skip AI tools until you have read the Science Hour AI box once. The child looks. You may use a tool on your account. The child does not sit alone with an open chat.

Records without drama

Date the notebook page. Title it **Science**. Keep the drawing, the spoken crumb you scribed, and the one-thing-changed note. That file is the record. Details live in the records chapter. You do not need a brand name on the spine.

If you only remember five things

Keep this page short. Pin it where you plan the week. When the hour feels scattered, return here before you add more materials.

1. Observe before explain

Put the ice, the leaf, the ramp, the flashlight, or the weather window on the table first. Looking is the lesson. The adult explanation waits. A fluent lecture with no lookable is not a Science Hour. Library, museum, and nature days motivate; they do not replace the object on the table. One lookable beats a dizzying tour of twenty demos.

2. Fair test light

When trying is the point, change one thing. Keep the rest the same. Ask *can we tell for sure?* A confounded “proof” — water *and* light *and* pot size all changed — is a chance to redesign, not a victory. At 5–7, you scaffold the same-list. At 8–10, the child names change and same. Fair test is a taught procedure, not a personality trait.

3. Ask one good question and wait

One authentic question about what they notice before a name, what the drawing shows, what changed and what stayed the same, what a living thing needs, what is same or different in the weather, what happened with light or a push. Then a slow three. Look at the materials, the drawing, or the notebook — not at the child — if

the silence is hard. Do not fill the silence with the right vocabulary. Do not grab the answer. Struggle before rescue: ask, wait, hint, then model.

4. The child's drawing or sentence is the data

At 5–7, the drawing *is* the notebook. At 8–10, picture-plus-sentence carries the record. A finger on the page can answer “what did you see?” An adult-filled observation is not the child's looking. A generated paragraph is not data. A generated image is not a result. Protect the record. Scribe only what they said.

5. Place by skill, not birthday

Use the bands 5–7 and 8–10 as planning help. Move a child up or down by what they can show: a notice sentence that points at the lookable; a drawing that answers what happened; one deliberate change named; a living-or-not sort by a need they can see. A path through elementary science looking is the promise. A diploma is not. A percentile is not. A culture-war win is not.

These five are the book in miniature.

AI honesty — a generated observation is not data; a model summary is not the observation; a generated image is not a result; the parent holds the account; unaided first on looking and trying — lives once in **The Science Hour**. Point back there when you need it. Do not let a sixth rule crowd the fridge list. Safety (adult present; no unauthorized mixing) lives there too.

The Science Hour

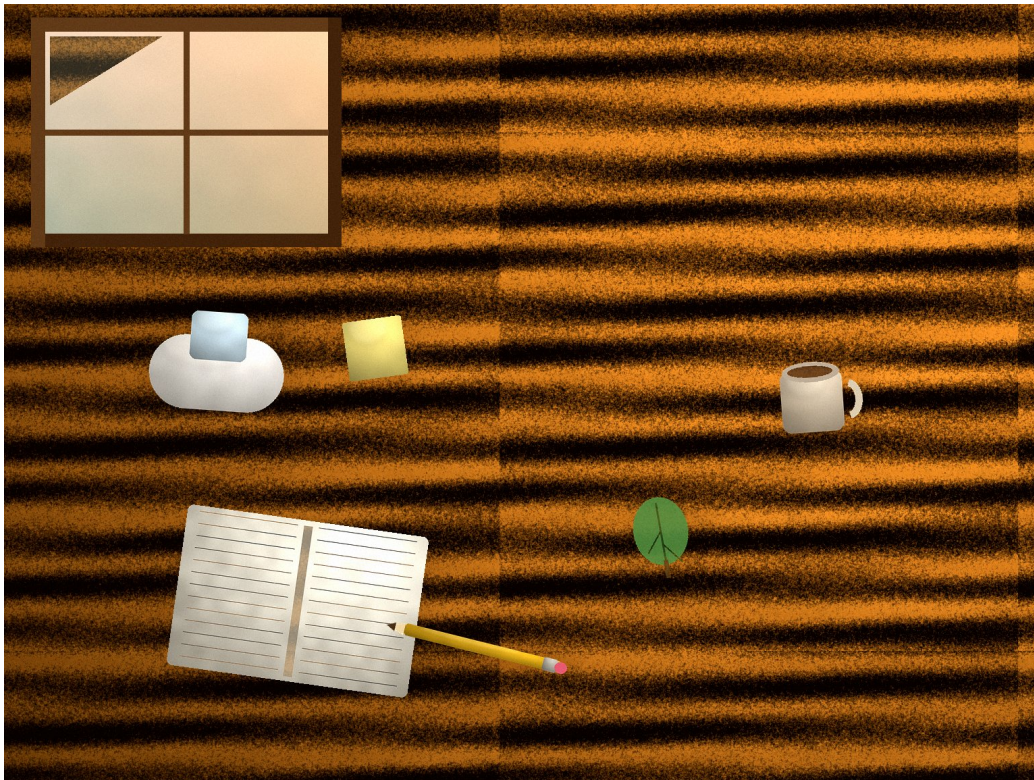


Figure 2: A kitchen table with an ice cup on a plate, a blank sticky, and an open notebook — still-life, no people, no logos.

Twenty to forty minutes. Five beats. Stop while a look remains.

This chapter is the session spine for the whole book. Later chapters say “run the

Science Hour” and point here. You do not rebuild the shape every week.

Parent warm-up (alone, about five minutes)

Before the child sits, put tomorrow’s lookable on the table. Practice one sentence aloud: “What do you notice?”

Do not rehearse a lecture. Cover any worksheet title that already names *melt*, *condensation*, or *force*. If the only “ice” you have is a generated picture of melting, put real ice in a cup instead.

Without this warm-up, the Hour collapses into explanation-first.

The five beats

1. Put the lookable down (3–8 minutes)

One object or setup: ice in a cup, a leaf, a board-as-ramp, a flashlight and a card, a weather window, a plant on the sill, a sink-and-float tray. One lookable, not a stack of kits.

Ask what they notice. Where does your eye go first? What do you see that you did not expect?

You do not dump the explanation yet. If they ask “what is this called?” right away, turn them back to what is on the table.

2. Notice / predict before naming (3–7 minutes)

One clear question in ordinary words. What do you notice? What do you think will happen if...? Then wait.¹

At 5–7, two-choice predict is enough (“stay the same / change”). At 8–10, ask for a reason tied to what they can already see. You still do not hang the vocabulary

¹Mary Budd Rowe, “Wait-Time and Rewards as Instructional Variables,” NARST / ERIC ED061103 (1972): elementary *science* tapes; typical wait near one second; extending wait to about three to five seconds changed student discourse. Robert J. Stahl, “Using ‘Think-Time’ and ‘Wait-Time’ Skillfully in the Classroom,” ERIC ED370885 (1994): classroom think-time convention near three seconds. Both are classroom-origin. They are not a kitchen-table science trial.

poster first.

3. Try / fair-test light (5–15 minutes)

The child looks, tries, or changes one thing. Adult present for safety. When a fair test is the point, the child helps decide what stays the same. Ask *can we tell for sure?* if more than one thing moved.

This is the child's try. A parent demo with no child's hands is a picture, not the hour's work.

4. Record (3–8 minutes)

At 5–7, draw what happened. You may scribe one sentence they said. At 8–10, picture-plus-sentence, or a simple two-row table when the try needs it.

The drawing is the data. A pretty picture that does not match the try is not the data. A generated observation is not the data.

5. Exit crumb / talk box (1–5 minutes)

One authentic question. Wait. The child leaves one durable trace: a spoken notice sentence that points at the lookable, a labelled drawing, a sticky, a one-thing-changed note. You note one diagnostic error if any.

Done-enough today is “right, or wrong-with-a-reason we can use tomorrow” — not a percentage. The exit should be able to stand without a chat window and without your summary sentence.

Optional sixth beat — siblings

After one child answers, wait so the next can speak. Share the lookable; split the try. Do not make the older child the unpaid tutor as their only role.

Bands inside the hour

5–7. Look → notice → short try → draw → oral or phrase label. You scaffold the “same” list when trying is the point. You read any print aloud. Prefer sits nearer

twenty minutes when attention is thin. No live open chat as the child's partner.

8–10. Look → predict with a reason → one-thing-changed try → picture-plus-sentence or simple table → a spoken claim that points at the record. They may name Change / Measure / Same in ordinary words. If looking habits are weak, stay on 5–7 observe longer — regardless of a ninth birthday.

Place by skill, not birthday. Move a child between bands by what they can show.

When the hour goes wrong

If the hour looks like...	Try this move
Only demo / only parent hands	Child's try first; adult scaffolds safety
Only worksheets / checklist of topics	One real lookable + one oral question
Name-first freeze	Cover vocabulary until after the look
Wait was actually zero	Count a slow three; look at the materials
"It stopped because the push ran out"	Another push-harder look; then name change-of-motion lightly
Confounded pride	"Can we tell for sure?" then redesign
Adult-filled notebook	Scribe only what the child said; protect the record
Generated observation filed as real	Label fiction; child's eyes first
Activity mania / five demos	One try + one record; stop while a look remains
Phone radar or a generated sky first	Window first; screen after as a check
Vinegar-soda as "chemistry credit"	Change picture; transcript still Science
Child alone with open chat	Close the tab; SCRIPT after the try; parent holds the account
Safety fails	Stop the hour; continue another day

Talk-box shape (once for the book)

Every teaching chapter carries a talk box. The shape is always the same.

Opening question. Authentic. About what they notice before a name, what the

drawing shows, what changed and what stayed the same, what a living thing needs, what is same or different in the weather, what happened with light or push/pull, what they know from what they saw, or what they can show two ways. Not “did you have fun.” Not “what is the vocabulary quiz word.”

Locked openings live in each chapter. Chapter 1 opens with notice before name. Chapter 2 opens with what the drawing shows — and what is still missing.

Three to five follow-ups. One for observe (“what do you *see* / *notice*?”). One for record (“what does the drawing / notebook show?”). One for reason tied to the try (“how do you know from *this*?”). One for check or alternative (“what else was different / can we tell for sure?” at 8–10). One that lets them revise (“you may change your mind” / “try another look”).

How to wait. After you ask, a slow three is a useful convention — drawn from classroom science wait-time research, not a kitchen-table science experiment. After they stop, wait again. If they are mid-look or mid-draw, do not cut them off. Look at the materials, the drawing, or the notebook if the silence is hard.

Stuck silence usually means: the question was vague; you did not wait; they are hunting a vocabulary word because you asked name-first; they are guessing what you want; the materials never landed; they offered a book-only guess; they think science means “repeat the parent’s explanation”; they think looking means “say what the screen already told me.”

Your job is to pose, wait, and point back at the lookable — not to fill the silence with the right vocabulary, and not to turn the table into a fight.

In-home try-it shape (once)

Every named in-home try-it in this book carries:

- **Time** (kitchen estimate)
- **Materials** (what is on the table)
- **Safety** (ordinary and non-negotiable — see the spine below)
- **The fun** (why a child might want to try)
- **The skill** (the looking move being practiced)

Typical in-home lookables: ice plate, leaf before name, sink/float tray, board-as-ramp, flashlight path, weather window card, plant one-thing, magnet bag.

Out-of-home try-it shape (once)

Same fields — time, materials, safety, the fun, the skill — plus ordinary outing care.

Typical outings: park bark rub, puddle or creek edge, grocery produce sort, museum one exhibit, night sky sit, garden insect count soft, library science shelf peek.

Safety and courtesy. Roads. Weather extremes. Wildlife. Museum hands-off rules. Grocery aisle courtesy. No lecture to a stranger's child. No harm to organisms. Night sky soft — no lasers. A plaque or a book picture is motivation; it does not overwrite the home trial.

Library, museum, and nature outings motivate. Those visits still do not replace the observe-before-explain move at home.

Safety spine (once for the book)

1. **Adult present** for water, heat, outdoors, small parts, kitchen substances, night work, tools, and magnets that pinch.
2. **Authorized materials only.** No unmarked bottles. No bleach mixes. No drain cleaners. No alcohol flames. No sealed pressurized fizz bottles.
3. **Kitchen chemistry picture is not Chemistry credit.** Vinegar and baking soda is a change picture. It does not unlock titration.
4. **No unauthorized mixing.** Curiosity is not a permit.
5. **Name the hazard before the try** — eyes, skin, slip, choke, allergy, traffic, wildlife, weather extremes.
6. Prefer cool LED flashlights. Lasers are refused in the default banks for these ages.
7. **Living things:** no harm; return organisms; wash hands.
8. **Mold cultivation:** refused by default. A bread-in-a-bag culture is not this book's try-it.
9. **Stop the hour when safety fails.** Investigation continues another day.

When in doubt, do not run it. A chatbot cannot be the safety officer. Later chapters point back here in one sentence.

Dose honesty

If a day only holds twenty minutes: lookable + notice + try + exit crumb. One named move done well beats a rushed medley. Three to five Science Hours plus one out-of-home try-it is a full week.

There is no nationally representative study of science minutes per day in U.S. homeschooled that this book can honestly quote. Do not invent one. Program lesson-length claims are program claims, not a national table. Do not shame a family whose week held three short sits.

Life of the habit (session rhythm — not a ninth chapter)

Revisit yesterday's drawing. Mix last week's one-thing try into a warm-up. Stop while a look remains.

That rhythm lives here, in later "done enough" checklists, and in the records you keep. This book does not invent a ninth teaching chapter titled "life of the habit."

AI at the science hour (ages 5–10) — said once

The child looks at a real object and tries one thing. You may use a tool on your account. The child does not sit alone with an open chat.

Ages 5–7: no live open chat as the child's partner. You may generate a four-line SCRIPT for after the attempt, and you may ask a tool to explain a named page *to you*. No child-facing chat. No child account.

Ages 8–10: still parent in the room. The child may sit next to you while you use a tool. They may hear a hint after they have tried. They may not have their own companion account. They may not be asked to "prompt the tutor until it agrees." Unaided first on looking and trying. SCRIPT after the try.

You may, after the child has tried (or before, for your prep):

1. Explain this week's idea *to you* from a named textbook or lab page.
2. Extra practice (blank observation lines, predict-then-look cards) with PARENT KEY held by you. The child never sees the key.

3. A short labelled SCRIPT you will say *after* they look and try — questions that keep *your* voice and the child's eyes on the object.
4. A hint after an attempt (a question or named idea — not the finished observation).
5. Diagnose work they already produced (crop to the paper; no child's face).
6. Discussion questions you will vet and ask yourself — the child answers you, not the tab.
7. A labelled schematic (not a result) to look at together, then put away before the try.

You may not:

1. Paste the worksheet or blank data table and take the filled key (photo-to-key).
2. Ask “what's the answer” or “write what I should have seen” during the attempt, or let the child ask that of a window.
3. Invent an observation or a data table and file it as the child's looking.
4. Put a model summary on the table as the observation instead of a real lookable.
5. Generate a gel, micrograph, sky photo, or “what I saw under the glass” and treat it as the student's result. **Generated image $\neq$ data.**
6. Park an unsupervised chatbot as the child's partner during the attempt — not at 5, not at 7, not at 10.
7. Write the student's observation, caption, or lab paragraph.
8. Invent research or trust a fluent citation without opening the paper.
9. Use TTS, narrated rewrite, or a voice-clone “scientist friend” to replace looking at today's object.
10. Detector-grade a child's “the ice got smaller.”²
11. Print a certificate of science, or hand the child an agent (Hermes Agent, Grok Bot, or any cloud-computer worker) as a science partner.

One crutch sentence, labelled OFF-AGE for science hour, then teach: in a **high-school math** trial (not this science band), an unguarded chatbot made practice look better and left students worse on the test with the window closed (Bastani et al., 2025)³. This is a useful study, not a promise that every home will see the

²Weixin Liang et al., “GPT Detectors Are Biased Against Non-Native English Writers,” *Patterns* 4, no. 7 (2023). William H. Walters and Esther Isabelle Wilder, “Fabrication and Errors in the Bibliographic Citations Generated by ChatGPT,” *Scientific Reports* 13 (2023): 14045.

³Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakçı, and Rei Mariman,

same result — and it is **not a 5–10 science experiment**. The kitchen-table rule it supports: the model may prepare the adult and the next looking page; it may not do the child’s looking, trying, or recording when those are the lesson.

COPPA. Ages 5–10 are under 13.⁴ Parent holds the account always. Parent in the room. The 22 April 2026 compliance date has passed. Keep accounts, logs, and keys where you can open them. A science tool is not a secret friend.

Names that stay separate. Hermes 4 is not Hermes Agent. Grok is not Grok Bot and is not a national “Grok-as-tutor” program. There is no Grok Bot education product this book can honestly name as a school SKU. Game artist makes pictures, not a science course. Agents are not chatbots.

Detectors and fake citations (parent-side). Detectors fail often enough that you should not grade a child’s “the ice got smaller” from a detector score. Fabricated citations are a parent-side problem. If you asked a tool for a paper on observation, open *Ready, Set, SCIENCE!* or *Taking Science to School* yourself.⁵

Later chapters point back here in one sentence. They do not reprint this box.

SCRIPT shapes (you say these; the child never chats)

After the child has looked and tried:

- Level 1: “What do you notice?”
- Level 2: “Point to one thing that changed.”
- Level 3: “Say it in your words. Look at the other cup.”

Put any “right” answers under PARENT KEY only. Skip invented observations. A model summary is not the lookable on the table.

“Generative AI without Guardrails Can Harm Learning: Evidence from High School Mathematics,” *Proceedings of the National Academy of Sciences* 122, no. 26 (2025): e2422633122. Turkish high-school math; OFF-AGE for science hour.

⁴COPPA: Children’s Online Privacy Protection Rule, 16 C.F.R. part 312; amended rule 90 Fed. Reg. 16918 (22 April 2025); general compliance date 22 April 2026.

⁵National Research Council, *Ready, Set, SCIENCE!* (Washington, DC: National Academies Press, 2008); National Research Council, *Taking Science to School* (Washington, DC: National Academies Press, 2007).

Ready to begin

You have the hour. You have the talk-box shape. You have in-home and out-of-home try-it shapes. You have the safety spine once. You have the AI box once.

Chapter 1 puts observe-before-explain on the table. Chapter 2 puts drawing as the notebook beside it.

Open tomorrow's ice cup. Practice "What do you notice?" once. Then sit down.

Chapter 1 — What science looks like at 5–10



Figure 3: An ice cube on a plate beside a blank sticky and pencil — still-life, no people, no logos.

Why this matters

Science at this age is not a waiting room and not a coverage slog.

It is notice before name. It is a lookable on the table before the adult's explanation. It is placement by skill, not by birthday.

When a five-year-old can point at ice in a cup and say “it's getting smaller” before anyone says *melt*, something real has begun. When an eight-year-old can hang three weather cards and name one thing that stayed the same and one that changed, the same habit has deepened. Neither child needs a laminated scientific-method poster to start. Neither child needs to wait until middle school for any science looking.

Many homes treat early years as “not yet science.” The shelf fills with vocabulary lists and kingdom charts. The child still never gets a quiet minute with one object and one question. That is the gap this chapter closes.

Coverage is a map. Depth is the hour. Light tours through mounds of facts can entertain and leave few traces.⁶ The mind wants pattern, built slowly, with repeated looks. Your job this week is not to finish a kingdom. Your job is to put the ice down, ask what they notice before you tell them what this is called, and wait.

This chapter unlocks the rest of the book. Without notice-before-name, later drawings float. Without lookable-before-explanation, every later chapter collapses into lecture. Without skill-not-birthday placement, you will either withhold looking from a capable six-year-old or dump a middle-school variables course onto a tired seven-year-old.

Worth the struggle: one week of ice, leaf, and shadow hours trains your ear for name-first rushes, demos with no child's record, and coverage lectures with nothing on the table. You will know what to say tomorrow. Your child will know that science looking means pointing at what they saw — not guessing the word you want to hear.

A path through elementary science looking is the promise. A diploma is not. A percentile is not. A culture-war win is not. Start here.

⁶Sam Wineburg, “Crazy for History,” *Journal of American History* 90, no. 4 (2004): 1401–1414 — coverage-versus-depth shape borrowed carefully; science depth in this book means this week's phenomenon.

If you take nothing else from this chapter into next week, take this: look before you name, put the object down before the explanation, place by skill before birthday. The rest of the manual assumes you can hear a name-first question and turn it back to the lookable without apology.

For the parent: understand it yourself

One everyday picture

Soup on the stove. Bubbles. Steam. You watch before anyone says *boil*.

That is science looking at 5–10 in one frame. The name can come after. The idea, if it comes at all this sitting, comes last — as a short ordinary sentence the looking can carry. “The ice got smaller. There is water on the plate.” Not a lecture on states of matter.

One precise picture

Researchers who mapped what children can do in science from kindergarten through grade 8 named four intertwined strands, not a single method poster.⁷ In kitchen language they are: (1) after looking, use a short idea without mistaking the word for the looking; (2) generate and evaluate evidence — draw, point, say what would change your mind; (3) treat ideas as revisable — “I think because I saw”⁸; (4) talk after the trial, wait, ask for evidence — not a recitation quiz alone. Each strand supports the others. A year of vocabulary without trials starves the looking. A year of activities without talk and an idea is activity mania.

Children already know a great deal. They enter with substantial, often unspoken, knowledge of the natural world — how solid objects behave, how living things seem different from spoons, how materials feel. The picture of the “concrete, simplistic child” who cannot reason until a later stage is outmoded. What a child can do at a particular age is an interplay of maturation, experience, and instruction.

⁷National Research Council, *Taking Science to School: Learning and Teaching Science in Grades K–8* (Washington, DC: National Academies Press, 2007): four strands of proficiency; children enter with substantial knowledge of the natural world; capability as maturation × experience × instruction; alternative conceptions including force-as-fuel and plants taking food through roots.

⁸California Department of Education, *2016 Science Framework for California Public Schools* (Sacramento, 2016), Chapter 3: K–2 practices in ordinary language, including drawing and “I think because I see.”

Place by skill. Do not map 5–7 and 8–10 onto a rigid stage calendar as if birthdays opened locked doors.

National science guidance — the *Framework for K–12 Science Education* and the Next Generation Science Standards — names K–2 moves that map cleanly onto this hour: observe, draw, needs of living things, local weather, push and pull, light path.⁹ Grade 3–5 edges add fair tests, life-cycle cards, and weather tables — light, and mostly for 8–10. Those are a ceiling and a shape. They are not national homeschool law. They are not a performance-expectation checklist to tape to the fridge.

A single linear “scientific method”¹⁰ (observe, hypothesize, deduce, conclude as the only story) is a textbook myth those same documents refuse. Practices are plural. A weather-window week may be almost pure observation. A ramp week may be a simple try. A plant week may be one change over days. None needs the full poster.

Sequence matters. Explore-before-explain, in the science-teaching sense, means look → try → talk → name.¹¹ It does not mean never explain. It does not mean unguided discovery as the only method. Put the phenomenon down first. Ask what they notice and what they predict. Then name the idea in kid language as far as the looking can carry. Then stop talking.

One more limit, said plainly: thinking well depends on knowing something about the world you are looking at. A child cannot run a fair test about a world she has no picture of. Give the lookable and a tiny try. Do not demand multi-variable design of ecosystems in week one of age six. Knowledge first; looking with that knowledge; analysis later when the picture is owned.

Prior ideas are the mind at rest, not a character flaw.¹² When a child says “it stopped because the push ran out,” do not scold. Put the ramp down again and ask

⁹NGSS Lead States, *Next Generation Science Standards* (Washington, DC: National Academies Press, 2013): K–2 observe / draw / needs / weather / push-pull / light-path moves; 3–5 fair-test and weather-table edges — used here as a map, not a mandate.

¹⁰National Research Council, *A Framework for K–12 Science Education* (Washington, DC: National Academies Press, 2012): practices, crosscutting concepts, and disciplinary ideas; the single linear “scientific method” as a textbook myth.

¹¹Patrick Brown and the explore-before-explain teaching sequence as used in NSTA-facing work: sequence (look, then name), not “never explain.”

¹²Daniel T. Willingham, “Critical Thinking: Why Is It So Hard to Teach?,” *American Educator* 31, no. 2 (2007): 8–19. Domain knowledge constraint — one sentence in the parent section.

what they saw when you pushed harder. When a child says plants eat through their roots, hear it. Repair with another look — water, light, a leaf — not a vocabulary sermon.

This is a useful set of studies and frameworks, not a promise that every home will see the same result. Then teach.

Wrong answers you should be able to hear

1. **“What is the science word?” before any looking.** Usually means science has been taught as a vocabulary quiz. Redirect: “Before the name — what do you *notice*?”
2. **“You’re only six — we’ll do science later.”** Usually means wait-until-middle-school. Refuse kindly: “We can look at this ice today.” Young children can notice, draw, and try. Withholding looking is the false story.
3. **Wow demo, empty notebook.** Usually means activity mania or a parent show. Repair: one try + one drawing before tidy-up. The child’s record is the hour’s product.
4. **Encyclopedia hour with no lookable.** Usually means coverage itch. Put one object down tomorrow and shorten the lecture. Depth is the hour; coverage is a map you glance at, not a train you ride for forty minutes of kingdoms.
5. **“You’re eight, so you must do fair tests now.”** Usually means birthday lock. Place by what they can show. If looking habits are still thin, stay on notice-before-name. Fair-test light is Chapter 3, and it waits until the child can look.

Five-minute parent warm-up

Alone with tomorrow’s ice cup for five minutes. Practice asking “What do you notice?” before any melt vocabulary.

Cover any worksheet title that already says *melting*. Check that you will not lead with the name. Lay out a blank page so the drawing is ready. Then stop rehearsing. The warm-up is looking practice for you, not a lecture script.

Four strands on the kitchen table (not a content dump)

You do not need to memorize strand numbers. You do need the kitchen translations:

- Know a short idea after looking → “The ice got smaller. There is water.” The word *melt* waits until the looking owns it.
- Generate evidence → a drawing, a pointing finger, a spoken notice.¹³
- Treat ideas as revisable → “You may change your mind when we look again.”
- Talk after the trial → one authentic question, then wait. Not “define force.”

Keep the investigation spirit: look, try, record, a small conclusion the child can say. Do not dump middle-school variables or high-school Biology as the spine of age six.

What this chapter refuses (kitchen examples)

Wait until middle school. A five-year-old notices ice getting smaller. That is science looking. Withholding it “until later” is the false story.

Coverage as the year. A parent reads twenty vocabulary words with no object on the table. Few traces remain. Tomorrow: one lookable, one question, one crumb.

Birthday placement. “You’re eight, so you get the method poster.” Skill says otherwise. Place by what they can show.

Demo with no child’s record. A parent performs; the child applauds; the notebook stays blank. Tomorrow: the child’s hands, then the child’s drawing.

Culture-war pamphlet as the hour. Safe objects only. Ice, leaf, shadow, ramp, window. Soft entry on living things. Graphic harm is not the default try box.

How to teach it this week

Session shape

Use the Science Hour beats from the front of the book. This week’s focus: notice before name, and lookable before explanation.

1. **Warm-up looking (2–3 min).** Yesterday’s drawing, the plant on the sill, or the window. “What do you notice today?” Keep it light. No new vocabulary.

¹³National Research Council, *Ready, Set, SCIENCE!: Putting Research to Work in K–8 Science Classrooms* (Washington, DC: National Academies Press, 2008): strands in teacher-facing language; activity mania versus structured investigation; talk moves including wait.

2. **New lookable down (3–8 min).** Ice, leaf, shadow, sink/float tray, or magnet bag. Cover any title that already names the idea. “What do you notice — before I tell you what this is called?” You do not dump the explanation yet.
3. **Notice / predict (3–5 min).** One clear question. At 5–7, two-choice predict is enough. At 8–10, ask for a reason tied to what they see. Wait.
4. **Short try (5–12 min).** Watch, sort, or test one object at a time. See the named try-its below. One try — not a packet. Adult present for water, small parts, outdoors.
5. **Record (3–8 min).** Draw what happened. At 8–10, add one sentence under the picture if decoding supports it. You may scribe what they said. You do not overwrite.
6. **Exit crumb (1–3 min).** One oral notice sentence that points at the lookable. Date the notebook page. Title: **Science**.

If the day only holds twenty minutes: lookable + locked opening + one spoken crumb. That still beats a rushed medley.

Exact wording you can say

- Opening (locked): “**What do you notice — before I tell you what this is called?**”
- “Point to one thing. I’ll wait.”
- “What do you *see*?”
- “What changed? What stayed the same?”
- “What do you think will happen if...?”
- “How do you know from *this* — not from the book?”
- “You may change your mind when we look again.”
- “We’ll name it after we look. First, your eyes.”
- “Take your time. We’ll wait.”

First try-it for the student

Put ice on a plate. Cover any *melting* title. Ask the locked opening. Wait. The child points at one thing that changed (or has not yet) and says one notice sentence. That is enough for day one. Draw it if there is time; oral is enough if there is not.

How to fade help

Day 1: You model one notice while pointing. Child echoes with a different notice.

Day 2: They point first; you only restate what they said so they hear their own looking.

Day 3: They look alone for two minutes, then tell you three notices. You still wait before any name.

Day 4–5: Mixed lookable (ice + shadow, or leaf + window); you ask one follow-up and wait.

If they stall, return to one object — do not add a worksheet packet. Struggle before rescue: ask, wait, hint (two choices tied to what is visible), then model one notice.

When to stop talking

When the lookable is down and they are mid-look. When you feel the explanation rising in your chest. When they have given one honest notice — celebrate it and move to the try, rather than stacking five more adult sentences. When sibling talk needs a “whose turn?” instead of your second lecture.

Age-band moves

5–7. Prefer ice, leaf, shadow, sink/float, window weather. Oral exit or one-word/phrase labels. You read any print aloud. Drawings carry the record. Avoid density, force-as-fuel sermons, and method-poster words. Keep sits nearer twenty minutes when attention is thin. You scaffold “what stayed the same” if a try needs it.

8–10. Same lookables, plus a reason tied to what they saw (“I think it will change because I already see water”). Optional simple table of notices. They may hear one short name after the look — *melt*, *shadow*, *float* — then stop. If looking habits are weak, stay on 5–7 notice longer. A caption sentence is fine if decoding supports it; if not, oral still counts.

Place by skill, not birthday. A careful six-year-old who notices with reasons may sit beside an eight-year-old who still hunts vocabulary — different exit crumbs, same lookable.

Sibling note inside the hour

Share the ice plate. Younger child names one notice. Older child adds a because-I-see reason or spots a name-first rush in their own talk. Post-response wait so each speaks. The older child still needs their own looking move — not only tutoring.

Practice that actually builds learning

Library, walk, museum, kitchen, and windowsill *are* the practice. They do not replace the looking-and-trying move. Pick from the menu below. Do not assign every try-it this week.

Talk box

Opening (locked): “What do you notice — before I tell you what this is called?”

Follow-ups:

1. What do you notice first?
2. What changed?
3. What stayed the same?
4. What would you try next?
5. You may change your mind when we look again.

How to wait. Ask. Count a slow three (a classroom science wait-time convention, not a kitchen-table science experiment)¹⁴. After they stop, wait again. If they are mid-look, do not cut them off. Look at the ice, the leaf, or the tray if the silence is hard.

Stuck silence usually means: you asked name-first; the materials never landed; they think you want a vocabulary word; they offered a book-only guess; they think science means “repeat the parent’s explanation”; they think looking means “say what the screen already told me.” Point back at one detail. Offer two choices tied to what is visible. Do not fill the silence with the right name.

¹⁴Mary Budd Rowe, “Wait-Time and Rewards as Instructional Variables,” NARST / ERIC ED061103 (1972). Robert J. Stahl, “Using ‘Think-Time’ and ‘Wait-Time’ Skillfully in the Classroom,” ERIC ED370885 (1994). Classroom-origin; not a kitchen RCT.

In-home try-its

OBS-02 — Ice Plate Look

- **Band:** 5–7 (stretch 8–10)
- **Time:** 20–30 minutes
- **Materials:** Ice cube; two plates or paper towels; timer optional; blank paper
- **Safety:** Adult present; no eating lab ice; wipe slips
- **The fun:** Watching something change while you sit still — kids like catching the first drip
- **The skill:** Notice before name; predict, then look; drawing as a first record
- **Steps:** (1) Place ice. Cover any “melting” worksheet title. (2) Ask the locked opening. Wait. (3) Notice shape, wetness, sound. Predict what changes in about ten minutes. (4) Watch. (5) Name *melting* only after evidence. (6) Draw start and after.
- **Exit crumb:** Drawing of ice at start and after + one word or sentence for the change
- **Error patterns:** Saying “melted” before looking; licking ice from unclean surfaces; adult narrating the whole change

OBS-03 — Leaf Before Name

- **Band:** both
- **Time:** 20–30 minutes
- **Materials:** 3–5 fallen leaves from a known non-toxic area; magnifier optional
- **Safety:** Adult present outdoors if you collect; wash hands; no mouthing; skip unknown berries and leaves that look edible
- **The fun:** Sorting by what the eye catches — edge, color, veins — before anyone plays “what tree?”
- **The skill:** Notice-before-name on a living-world object
- **Steps:** (1) Sort by what you notice. (2) Adult refuses to quiz “what tree?” until after sorting talk. (3) Two ways these leaves differ. (4) Optional: one science word after the sort, if needed.
- **Exit crumb:** “Two ways these leaves differ: ____ and ____.”
- **Error patterns:** Identify-app first; toxic plant handling; taking leaves from protected sites

OBS-05 — Shadow Notice First

- **Band:** both
- **Time:** 15–25 minutes
- **Materials:** Sunny patch or lamp; toy or hand; chalk or tape
- **Safety:** Adult present; no looking at the sun; lamp cord control; cool LED preferred
- **The fun:** Making a shadow grow, shrink, or swing — then tracing it
- **The skill:** Notice size, shape, and direction before any “light travels in straight lines” talk
- **Steps:** (1) Make a shadow. (2) Locked opening. Wait. (3) Trace once. (4) Wonder what would change the shadow. (5) Name ideas only as far as the look can carry.
- **Exit crumb:** “My shadow is ____ because the light is ____.” (child words even if incomplete)
- **Error patterns:** Adult lectures rays first; staring into sun; hot lamp touch

OBS-01 — Window Weather First

- **Band:** both
- **Time:** 15–25 minutes
- **Materials:** Clear window view; blank paper; optional outdoor thermometer with adult
- **Safety:** Adult present for outdoor thermometer; no climbing on sills
- **The fun:** Two silent minutes of sky and street — then a list that is theirs
- **The skill:** Notices before weather-word dump; saves a list for a later weather week
- **Steps:** (1) Look outside about two minutes in silence. (2) “What do you notice?” Wait. (3) Child lists about five notices (cloud, wind evidence, wet/dry, light). (4) Only then may you name weather words that fit. (5) Save the list.
- **Exit crumb:** “I noticed ____ before we named it.”
- **Error patterns:** Adult narrates first; child copies a forecast app; inventing thunder not present

OBS-06 — Sink or Float Tray (look first)

- **Band:** 5–7
- **Time:** 20–30 minutes

- **Materials:** Clear bin of water; six safe objects (cork, coin, crayon, grape, spoon, eraser)
- **Safety:** Adult present; water spill plan; no batteries or phones in water; dry floor; no tasting
- **The fun:** One-at-a-time surprises — “I thought the grape would ____”
- **The skill:** Predict, then look; record; talk about heavy/size only after trials — density is not this hour’s goal
- **Steps:** (1) Predict column on paper. (2) Test one at a time. (3) Record. (4) Then talk. Do not teach density vocabulary as the hour’s goal.
- **Exit crumb:** “I was surprised that ____.”
- **Error patterns:** Explaining density before trials; electrical objects near water; tasting

OBS-07 — Magnet Mystery Bag

- **Band:** both
- **Time:** 15–25 minutes
- **Materials:** Magnet; bag of safe objects (paperclip, plastic bead, aluminum foil ball, key, crayon)
- **Safety:** Adult present; choke-size check for under-7; keep magnets away from screens, cards, and pacemakers as relevant; prefer weak fridge magnets for 5–7
- **The fun:** Mystery bag — stick or not, from evidence, not from hoping
- **The skill:** Predict, test, sort into two piles; name *magnetic* / *not magnetic* after piles exist
- **Steps:** (1) Predict stick/not. (2) Test. (3) Sort from evidence. (4) Name after piles exist.
- **Exit crumb:** Rule in kid words: “Things that ____ stuck.”
- **Error patterns:** Forcing foil to “count”; magnets near screens/cards; swallowing small parts

OBS-04 — Sound Walk Sit (optional extra)

- **Band:** both
- **Time:** 15–25 minutes
- **Materials:** Clipboard or paper; pencil; quiet outdoor or porch spot
- **Safety:** Adult present outdoors; sidewalk rules
- **The fun:** Three silent minutes of tallying — bird, car, wind, voice

- **The skill:** Listen first; then ask which sounds need living things
- **Steps:** Sit about three minutes; tally; no naming “ecology” yet; then which sounds need living things.
- **Exit crumb:** Top two sounds + “I think ____ came from ____.”
- **Error patterns:** Filling tally from memory of yesterday; unsafe street edge without adult

OBS-08 — Jar Condensation Look (8–10 stretch)

- **Band:** 8–10 (stretch 5–7 with adult lead)
- **Time:** 20–30 minutes
- **Materials:** Clear jar; warm water; lid or plate; ice optional on the lid with adult
- **Safety:** Adult handles hot water; no boiling for 5–7; towel for drips
- **The fun:** Watching droplets appear as if from nowhere — then arguing, kindly, where they came from
- **The skill:** Describe before saying “water cycle”
- **Steps:** Observe empty jar; add warm water; cover; watch; describe; name only after looking.
- **Exit crumb:** “Water appeared on ____ ; I think it came from ____.”
- **Error patterns:** Calling it “sweat of the jar” as the final science without talk; boiling water unsupervised

TALK-01 — Three-Second Notice (pair lightly)

- **Band:** both
- **Time:** 10–15 minutes (nests in any hour)
- **Materials:** Whatever lookable is already out
- **Safety:** Matches the paired try-it
- **The fun:** The parent actually waits — children notice
- **The skill:** Wait after “what do you notice?”; uptake one phrase
- **Steps:** Ask the locked opening; silently count three; do not leap in; then uptake one phrase they said.
- **Exit crumb:** Child’s notice sentence repeated back accurately
- **Error patterns:** Answering your own question at one second; praising only “right” science words

Out-of-home try-its

OUT-01 — Park Tree Bark Rub (notice before name)

- **Band:** both
- **Time:** 20–30 minutes on site
- **Materials:** Paper; crayon; respect for trees
- **Safety:** Adult present; trail rules; poison-ivy awareness; wash hands; no peeling living bark
- **The fun:** Two rubs that look different even when the trees “just look like trees”
- **The skill:** Notice-before-name outdoors; texture as data
- **Steps:** Rub two barks; compare texture; talk after looking; no species quiz first.
- **Exit crumb:** Two rubs labelled (child words: rough / smoother / lines)
- **Error patterns:** Damaging trees; taking protected plants; a scavenger checklist of twenty items

OUT-07 — Library Science Shelf Peek (motivation)

- **Band:** 8–10 (5–7 with you reading)
- **Time:** 25–35 minutes at the library
- **Materials:** Library card; notebook
- **Safety:** Adult present as needed for the trip; street safety; aisle courtesy
- **The fun:** Choosing one book after a home try — not instead of one
- **The skill:** Compare one picture to your own trial drawing; the book does not overwrite home data
- **Steps:** Find one trade book on this week’s idea. Compare one picture to your home drawing. Note agreement and difference. Resist checking out twelve books as coverage.
- **Exit crumb:** “Book showed ; *my trial showed* .”
- **Error patterns:** Replacing own data with the book picture; unchecked internet printouts as “library”

Blocked practice for a new move

When notice-before-name is brand new, use one lookable only (ice or leaf). Do not mix ice, magnets, and weather in the same sit. Master one object. Blocked practice

means the child practices the new move without extra categories competing for attention.

Mixed practice

When the child can notice on one object, mix two lookables later in the week (ice + shadow, or leaf + window). Ask which notice is strongest. Mixed practice is for *after* the move has landed — not day one.

One incorrect example to diagnose

Illustration (not a reported family): A parent opens with “This is called melting. What is melting?” The child freezes, then recites a sentence from a video. The ice sits untouched. The parent fills the silence with a five-minute lecture on states of matter. The notebook stays blank.

What went wrong: Name-first; wait-time zero; explanation before lookable; demo with no child’s record; coverage itch.

Tomorrow’s repair: Cover any melt title. Put the ice down. Ask the locked opening. Wait. Celebrate one notice. Draw before tidy-up. Only later — if at all this sitting — offer *melt* as an optional label under a drawing already owned.

A sample week (menu, not a calendar)

This is grain only — not a research trial and not a required schedule.

- **Monday:** OBS-02 Ice Plate Look; locked talk box; oral crumb; optional TALK-01 wait nested in.
- **Wednesday:** Revisit Monday’s drawing two minutes; OBS-05 Shadow Notice First; draw the shadow.
- **Friday:** OBS-03 Leaf Before Name; diagnostic — listen for name-first; done-enough checklist; date the notebook.
- **One outing:** OUT-01 bark rub *or* library one picture if weather or energy blocks the park.

Mixed-age sibling version: same ice plate; younger names notices; older adds because-I-see; one shared outing with different exit expectations.

For the student

Science is looking and trying.

You look at something real. You notice. You try one small thing. You say what you saw — before anyone tells you the science word.

You do not have to wait until you are older to look. You do not have to memorize a poster to begin. You point. You talk. You may change your mind when you look again.

Tiny worked example

Ice on a plate. You look. You say, “The edges are wet.” You wait. Later you say, “It’s smaller. There is water.” You did science looking. The word *melt* can come after.

Another example: a leaf. You do not name the tree first. You say, “This edge is bumpy. This one is smooth.” Two ways they differ. That is looking.

Try 1

Put one object on the table with an adult. Point at one thing that changed — or one thing you notice. Say one sentence. Point while you talk. If writing is hard today, saying it is enough.

Try 2

Look at the same object later. What is the same? What is different? You may change your mind. Changing your mind after a second look is smart, not a mistake to hide.

Explain it back

Tell someone at home: “Science looking means I say what I notice — before we say the name.”

Challenge

On a walk (with an adult), find bark, a puddle edge, or a shadow. Say one notice before anyone names it. No talking to strangers for answers. Your eyes do the work. At home, stick the sketch or the bark rub in the notebook.

You are allowed to wait. You are allowed to change your mind when you look again. You can point at what you saw. That is enough for today.

If it isn't clicking

Diagnostic 1 — They only hunt names

What you hear: “Is it melting?” before they look; freeze when asked “what do you notice?”; hunting for the word you “want.”

Next move: Remove vocabulary cards from the table for three sits. Locked opening only. Praise notices by pointing (“you noticed the wet edge”). Add a name later as an optional label under a drawing they already own. If an older sibling keeps name-quizzing, give them a because-I-see job so the younger child can look.

Diagnostic 2 — Materials never landed

What you hear: Vague talk; looking at you instead of the ice; “I don’t know” immediately.

Next move: Put the lookable in their sight line. Move your face out of the way. Cover worksheet titles. Give two silent minutes of looking before any question. If the object is too far, too small, or still in a bag, the hour cannot start.

Diagnostic 3 — Demo hour, empty record

What you see: You did all the hands. They watched. Nothing in the notebook. “It was fun” is the only crumb.

Next move: Tomorrow, their hands first. One try. One drawing before tidy-up. You may help with safety. You may not replace their looking. If drawing shame shows up (“I’m bad at art”), model a three-line drawing badly on purpose. Science drawing is data, not art class. Chapter 2 goes deeper on that repair.

When to slow down. If they still cannot point at one notice after several ice or leaf sits, stay here. Do not add fair-test language yet.

When to go ahead. If they can point at a drawing and say what happened, and they tolerate a short wait after a question, move into Chapter 2’s drawing-as-notebook with them.

When to get a human tutor. If looking will not start after weeks of visible materials, short sits, and no name-first quiz — and especially if attention, vision, or language concerns sit underneath — a human teacher who can sit beside the child is worth more than a new kit. No shame. The book is a path, not a verdict.

Tools, including AI

Optional. Parent-hosted only. Full allow/ban list lives once in **The Science Hour**. Point back there.

You may ask a tool, on your account, to explain notice-before-name from a **named** page, to draft a blank “I notice ____” card with PARENT KEY, or to write a four-line SCRIPT you will say *after* they look. Unaided first on looking. No live open chat as the child’s partner at 5–7. Parent in the room at 8–10. Do not put a model summary on the table as the observation. Do not invent what the ice did. Do not generate a “microscope photo” as today’s result. Do not use TTS to replace eyes on the ice that is today’s lesson. Detectors are not for grading a child’s “the ice got smaller.” Hermes Agent and Grok Bot are not science partners for the child.

Sample SCRIPT request you might type *to your tool, for you*: “I am teaching notice-before-name with ice in a cup. Write a four-line SCRIPT I will say after my child looks. Level 1: what do you notice? Level 2: point to one thing that changed. Level 3: say it in your words. PARENT KEY only. Do not invent an observation. Do not summarize melting as the looking.”

Analog tool this week: a magnifier and a blank page. That is enough.

What “done enough” looks like

Before you leave this chapter’s focus for a while, check:

- ☐ Child can point at one notice on a real lookable without hearing the name first.
- ☐ Child can say one thing that changed and, with help if needed, one thing that stayed the same.
- ☐ You put a lookable down before the explanation at least three times this week.
- ☐ You heard at least one wrong turn (name-first, demo with no record, or coverage itch) and asked a better question without grabbing the answer.
- ☐ One out-of-home or library noticing happened, or is scheduled.
- ☐ Exit crumb exists: spoken, drawn, or sorted — dated in the notebook under **Science**.
- ☐ Child can explain notice-before-name back in one plain sentence (oral is enough).

Showable two ways; holds after a night. Tomorrow, without reteaching the whole lecture, can they still point at yesterday's ice drawing and say what happened? If yes, the move is landing. If no, revisit — do not add a kingdom chart.

Placement by skill, not birthday. If a younger child shows the checklist, move into Chapter 2's drawing-as-notebook with them. If an older child still hunts names only, stay on ice and leaf. A publisher's "grade 2 science workbook" is a scope, not a legal grade, and not a transcript line.

You understand observe-before-explain as a sequence. You know what name-first sounds like. You know what a demo with no record feels like. You can put the lookable down before the explanation. That is done enough for Chapter 1.

Chapter 2 — Looking and drawing what you saw

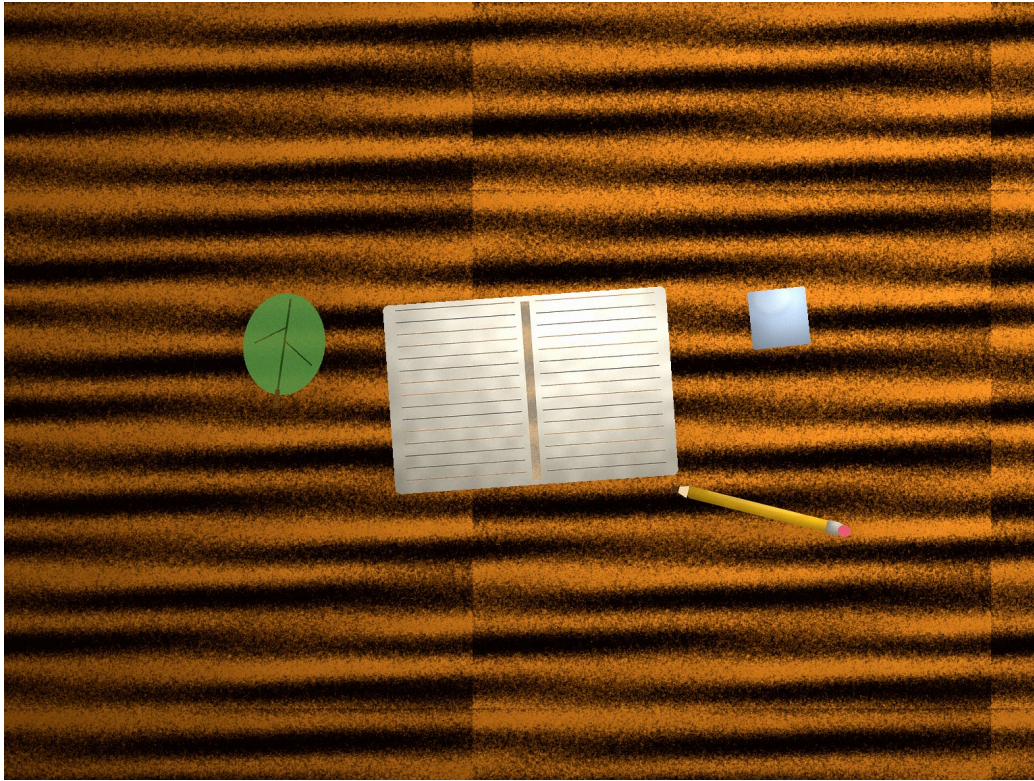


Figure 4: A leaf beside an open notebook with a three-line drawing and pencil — still-life, no people, no logos.

Why this matters

Drawing is not cute fluff.

At 5–7 the drawing *is* the notebook. At 8–10, picture-plus-sentence carries the record. A finger on the page can answer “what did you see?” Without a record, talk collapses into “it was fun.” Without talk, the record is a coloring page.

Adult-filled observations are not the student’s data. Generated observations are not the student’s data.¹⁵ A pretty picture that does not match the try is not the student’s data. A generated “microscope photo” is not a result.

This chapter is the second half of the book’s looking spine. Chapter 1 gave you notice-before-name and lookable-before-explanation. Chapter 2 gives you the record: what the child saw, in marks they made, in words they said.

Worth the struggle: five minutes alone drawing tomorrow’s leaf badly on purpose changes the hour. You will hear “I’m bad at art.” You will catch yourself rewriting their seeing while they look at the dog. You will recognize a worksheet that already lists condensation. You will know what to ask next without grabbing the pencil.

Your child will learn to point at a drawing — and to say what is still missing — before anyone grades the art. That habit travels into fair tests, weather tables, light paths, and captions later. Skip it, and every later chapter fights an empty notebook.

If you take nothing else from this chapter into next week, take this: protect the child’s record. Scribe only what they said. Do not overwrite. Science drawing is data, not art class.

This week you can learn to hear an adult-filled page and a generated observation filed as real. Today your child can draw before-and-after, or look-label one object, and point at the drawing to answer one question. That is the week’s work. A path through elementary science looking still does not require a diploma, a percentile, or a pretty portfolio.

¹⁵On generated images and invented observations as non-data: parent-side honesty from the Science Hour AI box in this book; publisher norms that a generated figure is not a result. See also William H. Walters and Esther Isabelle Wilder, “Fabrication and Errors in the Bibliographic Citations Generated by ChatGPT,” *Scientific Reports* 13 (2023): 14045 — fluency is not evidence.

For the parent: understand it yourself

One everyday picture

A grocery list with a sketch of the missing item — a bottle shape, a leaf, three circles for eggs. Nobody grades the art. The sketch carries the information. You can shop from it.

That is science drawing at 5–10 in one frame. Three lines can still be data. Labels beat decoration. The question is not “is this pretty?” The question is “does this page show what happened?”

One precise picture

Recording at this age is primarily visual and oral, then lightly verbal. National K–2 science practice language asks children to draw what they observe and to say “I think ____ because I see ____.”¹⁶ Performance expectations in the early elementary band repeatedly ask for observations and drawings.¹⁷ Those are a map, not a kitchen law. They tell you the record is part of the science, not a garnish after the “real” work.

Sister English craft in this house is simple: oral language is the fuel tank before heavy print. For science at 5–7, the child may not yet write a procedure. You may scribe a caption the child said. The child’s drawing carries the evidence strand. Print load rises at 8–10 as decoding strengthens — still not a lab-report genre dump.

A finger on the drawing is the honesty check. If you cover the caption and ask “what did you see?”, can they point to the part that shows the change? If a stranger saw only this page, could they guess what you tried? Those two questions keep the page honest.

What does not count as the child’s data:

- You wrote the “right” seeing while they looked away.

¹⁶California Department of Education, *2016 Science Framework for California Public Schools* (Sacramento, 2016), Chapter 3: K–2 practices include drawing diagrams and the language “I think ____ because I see or know ____.”

¹⁷NGSS Lead States, *Next Generation Science Standards* (Washington, DC: National Academies Press, 2013): K–2 performance expectations that ask for observations and drawings — used here as a map, not a mandate.

- A chat window summarized “what happens when ice melts” and that paragraph went in the folder.
- A generated gel, micrograph, sky photo, or “what I saw under the glass” filed as the result.
- A beautiful leaf copied from a textbook that does not match the leaf on the table.

Labelled illustration for two minutes of looking together is a different design. Filing fiction as evidence is not.

This is a useful set of practices, not a promise that every home will see the same result. Drawing quality is not a science percentile. Then teach.

Wrong answers you should be able to hear

1. **You write the “right” seeing while they look at the dog.** Usually means the adult filled the observation. Repair: protect the record. Scribe only what they said. If they said “wet,” write *wet* — not *condensation formed on the surface*.
2. **A worksheet that already lists condensation.** Usually means a model summary sat down as the observation. Repair: blank page. Child’s eyes first. Label any generated text as fiction and put it away.
3. **Pretty picture, wrong try.** Usually means art-class habits swallowed the record. Ask: “What does your drawing show that happened — and what is still missing?” Point back at the ice, the leaf, or the shadow.
4. **“I’m bad at art.”** Usually means they think the grade is beauty. Repair: you draw the leaf badly on purpose in three lines. Labels. Date. “This is data, not art class.”
5. **Refusing to draw; only talking.** Talking is welcome. The record still needs a mark on the page. A scribble with one label beats a fluent oral summary that vanishes by tomorrow.

Five-minute parent warm-up

Alone with tomorrow’s leaf. Draw it yourself badly on purpose. Three lines. Edge, stem, one spot. Label three parts in ordinary words.

Notice how little drawing still carries data. Practice the locked opening aloud once: “What does your drawing show that happened — and what is still missing?”

Then stop rehearsing. You are practicing a data drawing, not a still-life lesson.

What the record is for

Without a record, Friday has nothing to revisit.¹⁸ Without a date, a stranger cannot tell when the look happened. Without a label, a shadow trace is a doodle.

The notebook title a stranger can read is **Science**. Not a brand. Not Little Thinkers Science I.

At 5–7, done-enough for the record is: finger on the drawing can answer “what did you see?” At 8–10: a sentence points at the picture; a stranger could guess the trial.

Chapter 3 will ask the record to show one change and one same. Chapter 7 will ask a caption and a short claim. This chapter only asks that the page show the look.

Scribe versus overwrite

Scribing is the child’s words in your handwriting, labelled as such. Overwriting is your seeing in their notebook.

If they said “wet,” write *wet*. If they said “the round part is gone,” write that. You may add, in a different color, one science word *after* they have owned the look — and you say it is your add-on, not their noticing.

If writing is painful, oral plus your scribing still counts. Science looking does not wait for fluent handwriting. Sister English craft in this house is the same idea: talk first; print load later. The page still needs a mark they can point to tomorrow.

What is still missing is an honest science question. A missing puddle on an after-ice drawing is data about the look, not a failure of art. A missing stem on a leaf drawing is a chance to look again. A missing date is a chance to write one. The locked opening trains that honesty: the page shows something, and it does not show everything yet.

¹⁸National Research Council, *Taking Science to School* (Washington, DC: National Academies Press, 2007), Strand 2: generate and evaluate evidence. National Research Council, *Ready, Set, SCIENCE!* (Washington, DC: National Academies Press, 2008): record and talk after investigation; activity without record is not the hour.

How to teach it this week

Session shape

Use the Science Hour beats from the front of the book. This week's focus: drawing as the notebook, and protecting the child's record.

1. **Warm-up looking (2–3 min).** Yesterday's drawing. "What does this page show?" No new object yet. Finger on the page.
2. **Lookable down (3–8 min).** Leaf, ice, shadow, cloud, or a ramp setup at 8–10. Notice first — Chapter 1's move still holds. Cover any title that already names the result.
3. **Look / try (5–12 min).** The child looks or runs the short try. Adult present for safety. You do not draw for them during the look.
4. **Record (5–10 min this week — a little longer than usual).** Draw what they see. Labels. At 8–10, one sentence under the picture. You may scribe what they said. You do not overwrite.
5. **Talk box (2–5 min).** Locked opening. Wait. Point at what is still missing. They may add one mark. Date the page. Title: **Science**.

If the day only holds twenty minutes: lookable + drawing + locked opening. Skip the pretty-up. The three-line drawing is the hour.

Exact wording you can say

- Opening (locked): **"What does your drawing show that happened — and what is still missing?"**
- "Point to the part that shows the change."
- "What did you leave out?"
- "If a stranger saw only this page, what would they guess we tried?"
- "Add one sentence under the picture." (8–10)
- "I'll write the words you said. Tell me what to write."
- "This is data, not art class."
- "You may change the drawing when we look again."
- "Show me with your finger — not with a new speech."

First try-it for the student

Look-label one leaf. Draw what you see (edge, stem, spots). Label three parts with child words. Point at the drawing to answer one question. That is enough for day one. Oral labels you scribe still count.

How to fade help

Day 1: You model a three-line data drawing badly on purpose. Child draws the same object. You do not fix their lines.

Day 2: They draw first. You ask the locked opening. They add one missing part.

Day 3: Before-after boxes (ice). They draw both. Arrow for change. You scribe one sentence they said.

Day 4–5: They point at yesterday’s page and retell. Then a new drawing with less prompting. At 8–10, they add the sentence without hearing yours first.

If they stall, return to three lines and three labels — do not add an art lesson. Struggle before rescue: ask, wait, hint (“where is the stem on the leaf *and* on the page?”), then model one label.

When to stop talking

When they are mid-draw. When you feel the urge to “fix” a line. When they have pointed at one true part of the page — celebrate it rather than stacking five adult labels. When “I’m bad at art” needs one calm sentence, not a pep talk that becomes the hour.

Age-band moves

5–7. Drawing is the notebook. Parent may scribe one sentence the child said. Three labels in child words. Oral exit is enough. You read any print aloud. Skip procedure writing. Skip grading beauty. Keep sits nearer twenty-five minutes when drawing is new.

8–10. Picture-plus-sentence. Simple two-box before/after. Ramp setup sketch with Change / Same labelled when they are ready for Chapter 3’s edge. “If a stranger saw only this page...” is a fair question. If decoding is hard, you still

scribe; science looking does not wait for fluent writing. If looking habits are weak, stay on 5–7 three-line drawings longer.

Place by skill, not birthday. A careful six-year-old whose drawing answers “what happened?” may sit beside a nine-year-old whose pretty leaf does not match the table — different next moves, same object.

Sibling note inside the hour

Share the leaf. Younger child draws and labels in child words. Older child adds a sentence or spots what is missing on their own page. They do not “correct” the younger drawing into a textbook leaf. Post-response wait so each points at their own page.

Practice that actually builds learning

Library, walk, museum, kitchen, and windowsill *are* the practice. They do not replace the looking-and-drawing move. Pick from the menu below. Do not assign every try-it this week.

Talk box

Opening (locked): “What does your drawing show that happened — and what is still missing?”

Follow-ups:

1. Point to the part that shows the change.
2. What did you leave out?
3. What do you *see* on the table that is not yet on the page?
4. If a stranger saw only this page, what would they guess we tried? (8–10)
5. You may change your mind — add one mark, or add one sentence under the picture.

How to wait. Ask. Count a slow three (a classroom science wait-time convention, not a kitchen-table science experiment)¹⁹. After they stop, wait again. If they are

¹⁹Mary Budd Rowe, “Wait-Time and Rewards as Instructional Variables,” NARST / ERIC ED061103 (1972). Robert J. Stahl, “Using ‘Think-Time’ and ‘Wait-Time’ Skillfully in the Classroom,” ERIC ED370885 (1994). Classroom-origin; not a kitchen RCT.

mid-draw, do not cut them off. Look at the page and the object, not at the child, if the silence is hard.

Stuck silence usually means: they think you want a pretty picture; they are ashamed of the drawing; you already filled the page; they are hunting a vocabulary word; the object never landed; they offered a book-only leaf; they think science means “repeat the parent’s explanation.” Point back at one detail on the table. Offer two choices (“is the missing part the stem, or the wet edge?”). Do not fill the silence by redrawing it for them.

In-home try-its

DRAW-01 — Look-Label Leaf

- **Band:** both
- **Time:** 15–25 minutes
- **Materials:** One leaf; pencil; colored pencils optional
- **Safety:** Non-toxic leaf source; wash hands
- **The fun:** Three labels that belong to *this* leaf — spots, tears, and all
- **The skill:** Drawing as data; labels beat decoration
- **Steps:** (1) Draw what you see (edge, stem, spots). (2) Label three parts with child words. (3) Adult adds one science word after, if needed. (4) Locked opening. Wait.
- **Exit crumb:** Three labels on the page
- **Error patterns:** Copying a textbook leaf; inventing insects not present; skipping labels; adult redrawing

DRAW-02 — Before-After Ice

- **Band:** 5–7 (useful at 8–10 too)
- **Time:** 20–25 minutes
- **Materials:** Ice; two boxes on paper labelled Before / After
- **Safety:** Adult present; no eating lab ice; wipe slips (same as Ice Plate Look)
- **The fun:** Catching the change in two pictures with an arrow
- **The skill:** Record of what happened over time; sentence under the drawings
- **Steps:** (1) Draw before melting progresses. (2) Wait. (3) Draw after. (4) Arrow for change. (5) One sentence — child’s words, scribed if needed.
- **Exit crumb:** One sentence under the drawings

- **Error patterns:** Drawing “puddle” before it exists; adult redrawing the child’s after; name-first *melt* title on the page

This try-it is also an OBS-02 record sit: same ice look, now with the drawing as the product.

DRAW-03 — Shadow Trace Pair

- **Band:** both
- **Time:** 20–30 minutes
- **Materials:** Chalk or tape; object; sun or lamp
- **Safety:** Adult present; no looking at the sun; lamp cord control; cool LED preferred
- **The fun:** Two colors — first trace and later trace — and the story of what moved
- **The skill:** Pair of records; label which was first
- **Steps:** (1) Trace shadow. (2) Move the light, or wait for the sun to move. (3) Trace again in a second color. (4) Label which was first. (5) Locked opening.
- **Exit crumb:** “The shadow moved when ____.”
- **Error patterns:** Traces from imagination; sun-staring; adult moving the object while calling it “the sun”

This try-it is also an OBS-05 record sit: same shadow notice, now with traces as data.

DRAW-04 — Magnifier Bugs Soft

- **Band:** both
- **Time:** 15–25 minutes
- **Materials:** Magnifier; safe insect photo *or* insect in a clear viewer (no harm)
- **Safety:** Adult present; no bees/wasps handling; bite/sting plan; wash hands; release outdoor insects after
- **The fun:** Counting legs under glass — then letting the visitor go
- **The skill:** Draw what is seen, not a cartoon; care for living things
- **Steps:** Draw legs, wings, segments actually seen. Refuse squishing. Release outdoor insects after.
- **Exit crumb:** “I counted ____ legs.”

- **Error patterns:** Cartoon bug instead of looking; collecting protected species; bee handling; generated “bug photo” filed as the look

DRAW-05 — Ramp Setup Sketch (8–10)

- **Band:** 8–10
- **Time:** 15–20 minutes
- **Materials:** Ramp materials; notebook
- **Safety:** Adult present for tall ramps; eye protection if anything can fly off
- **The fun:** A plan on paper before the race — like a map of the try
- **The skill:** Sketch the setup; mark what will change and what stays same (links to Chapter 3)
- **Steps:** Before racing, sketch. Labels: Change / Same on the sketch. Then, if time, one run — still one change.
- **Exit crumb:** Labels Change / Same on the sketch
- **Error patterns:** Pretty picture with no labels for what will change; skipping the sketch then arguing about setup

DRAW-06 — Cloud Shape Record

- **Band:** both
- **Time:** 15–20 minutes
- **Materials:** Sky view; paper; gray crayon
- **Safety:** No sun stare; adult for outdoor
- **The fun:** Today’s sky as a dated picture — not a poster of cloud types
- **The skill:** Record now; names later, after several days of drawings
- **Steps:** Draw today’s cloud look. Note time. Do not force type names until after several days of drawings.
- **Exit crumb:** Time + “clouds looked ____.”
- **Error patterns:** Labelling “cumulonimbus” from a poster without matching look; drawing from memory indoors; a generated sky filed as today’s weather

WRITE-01 — Object Label Three (pair lightly)

- **Band:** both
- **Time:** 10–15 minutes
- **Materials:** Today’s drawing; pencil

- **Safety:** None beyond the paired try-it
- **The fun:** Three words that belong to *this* page
- **The skill:** Labels as data, not decoration
- **Steps:** Three labels in child words. You may scribe. One optional science word after.
- **Exit crumb:** Three labels
- **Error patterns:** Adult labels only; copying a diagram from a book

WRITE-02 — Caption Under Trial (pair lightly)

- **Band:** 8–10 (scribe at 5–7)
- **Time:** 10 minutes
- **Materials:** Today's drawing
- **Safety:** None beyond the paired try-it
- **The fun:** One sentence that could only come from this try
- **The skill:** Picture-plus-sentence
- **Steps:** One sentence under the picture. Child's words. Because-the-drawing, not because-the-video.
- **Exit crumb:** The caption itself
- **Error patterns:** Generated paragraph filed as the caption; parent's summary replacing their words

Out-of-home try-its

OUT-01 into a labelled drawing — Park Tree Bark Rub

- **Band:** both
- **Time:** 20–30 minutes on site, plus a few minutes to label
- **Materials:** Paper; crayon; pencil for labels
- **Safety:** Adult present; trail rules; poison-ivy awareness; wash hands; no peeling living bark
- **The fun:** Two rubs that become a labelled page, not a craft pile
- **The skill:** Outdoor noticing into a dated record
- **Steps:** Rub two barks. Compare texture. At home or on a clipboard, label each rub in child words. Locked opening on the page.
- **Exit crumb:** Two rubs labelled
- **Error patterns:** Damaging trees; twenty-item scavenger packet; leaving the rubs unlabelled so they cannot be read next week

OUT-04 — Museum One Exhibit (postcard sketch)

- **Band:** both
- **Time:** 30–40 minutes in the museum (fits Science Hour plus a little)
- **Materials:** Notebook; pencil; optional postcard of the *same* exhibit after looking
- **Safety:** Adult present; museum rules; lost-child plan; hands-off
- **The fun:** One exhibit, ten quiet minutes, a sketch that is yours
- **The skill:** Observe before plaque; drawing as the record of one thing
- **Steps:** Choose **one** exhibit. Observe about ten minutes. Draw. One question. Postcard (if you buy one) compared *after* the sketch — the postcard does not replace the looking. Refuse a whole-museum worksheet dump.
- **Exit crumb:** Drawing + one question
- **Error patterns:** Racing galleries; touching forbidden objects; copying the plaque before looking; photographing every hall as “the record”

OUT-06 — Garden Insect Count Soft (observe, do not harm)

- **Band:** both
- **Time:** 20–25 minutes
- **Materials:** Tally sheet; agreed square of garden or park; optional magnifier
- **Safety:** Adult present; sting plan; allergy; no bees-nest approach; do not kill; wash hands
- **The fun:** A five-minute count in a small square — ants, flies, other
- **The skill:** Tally plus a quick sketch of one visitor; care
- **Steps:** Count what you see in five minutes. Classify roughly (ant / fly / other). Sketch one. Release any visitor. Date the tally.
- **Exit crumb:** Tally totals + one sketch
- **Error patterns:** Stomping; collecting protected species; generated tallies; cartoon insects with no look

Blocked practice for a new move

When drawing-as-data is brand new, use one object only (leaf or ice). One page. Three labels. Do not mix leaf, shadow, and clouds in the same sit. Master the “finger on the page answers the question” move.

Mixed practice

When the child can look-label one object, mix a before-after ice with a shadow trace later in the week. Ask which page would tell a stranger more. Mixed practice is for *after* the data-drawing move has landed — not day one.

One incorrect example to diagnose

Illustration (not a reported family): A parent photographs the ice, pastes the photo into a chat window, and asks the tool to “write what we should have seen.” The printed paragraph goes in the notebook. The child’s drawing is thrown away because “it didn’t look right.” The child learns that pretty machine words beat their eyes.

What went wrong: Invented observation; model summary as the record; generated prose filed as looking; adult-filled notebook; art-shame.

Tomorrow’s repair: Blank page. Real ice. Child draws. You scribe only their words. Ask the locked opening. Celebrate one true mark. Put the machine paragraph in a “fiction / parent prep” pile if you still want it — never in the child’s data.

A sample week (menu, not a calendar)

This is grain only — not a research trial and not a required schedule.

- **Monday:** DRAW-02 Before-After Ice (OBS-02 as a record sit); locked talk box; scribed sentence.
- **Wednesday:** DRAW-01 Look-Label Leaf; WRITE-01 three labels; revisit Monday’s ice page for two minutes.
- **Friday:** DRAW-03 Shadow Trace Pair; diagnostic — listen for adult-filled obs or “I’m bad at art”; done-enough checklist; date the notebook.
- **One outing:** Bark rub into a labelled drawing, *or* museum one-exhibit sketch, *or* garden insect count soft.

Mixed-age sibling version: same leaf; younger three labels; older caption sentence; one shared outing with different exit expectations.

Chapter 1 lookables, now as records

Ice Plate Look and Shadow Notice First still belong this week — not as new noticing lessons, but as record sits. Use the same ice and the same shadow. This time the product is the page: before/after boxes, or a two-color trace, dated, labelled, able to answer the locked opening.

If Monday’s ice hour produced only talk, Tuesday’s job is the drawing. If Wednesday’s shadow hour produced only a demo, Thursday’s job is the trace. The lookable does not have to be new. The record does.

For the student

A science drawing is a record of what you saw.

It does not have to be pretty. It has to be true to the look. Three lines can still be data. Labels help. A sentence under the picture helps when writing is ready.

Your finger on the page can answer “what did you see?” That is the job.

Tiny worked example

A leaf on the table. You draw the edge, the stem, one spot. You write *bumpy*, *stem*, *spot* — or an adult writes the words you said. Someone asks, “What does your drawing show?” You point at the spot. You did science recording.

Another example: ice. Two boxes: Before and After. Before is a white square. After is a smaller shape with a puddle. An arrow. You say, “It got smaller. There is water.” The page shows what happened.

Try 1

Draw one object from looking — not from a book picture. Label three parts. Point at the drawing to answer one question.

Try 2

Make a before-and-after page (ice, or a shadow at two times). What does the after drawing show that the before does not? What is still missing?

Explain it back

Tell someone at home: “My drawing is the notebook. It shows what I saw. It does not have to be art.”

Challenge

On a walk or at a museum (with an adult), sketch one thing you actually looked at. Label it. At home, point to the part that shows what happened or what you noticed. No copying a poster first. Your eyes, then the page.

You are allowed to wait. You are allowed to add a mark when you look again. You can point at what you saw. That is enough for today.

If an adult writes your words, they should be *your* words. If a screen writes a paragraph, that paragraph is not your looking. Your page is the one with your marks on it.

If it isn’t clicking

Diagnostic 1 — Drawing shame

What you hear: “I’m bad at art.” Crumpled pages. Refusal to pick up the pencil. Waiting for you to draw it “right.”

Next move: You draw badly on purpose in three lines, in front of them. Labels. Date. Say once: “This is data, not art class.” For three sittings, the only grade is “can a finger on the page answer what did you see?” If an older sibling still beauty-grades, give them their own page and a stranger-test (“what would they guess we tried?”).

Diagnostic 2 — Adult-filled or generated record

What you see: Your handwriting is the whole observation. Or a fluent paragraph that does not sound like the child. Or a worksheet that already lists the result.

Next move: New blank page. Child’s marks only. You may scribe *their* words in a different color, labelled “words I said.” Put machine text away. If you used a tool to prepare, keep that on your account — not in their notebook. Point back to

the Science Hour AI box: unaided first; no invent-observation; generated image is not data.

Diagnostic 3 — Pretty page, no look

What you see: A textbook leaf. A cartoon sun. A puddle drawn before the ice changed. The object is still in the bag.

Next move: Put the lookable in the sight line. Cover book pictures. Two silent minutes of looking. Then three lines only. Ask what is still missing. If they invent insects that were not there, walk back to the leaf together. “Show me on the leaf.”

When to slow down. If they still cannot point at one true part of a drawing after several sits, stay here. Do not add fair-test sketches yet.

When to go ahead. If a finger on the page answers “what did you see?”, and an 8–10 child can add one sentence that points at the picture, move into Chapter 3’s one-change tries with the record habit already warm.

When to get a human tutor. If fine-motor work or vision makes drawing painful, a human teacher or occupational-therapy conversation is worth more than pushing prettier leaves. Oral record plus your scribing still counts as science looking. No shame. Adapt the page; do not abandon the look.

Tools, including AI

Optional. Parent-hosted only. Full allow/ban list lives once in **The Science Hour**. Point back there.

You may ask a tool, on your account, to explain drawing-as-record from a **named** page, to draft a blank before/after card with PARENT KEY, or to write a four-line SCRIPT you will say *after* they draw. Unaided first on looking and drawing. No live open chat as the child’s partner at 5–7. Parent in the room at 8–10. Do not invent an observation. Do not put a model summary on the table as the record. Do not generate a leaf, a gel, a micrograph, or a sky and file it as the student’s result. Generated image $\neq$ data. Do not use TTS to replace eyes on today’s object. Do not detector-grade a child’s “the ice got smaller.” Hermes Agent and Grok Bot are not science partners for the child.

Sample SCRIPT request you might type *to your tool, for you*: “I am teaching drawing as data with a leaf. Write a four-line SCRIPT I will say after my child draws. Level 1: what does your drawing show? Level 2: point to one part that matches the leaf. Level 3: what is still missing? PARENT KEY only. Do not invent labels. Do not redraw the leaf for them.”

Analog tools this week: a pencil, a magnifier, two boxes labelled Before / After. That is enough.

What “done enough” looks like

Before you leave this chapter’s focus for a while, check:

- ☐ Child can point at a drawing and say what happened (5–7) or add one sentence that points at the picture (8–10).
- ☐ Three labels or a before/after pair exist in the dated notebook under **Science**.
- ☐ You scribed only what they said — or they wrote it — at least three times this week.
- ☐ You heard at least one wrong turn (adult-filled obs, generated observation, pretty-but-false page, or “I’m bad at art”) and asked a better question without grabbing the pencil.
- ☐ One out-of-home labelled drawing, postcard sketch, or insect tally happened, or is scheduled.
- ☐ Exit crumb exists: finger on the page answers “what did you see?”
- ☐ Child can explain drawing-as-notebook back in one plain sentence (oral is enough).

Showable two ways; holds after a night. Tomorrow, without reteaching, can they still point at yesterday’s page and tell what happened? If yes, the move is landing. If no, revisit — do not add a lab-report template.

Placement by skill, not birthday. If a younger child shows the checklist, take the record habit into Chapter 3’s one-change tries. If an older child still produces pretty pages that do not match the try, stay on look-label and before/after. A publisher’s “grade 2 science workbook” is a scope, not a legal grade, and not a transcript line.

You understand drawing as data. You know what an adult-filled observation sounds like. You know what a generated page filed as real looks like. You can protect the child’s record. That is done enough for Chapter 2.

Chapter 3

Try it: one change at a time



Figure 5: A kitchen-table still-life from a high three-quarter view: a board-as-ramp with two surface strips (smooth paper and a towel), one toy car ready at the top, a short checklist card labelled Change / Same, a dated notebook open to a blank line, a soft ball and floor marks nearby. No people. No logos. No readable package branding.

Why this matters

You changed the surface. You also changed the height. You also used a different ball. Then you announced that rough makes it stop.

That announcement is not a character flaw. It is what happens when a try feels busy and the mind grabs the first difference it likes. The hour's job is smaller and kinder: name the one thing you meant to change, name what you kept the same on purpose, and ask — after the try — whether you can tell for sure.

Fair test is a taught procedure, not a personality trait that appears after enough hands-on mess. Children this age can predict, try, and compare. They can help choose one deliberate change. At 8–10 they can redesign a messy try once. They do not become fair-minded because a parent bought a kit, and they do not invent control-of-variables by staring at a pile of ramps alone.²⁰

What this idea unlocks is every later looking claim that points at a try. A plant that got more water *and* more sun cannot teach you about water alone. Ice that sat in a sunny window *and* on a warm palm cannot teach you about place alone. A vinegar-and-soda fizz is a change picture — adult present, authorized pair only — and it still is not a Chemistry credit.²¹ One deliberate change is how looking turns into a claim you can check.

Two refusals sit beside the idea, or the week becomes noise.

Discovery-only is not the program. A useful study of third- and fourth-graders learning to design unconfounded experiments found that children who received clear teaching of the control-of-variables rule produced far more “masters” than children left to unguided discovery on the same materials — and both groups manipulated the materials. The difference was explanation of the rule. That study was a classroom trial, not a kitchen RCT, and it is not a promise that every home will see the same result.²² Use it as a move: show a confounded try and a clean

²⁰Fair test / control-of-variables as a taught procedure: National Research Council, *Taking Science to School* (Washington, DC: National Academies Press, 2007); David Klahr and Milena Nigam, “The Equivalence of Learning Paths in Early Science Instruction,” *Psychological Science* 15, no. 10 (2004): 661–667 — children do not invent unconfounded design from materials alone.

²¹Kitchen vinegar-and-soda as a change picture, not Chemistry credit: parent safety spine in this book; sister *Science for Homeschooling* kitchen-chemistry fence. Adult present; authorized pair only; no sealed pressurized bottles.

²²Klahr and Nigam (2004): n=112 third- and fourth-graders after exclusions; ramps; direct instruction on control-of-variables vs unguided discovery; masters roughly 77% vs 23%. Classroom

try; ask “can we tell for sure?”; then let the child design.

Worksheet-only is not the program either. A fair-test sheet with no apparatus on the table is vocabulary without a lookable. Activity mania — ten demos, no record, no “what changed?” — is entertainment. One try with predict → try → compare, a child’s drawing or short table, and one authentic question beats both.

You do not need to be a scientist. You do need to put the materials down, ask what they will change and what they will keep the same, wait, and point back at the setup when the claim gets ahead of the evidence.

A third picture helps parents who learned “the scientific method” as a poster. The poster is not wrong as a map. It becomes wrong as an hour when the child never touches a lookable, or when “hypothesis” is demanded before eyes have had three quiet seconds. This chapter’s hour is smaller: one deliberate change, a record, one good question. Coverage maps belong on the shelf. Depth is this week’s phenomenon.²³

This week you can learn to hear confound pride and to answer it with a redesign, not a scolding lecture. Today the child can name one change and one same — or catch an obvious confound with help and redesign once.

For the parent: understand it yourself

Many adults feel rusty on “variables” and laminated scientific-method posters. That is ordinary. Five minutes here, then the warm-up, is enough for tomorrow. The child still does the trying.

Everyday picture. Two cookies from different batches. One was baked longer. One used a different pan. Someone asks which change made the cookie browner. You cannot tell for sure — two things changed. That kitchen moment is fair-test light in ordinary clothes. You do not need the word *variable* to feel the problem.

Precise picture. Three kid words carry the procedure:

- **Change:** the one thing we decide to make different on purpose.

trial, not a kitchen RCT; not a promise every home will see the same result.

²³Coverage maps on the shelf; depth is this week’s phenomenon: Sam Wineburg, “Crazy for History,” *Journal of American History* 90, no. 4 (2004): 1401–1414 — coverage-versus-depth shape borrowed carefully for science looking.

- **Measure / watch:** what we look at, count, or mark to see what happened.
- **Same:** everything else we keep alike on purpose.

Then the check question: **Can we tell for sure?** If two things changed, we cannot blame one.²⁴

At 5–7, keep those plain words. The adult often holds the “same” list while the child helps choose the one change. At 8–10, you may teach Change / Measure / Same as a short chorus — some programs use a mnemonic such as “Cows Moo Softly” for Change, Measure, Same. Treat that mnemonic as a pedagogical tool, not as a U.S. trial result.²⁵ Still refuse full lab reports. Still refuse Chemistry credit for a kitchen fizz.

A second precise picture: a confounded try next to a clean try. Same question — does the surface change how far the ball goes? Confounded: steep-smooth-short with a golf ball against shallow-rough-long with a rubber ball. Clean: same height, same ball, same start line; only the surface differs. Run at least two trials. Record distances. Ask what a stranger would need to believe the claim. That hour is the rule with examples, then the child’s redesign. It is not a sermon that discovery is dead.

Wrong answers you should be able to hear

1. *Confound pride.* “We changed the water *and* the light *and* the pot — and look, it grew!” Hear the pride. Do not crush the look. Ask: “Can we tell for sure it was the water?” Point at the two other changes. Redesign so only one thing differs.
2. *“Discovery means you never explain.”* Hear the slogan. The move is sequence, not silence forever: look, try, talk, then name the rule. Unguided discovery alone left most children in that classroom study unable to design unconfounded experiments. Teaching the rule with materials in hand worked better. Label the study classroom-origin. Then teach.²⁶

²⁴“Can we tell for sure?” after a confounded contrast: Klahr and Nigam (2004) CVS teaching move; Primary Connections fair-test language adapted for home.

²⁵“Cows Moo Softly” (Change / Measure / Same): Primary Connections pedagogical mnemonic — not a U.S. trial result.

²⁶Teaching the rule with materials in hand outperforms unguided discovery alone on the same materials (Klahr and Nigam 2004); both groups manipulated apparatus — the difference was explanation of the rule.

3. *A fair-test worksheet with no apparatus.* Columns for Change / Same / Result, already filled or waiting for invented numbers. Hear coverage dressed as investigation. Put the ramp, the ice, or the cups down first. The sheet is a blank record after looking, not a substitute for looking.
4. *Vinegar-and-soda as Chemistry credit.* Fizz, foam, excitement — then “we did chemistry.” Hear the credit grab. Name it plainly: this is a change picture. Adult measures. Authorized pair only. No sealed bottle. No other mixes. It does not unlock titration or a Chemistry transcript line.²⁷
5. *One trial as truth.* One throw, one melt, one bounce — announced as the law of the house. Hear overclaim. Ask for a second try with the same setup. Write both. Disagreement is data, not failure.

A sixth you will also hear: “Just tell me if I’m right.” The child wants the answer key instead of the redesign. Point at the materials. Ask what they would keep the same next time. Do not grab the answer. Do not fill the silence with the explanation.

Transfer, lightly. In the classroom study above, children who mastered the rule later looked alike on a transfer task that asked them to judge someone else’s experiment poster. Masters transferred; non-masters did not look the same.²⁸ At home, transfer is ordinary and small: after a ramp fair test, ask the same three questions about ice place or plant water. Do not print classroom percentages as a kitchen promise. Do not claim a five-year-old must master control-of-variables before weather looking. Fair-test light at 5–7 is the oral habit — one change, keep sames, ask can we tell — not a gate that locks the rest of the book.

Five-minute parent warm-up

Do this alone with tomorrow’s lookable — a board, two surface scraps, one ball — before the lesson.

Minute 1. Set a deliberately messy try: different height *and* different surface. Say aloud: “Can we tell for sure?” Answer yourself: no.

Minute 2. Reset so only the surface changes. Touch each “same”: ball, height, start line, floor. Say the change once.

²⁷Vinegar-soda does not unlock titration or a Chemistry transcript line: records chapter fence; never brand the hour as Chemistry credit at 5–10.

²⁸Klahr and Nigam (2004) transfer: masters later looked alike judging someone else’s experiment poster; non-masters did not. Do not print classroom percentages as a kitchen promise.

Minute 3. Practice the opening you will ask: “What did we change — and what did we keep the same?” Count a slow three after you ask. Look at the ramp, not at an imaginary child.

Minute 4. Write on a scrap: Change . **Same:** , , . Leave Result blank for the child.

Minute 5. Put the scrap face down. Tomorrow the child’s words fill it.

How to teach it this week

Science Hour stays the five beats you already know from the front of the book: materials down; notice / predict before naming; try; record; exit crumb / talk box. This week the try beat owns fair-test light. Total often ~20–40 minutes. Stop while a look remains.

Warm-up looking (3–8 min). Put the materials in view. Cover any worksheet title that already says “variables” or “hypothesis.” Ask what they notice about the setup — before any vocabulary sentence. Wait.

Predict before the rule dump (3–7 min). “What do you think will happen if we change only ____?” At 5–7, offer two choices tied to visible clues. At 8–10, ask for a reason that points at the materials. Write the prediction so it can be wrong in public without shame. Wrong predictions are useful.

Try / fair-test light (5–15 min). Child helps decide the one change. Adult scaffolds the same-list at 5–7; child names Change / Measure / Same at 8–10. Adult present for water, heat, ramps, kitchen substances, outdoors. Authorized materials only. No unmarked bottles. No bleach mixes. No drain cleaners. No alcohol flames. No sealed pressurized fizz bottles.²⁹

Record (3–8 min). Drawing at 5–7. Picture-plus-sentence or a two-row table at 8–10. The child’s numbers and words are the data. Do not overwrite. Do not paste a generated observation as theirs.

Talk and name (inside the exit / talk box). Now the short naming: one change, keep the rest the same, ask can we tell for sure. Point at the record. If two results disagreed, that is data.

²⁹Safety spine for fair-test tries: authorized materials only; no unmarked bottles; no bleach mixes; no drain cleaners; no alcohol flames; no sealed pressurized fizz bottles. Adult present for water, heat, ramps, kitchen substances, outdoors.

Exact wording you can say

- “What did we change — and what did we keep the same?”
- “What did you predict?”
- “What happened?”
- “Can we tell for sure it was the thing we named — what else was different?”
- “How would you redesign?”
- “You may change your mind.”

First try-it for the student (pick by skill, not birthday)

- **5–7 default:** FAIR-08 Same Same Checklist paired with a simple melt or bounce — child lists three “keep same” items before the try.
- **8–10 default:** FAIR-01 Ramp Surface Only — show a confounded setup first, ask can we tell, redesign so only surface changes.
- Soft alternate either band: FAIR-02 Melt Place Only (equal ice; sunny sill vs shade; same dish type).

How to fade help. First sitting: you hold the same-list and the child chooses the change. Next sitting: child names two sames before you add the third. Later: child runs the fair-box chorus (What changed? What stayed same? Can we tell for sure?) with you only pointing at the setup when they skip a step.

When to stop talking. After you ask, wait a slow three (a classroom-origin convention from elementary science wait-time research — useful at the table, not a kitchen RCT).³⁰ If they are mid-setup or mid-draw, do not cut them off. Look at the materials. Do not fill the silence with the right vocabulary.

Age-band moves

Band	This week’s ambition
5–7	Adult scaffolds the “same” list; child helps choose one change; oral “can we tell?”; drawings and labels; short oral exit

³⁰Mary Budd Rowe, “Wait-Time and Rewards as Instructional Variables,” NARST / ERIC ED061103 (1972); Robert J. Stahl, “Using ‘Think-Time’ and ‘Wait-Time’ Skillfully in the Classroom,” ERIC ED370885 (1994). Classroom-origin; useful at the table; not a kitchen RCT.

Band	This week's ambition
8–10	Child names change / measure / same; redesigns a confounded try once; simple tables; still refuse full lab reports

If looking habits from Chapters 1–2 are still thin — notice before name, draw what happened — stay longer on observe and draw before FAIR-01 becomes the gate. Place by skill, not birthday.

This-week shape (hypothetical illustration, not a reported family)

Day	Beat	Named try
Mon	Same-list scaffold or ramp fair test	FAIR-08 (5–7) or FAIR-01 (8–10)
Wed	Place-only melt	FAIR-02 Melt Place Only
Fri	Meta catch + redesign	FAIR-07 Confounded Catch + one redesign
Optional OUT	One focused outdoor look	OUT-02 puddle edge noticing, or soft-ball bounce outdoors with adult (FAIR-06 shape, clear space)

One focus sentence on Monday: “We change only one thing on purpose.” Friday: exit crumb plus one diagnostic error noted privately for next week — confound pride, discovery-only slogan, or vinegar-soda-as-credit.

Practice that actually builds learning

Named try-its below carry **time, materials, safety, the fun, the skill**. Pick one per Science Hour. Pair lightly with talk when needed. Library, walk, and windowsill *are* the practice — they do not replace the one-change move.

Talk box (locked opening)

Opening question: “What did we change — and what did we keep the same?”

Follow-ups (use three to five):

1. What did you predict? (*observe / wonder*)
2. What happened — what does the drawing or notebook show? (*record*)
3. How do you know from *this* try — point at the setup? (*reason*)
4. Can we tell for sure it was the thing we named — what else was different? (*check; lean 8–10*)
5. How would you redesign? You may change your mind. (*revise*)

How to wait. After you ask, a slow three. After they stop talking, wait again. If they are mid-look or mid-draw, do not cut them off. Look at the ramp, the ice, or the notebook — not at the child — if the silence is hard.

Stuck silence usually means: the question was vague; wait-time was zero; they are hunting a vocabulary word because you asked name-first; they are guessing what you want; the materials never landed; they offered a book-only guess; they think science means “repeat the parent’s explanation.” Pose, wait, point back at the lookable. Do not fill the silence with the right word, and do not turn the table into a fight.

In-home try-its

TRIAL-01 Paper Towel Sip

- **Time:** 20–30 min
- **Materials:** Two cups of water; paper towel strip; optional food color with adult
- **Safety:** Adult present; food-safe color only; no drinking dyed water; cover the table
- **The fun:** Watching water climb when it “shouldn’t”
- **The skill:** Predict → try → compare; honest prediction kept even if wrong
- **Steps:** Predict whether water will climb; set the strip as a bridge; watch 10+ minutes; compare to the prediction; record
- **Error patterns:** Declaring “capillary action” before watching; spilling dye without a cover

TRIAL-02 Warm Hands Melt

- **Time:** 15–25 min

- **Materials:** Two equal ice chips; plate; child's hands vs shade
- **Safety:** Adult present; short palm contact only; no frostbite play; wipe water; no eating lab ice
- **The fun:** Feeling ice change in a warm palm
- **The skill:** Predict → compare; heat as an ordinary energy picture — not a thermodynamics lecture
- **Steps:** Predict which melts faster; one brief palm hold, one on the plate; observe; talk
- **Error patterns:** Unequal ice sizes; ignoring numb hands

TRIAL-05 Vinegar-Soda Picture (change only)

- **Time:** 15–20 min
- **Materials:** Baking soda; vinegar; clear cup; tray; spoon — **adult measures**
- **Safety:** Adult present; authorized pair only; eye protection optional good practice; no other mixes; no sealed bottle; stop if curiosity reaches for unmarked bottles
- **The fun:** Fizz and foam as a change you can see
- **The skill:** Predict → observe a change; explicitly **not** Chemistry credit
- **Steps:** Predict; combine on a tray; record what you saw; say aloud: we did not earn Chemistry credit
- **Error patterns:** Calling it a bomb; mixing other kitchen chemicals; repeating without a tray

TRIAL-07 Flashlight Through Stuff

- **Time:** 20–30 min
- **Materials:** Flashlight (cool LED preferred); cardstock; wax paper; clear plastic; fabric scrap
- **Safety:** Adult present; never shine in eyes; no lasers
- **The fun:** Sorting what light passes, softens, or stops
- **The skill:** Predict → test → sort from evidence before vocabulary dump
- **Steps:** Predict which materials let light through; test in a dim room; sort opaque / translucent / transparent in kid words after evidence
- **Error patterns:** Naming vocabulary before tests; shining in eyes

FAIR-01 Ramp Surface Only

- **Band:** 8–10 (stretch 5–7 with adult scaffold)
- **Time:** 25–40 min
- **Materials:** Board ramp; two surfaces (paper vs towel); same ball; tape measure or floor marks
- **Safety:** Adult present; clear landing zone; no hard balls at faces

- **The fun:** Racing the same ball on two “roads”
- **The skill:** Spot a confound; redesign so only surface changes; record ≥ 2 trials
- **Steps:** Show a confounded setup on purpose; ask can we tell?; redesign; run trials; record distances
- **Error patterns:** Changing height and surface together; one trial as truth; adult overwriting numbers

FAIR-02 Melt Place Only

- **Band:** both
- **Time:** 25–35 min
- **Materials:** Equal ice; sunny sill vs shade; same dish type
- **Safety:** Adult present; no sun-amplified fire hazards; water wipe; no tasting
- **The fun:** A race the sun almost joins
- **The skill:** Only place differs; fair-box answers (change / same / result)
- **Steps:** Equal starting ice; only place differs; time to melt or compare size after a fixed time
- **Error patterns:** Different ice sizes; leaving glass in a hot car unsupervised

FAIR-04 Plant Water Amount

- **Band:** both (multi-day)
- **Time:** 20 min setup + daily ~5 min
- **Materials:** Two similar seedlings; same light / soil / pot; different water amounts marked
- **Safety:** Adult present; non-toxic plants; no fertilizer experiments without authorization; wash hands
- **The fun:** A living diary on the sill
- **The skill:** Change only water amount; refuse fake growth stories and generated progress photos
- **Steps:** Set the two plants; keep light and soil the same; photo or draw diary; week claim with evidence
- **Error patterns:** Moving one plant to better light; inventing heights; overwatering as “kindness” then denying it was the change

FAIR-06 Bounce Drop Height

- **Band:** both
- **Time:** 20–30 min
- **Materials:** One ball; floor mark; ruler; same floor
- **Safety:** Adult present; clear space; avoid glass-risk floors; outdoors only with soft balls and clear zone

- **The fun:** Higher drops, higher bounces — pattern talk
- **The skill:** Change only drop height; keep ball and floor the same
- **Steps:** Mark drop heights; measure bounce roughly; record pattern sentence
- **Error patterns:** Different balls mid-test; measuring on carpet then tile without noticing

FAIR-07 Confounded Catch (meta)

- **Band:** 8–10
- **Time:** 15–25 min
- **Materials:** Photos or quick demos of two bad setups (adult scripts)
- **Safety:** Ordinary talk safety
- **The fun:** Detective work without a worksheet packet
- **The skill:** Spot what changed twice; redesign orally
- **Steps:** Child spots the extra change; fixes in one sentence; still schedule real trials other days
- **Error patterns:** Treating meta talk as enough forever

FAIR-08 Same Same Checklist

- **Band:** 5–7
- **Time:** 15–20 min
- **Materials:** Printed or drawn checklist blanks; any simple trial (ice or ramp)
- **Safety:** Matches the paired trial's safety
- **The fun:** Checking boxes that actually match the table
- **The skill:** Before trying, list three “keep same” items; check them during setup
- **Steps:** Child (with adult) lists three sames; check off while setting up; then try
- **Error patterns:** Empty checklist; checking without looking

FAIR-03 Paper Plane Fold Only and **FAIR-05 Sugar Stir Time** are strong 8–10 alternates when you want indoor calm: same paper / same thrower height for planes; adult-handled warm water for sugar — change only fold design *or* only stir time *or* only temperature, never two at once. Safety: no aiming at faces; adult handles hot water; no tasting many cups.

Out-of-home try-its

OUT-02 Puddle or Creek Edge Soft

- **Time:** 20–30 min
- **Materials:** Clear cup; optional magnifier
- **Safety:** Adult present; no flood or fast current; no drinking; no contaminated

flood-water play; local rules; waterproof plan

- **The fun:** One focused look at water's edge
- **The skill:** Deliberate noticing (depth vs width, clarity, tiny life) — same observe-before-explain grain; not a scavenger mania packet
- **Steps:** Observe without removing vertebrates; return scoops; record one sentence: "I saw ____ in the water."

Outdoor bounce (FAIR-06 shape, careful): Soft ball only; adult present; clear space away from traffic and windows; change only drop height; keep ball and surface the same; stop if the second change (wind, slope) sneaks in — name it and redesign or bring the try indoors.

Blocked for a new move / mixed for when to use it

Blocked (one new move): FAIR-08 or FAIR-01 alone, with talk box, until the child can name change + same without hunting your face for the "right" word.

Mixed (when the move holds): Pair FAIR-02 with a drawing; or FAIR-01 with TALK-05 Fair Box Chorus; or Friday FAIR-07 with TALK-07 Disagree Soft Redesign when two results conflict.

One incorrect example to diagnose

Illustration (hypothetical, not a reported family). Parent and child set a plant try. Child moves one seedling to the bright sill *and* gives it extra water *and* a bigger pot. On Friday the child announces, "Extra water makes plants grow — look!" The notebook shows a taller plant. The adult feels the urge to say "wrong" or to overwrite the page with the "correct" conclusion.

Diagnose: Confound pride. Three things changed. The taller plant is real looking. The water claim is not yet tellable for sure.

Next move: "Can we tell for sure it was the water?" Wait. Point at light and pot. Redesign next week with FAIR-04: same light, same pot, same soil; only water amount differs. Keep the child's original words in the notebook. Add a new dated page for the redesign. Do not paste a generated growth chart as "our" data.

Exit crumb

Name one change and one same aloud — or catch a confound with help and say one redesign sentence. At 8–10, optional two-row table: condition / result.

More practice notes (dose and safety)

Three to five Science Hours in a week plus one OUT try-it is a full week for this focus. One deep fair-test hour beats five manic demos. There is no national table of kitchen science minutes.

When you pair a trial with a fair test, keep one idea per hour. Monday can be “we only change the surface.” Wednesday can be “we only change the place the ice sits.” Do not stack ramp, melt, fizz, and flashlight into one afternoon and call it mastery.

Safety stays ordinary and non-negotiable every sitting:

1. Adult present for water, heat, outdoors, small parts, kitchen substances, night work, and tall ramps.
2. Authorized materials only — if you cannot name the hazard and the adult control, do not run the try-it.
3. Kitchen chemistry picture $\neq$ Chemistry credit.
4. No unauthorized mixing. Curiosity is not a permit.
5. Prefer cool LED flashlights; refuse lasers for these ages in the default bank.
6. Stop the hour when safety fails; investigation continues another day.

Sibling and mixed-band notes

If two children share the hour, use one shared phenomenon and parallel jobs: the younger names one same from a checklist; the older names Change / Measure / Same and records a two-row table. Do not crown a winner. If results disagree, keep both lines in the notebook and plan one redesign (TALK-07 shape). Mocking ends the talk box.

A nine-year-old who still rushes to name-first vocabulary is not “behind.” Stay on Same Same Checklist until the looking habit holds. An early fair-test master still needs Chapters 1–2 looking moves on new domains — plants, weather, light — because scientific reasoning is tangled with what they know about *this* materials

set.³¹

What “fair” does not mean this week

Fair does not mean “nice.” Fair does not mean “everyone gets a trophy prediction.” Fair means: the try is set so a stranger could see why you blame the one change. Unfair means: two things moved, and the claim pretends only one did.

Fair also does not mean you never explain. After evidence is on the table, name the rule in kid language. Explore-before-explain is a sequence: look and try first, then enhance with the short explanation.³² It is not “never tell.”

For the student

You are allowed to try one thing on purpose.

Science looking this week is not “guess what the grown-up wants.” It is not “change everything and hope.” It is this: pick **one** change. Keep the other things the **same** on purpose. Try. Look. Then ask, “Can we tell for sure?”

Tiny worked example

Two ramps. Same board height. Same toy car. Same start line. One ramp road is smooth paper. One is a towel.

You changed: the surface.

You kept the same: car, height, start.

You watch how far the car goes.

You write or draw what happened.

If you also tip one ramp steeper, you changed two things. Then you cannot tell for sure that the towel was the reason.

³¹Scientific reasoning is tangled with domain knowledge for *this* materials set: National Research Council, *Taking Science to School* (2007); Daniel T. Willingham, “Critical Thinking: Why Is It So Hard to Teach?,” *American Educator* 31, no. 2 (2007): 8–19 — one sentence grain.

³²Explore-before-explain as sequence (look, try, talk, then name), not “never tell”: Patrick Brown / NSTA *Instructional Sequence Matters* grain; National Research Council, *A Framework for K–12 Science Education* (2012).

Try 1

Before you start, say out loud:

- What will I change?
- What will I keep the same? (Name at least two.)

Then try. Draw what you see, or write one sentence under a picture.

Try 2

Ask someone to show you a *messy* try on purpose — two things different at once. Your job: spot the extra change. Say how you would fix it. You may change your mind when you look again.

Explain it back

Tell a grown-up, in your own words: “We changed . **We kept** the same. We saw . **Can we tell for sure?** .”

A note about being wrong

If your prediction was wrong, you still did science looking — if you wrote what you *saw*. Wrong guesses that meet real ice, real ramps, and real water are useful. Erasing the prediction to look “smart” hides the learning. Keep the old prediction. Draw a line. Write what happened next.

If a grown-up asks “can we tell for sure?” and the honest answer is no, saying “no” is brave looking. Then you redesign. That is the whole game this week.

Challenge

Design a bounce try where only the drop height changes. Same ball. Same floor. Two heights. Two results. If wind or a slope sneaks in, name the sneak — that is good looking too.

If it isn't clicking

No shame. Correct a wrong reason without crushing the look. “Never correct the child” is not this book’s rule — and a scolding lecture is not teaching either.

Diagnostic 1 — Too many changes at once.

You hear: Water and light and pot all different; one proud claim.

Next move: FAIR-08 Same Same Checklist. Write three sames before any try. Redesign with only one change. Keep their original page; add a new dated page.

Diagnostic 2 — No apparatus; all worksheet.

You hear: Perfect vocabulary, empty table, no ramp or ice in sight.

Next move: Put materials down first. Cover the caption. Ask what they notice. The sheet becomes a blank record after looking.

Diagnostic 3 — “Just tell me if I’m right.”

You hear: Answer-key hunger; no appetite for redesign.

Next move: Redesign prompt, not answer key. “What would you keep the same next time?” Wait. Point at the setup. If needed, model one fair try, then they copy the *procedure*, not your conclusion.

When to slow down. Looking and drawing from Chapters 1–2 still thin → return to notice-before-name and draw-what-happened for a few hours before FAIR-01.

When to go ahead. Child names change + same without hunting your face; re-designs a confound once with help → add a simple table at 8–10; try FAIR-03 or FAIR-05.

When to get a human tutor. Persistent overwhelm with multi-step setup *and* distress that does not ease with shorter tries; or safety judgment calls you do not want to hold alone (outdoor water, heat). A tutor who will sit beside a ramp and ask “can we tell for sure?” beats a packet dump. This book is not medical or legal advice.

Tools, including AI

Optional helpers for the adult. Keep this short. Point back to the Science Hour AI box in the front of the book.

You may, on **your** account, after the child has tried (or before, for your prep): ask

a tool to explain fair-test light *to you* from today's lookable; generate a four-line SCRIPT for *you* to say after the try; get isomorphic blank record lines with the answer key held by you; diagnose work the child already produced (crop to the paper; do not upload the child's face).

You may not: invent an observation or data table and file it as the child's looking; put a model summary on the table as the observation; generate a "result photo" of a melt or ramp and treat it as data; park an unsupervised chatbot as the child's partner; write the student's lab paragraph; ask "what should I have seen" during the attempt. Ages 5–10 are under 13 — you hold the account. Unaided first on looking and trying. SCRIPT after the try.³³

A fluent filled blank from a machine is performance, not the child's science looking.

What “done enough” looks like

Check these before you move the fair-test light focus to the next chapter's domain work. Place by skill, not birthday. A “grade 2 science workbook” is a publisher's scope, not a legal grade, and not a transcript line.

- ☐ Child can name **one change** and **at least one same** for a try they helped set up.
- ☐ Child can answer “can we tell for sure?” after a deliberately messy try — with help at 5–7; more independently at 8–10.
- ☐ At least one in-home fair or trial try is in the dated notebook (drawing at 5–7; picture-plus-sentence or simple table at 8–10).
- ☐ At least one out-of-home or windowsill transfer sit happened (puddle edge, outdoor soft bounce, or sill melt) with adult present as required.
- ☐ Parent can hear confound pride and answer with redesign, not overwrite.
- ☐ Vinegar-soda, if used, was labelled a change picture — not Chemistry credit; adult measured; no unauthorized mixing.
- ☐ No generated observation filed as the child's data.

Showable two ways (soft check): say the change/same aloud *and* point at the setup or notebook. Holds after a night: next day they still remember what stayed

³³COPPA: Children's Online Privacy Protection Rule, 16 C.F.R. part 312; amended rule 90 Fed. Reg. 16918 (22 April 2025); general compliance date 22 April 2026. Ages 5–10 under 13 — parent holds the account. Science Hour AI box: unaided first; SCRIPT after the try.

the same.

Not done-enough: ten demos with no record; a perfect fair-test worksheet with no apparatus; a portfolio page the child did not see happen.

When this holds, living-or-not and weather-window chapters can reuse the same one-change muscle on plants and window cards. Revisit a ramp or melt next Tuesday if the skill wobbles — mix last week's try; stop while a look remains.

Chapter 4

Living or not; needs and life cycles light



Figure 6: A kitchen-table still-life from a high three-quarter view: a sort tray with a seed, a fresh leaf, a rock, a spoon, and a dried leaf; a small plant on a windowsill with two marked water cups nearby; a dated notebook open; a magnifier. No people. No logos. No readable package branding.

Why this matters

Is the rock alive because mud stuck to it and it “grew”?

Is the dried leaf alive, once-living, or not living — and what makes you say that?

Those are not vocabulary quiz tricks. They are looking questions. Children this age can sort by a need they can see — water, food, air, light for a plant on the sill, care that keeps a living thing going. They can start a one-thing plant or seed sit. At 8–10 they can place a familiar organism on a light life-cycle card set: birth, growth, reproduction, death as a common pattern, not a lecture on every species on Earth.³⁴

What this idea unlocks is every later life-science looking claim that points at needs and evidence. A wilted plant that perks after water teaches a visible need. A seed that sprouts under care teaches change over days the child recorded. A worm watched gently and returned to soil teaches living care without a collection jar for keeps. Sorting at the grocery — leaf, root, fruit in kid words — trains once-living versus living versus not, without a lab coat.

Two refusals sit beside the idea, or the week becomes either a cartoon or a hazard.

Mold cultivation is refused by default. Open mold jars, sniffing projects, and “let’s grow the fuzzy bread” are not this book’s Science Hour. Spore exposure is a real household concern. Prefer a sealed store photo labelled as a picture, or a school-safe sealed kit only with a clear adult protocol — and even then, many families will simply skip. The refuse card is part of the teaching.³⁵

Cells-as-tiny-mouths dumps and deep-time lectures are not the hour either. Needs the child can see come first. Plants reclassified as living, then overgeneralized as “eating through roots,” are a prior idea you will hear — repair with another look (water, light, the celery dye path), not a scolding lecture about organelles.³⁶ No ecosystem multi-variable design at age six. No meiosis. No climate-change causal

³⁴Life-cycle common-pattern light and needs looking at 5–10: NGSS Lead States, *Next Generation Science Standards* (Washington, DC: National Academies Press, 2013) K–2 / 3–5 maps used as trail, not mandate; National Research Council, *A Framework for K–12 Science Education* (2012).

³⁵Mold cultivation refused by default: household spore/safety concern; prefer sealed photo or skip. Refuse card is teaching.

³⁶Plants-as-eating-through-roots and related prior ideas: National Research Council, *Taking Science to School* (2007) — repair with another look, not an organelles lecture.

modeling smuggled into needs talk.

Standards maps for the earliest grades ask children to notice patterns in what plants and animals need, to run simple plant tests one variable at a time, and — a little later — to recognize life cycles as a common pattern and traits that resemble parents as an edge idea for older children in this band.³⁷ Use those maps as a trail, not as a 180-day dump and not as a PE checklist for kindergartners. This book's hour is still one tray, one question, one record.

You do not need to be a biologist. You do need to put a sort tray or a plant on the table, ask what it needs to stay alive and what makes them say that, wait, and refuse the proud open mold jar.

This week you can learn to hear rock-as-living and plants-eat-dirt, and to answer with another look. Today the child can sort five objects living / once-living / not by a visible need — or start a one-thing seed or plant sit.

For the parent: understand it yourself

Many adults feel rusty on the life science they last met as a labelled diagram. That is ordinary. Five minutes here, then the warm-up, is enough for tomorrow. The child still does the sorting and the care.

Everyday picture. A wilted basil on the sill. You give it water. By evening the leaves look less floppy. You did not need a textbook sentence to see a need met. That kitchen moment is living-by-needs in ordinary clothes. The precise work is to ask the child what they see that shows a need — and to keep plant trials to **one** deliberate change when the question is “does water amount matter?”

Precise picture. Living / nonliving at 5–10 is sorted by **visible needs and evidence**, not by a memorized list of kingdom names. Kid criteria you can practice:

- Does it need water, food, air, or light (for green plants) in a way we can notice over time?
- Does it grow or change as a living thing does — not only “get mud stuck on”?

³⁷Early-grade standards maps on needs, simple plant tests, life cycles, and traits-like-parents as an older-band edge: California Department of Education, *2016 Science Framework*; NGSS Lead States (2013) — trail, not 180-day dump.

- Is it living now, was it once living (dried leaf, wooden spoon from a tree, apple slices), or was it never living (rock, metal spoon, plastic toy bug)?

Life-cycle cards at 8–10 are a **common pattern** light: many animals and plants go through birth or sprouting, growth, reproduction, death. Use familiar organisms (bean, butterfly picture set, chicken egg sequence from a named book after looking). Do not demand every organism’s complete natural history in week one of age six.³⁸

A second precise picture: one-variable plant assessment. Standards language at the Grade 2 trail limits plant investigation assessment to testing one variable at a time — water, light, or soil, not all three at once as a “proof.”³⁹ That matches Chapter 3’s fair-test light. FAIR-04 Plant Water Amount is the bridge: same light, same soil, same pot; only water amount differs.

Wrong answers you should be able to hear

1. *Rock is living because it “grows” with mud stuck on.* Hear the growth metaphor. Ask what the rock needs to stay alive. Wait. Compare to a seed that sprouts under care. Mud stuck on is not growth like a seedling.
2. *All plants “eat dirt.”* Hear the prior idea after plants get reclassified as living. Repair with another look: water, light, a pale flower or celery in food color (TRIAL-06) showing a path — transport picture, not a roots-are-mouths lecture. Keep their words; add a new dated look.
3. *Mold jar pride.* “We grew mold!” as the week’s trophy. Hear the hazard dressed as inquiry. Run LIFE-07 Refuse Card. Prefer sealed photos. Do not cultivate open mold at home for this book’s default.
4. *Cells as tiny mouths dumped at age six.* Hear coverage itch. Needs and care come first. Microscopic cell stories without a lookable the child can hold are a different hour — usually a later year.
5. *Fire is alive because it moves and eats.* Hear motion-as-life as the only rule. Sort with needs talk: does fire need what living things need in the same way? Motion alone is not the criterion. Stay gentle; stay on the tray in front of you.

³⁸Life-cycle cards as common pattern, not every species’ natural history in week one of age six: NGSS / Framework life-cycle grain light.

³⁹Grade 2 plant investigation assessment limited to one variable at a time (water, light, or soil): NGSS Lead States (2013) performance-expectation trail — matches fair-test light.

A sixth you will also hear: keeping a wild worm or insect “for science” with no care plan and no return. Hear collection without care. LIFE-03 and OUT-06 are observe-and-release. Wash hands. Return organisms.

Five-minute parent warm-up

Do this alone with a sort tray — seed, fresh leaf, rock, spoon, dried leaf — before the lesson.

Minute 1. Sort the five yourself into living / once-living / not. Notice where the dried leaf goes. Say aloud why.

Minute 2. Practice the locked opening: “What does this need to stay alive — and what makes you say that?” Count a slow three. Look at the tray.

Minute 3. Check your plant sill: non-toxic? Clear water plan? If you will start FAIR-04, mark two cups now.

Minute 4. Decide your refuse: no open mold grow this week. Write that on a scrap if you need the reminder.

Minute 5. Put the scrap away. Tomorrow the child’s reasons fill the talk.

Transfer from Chapter 3. When the plant question is about water amount, keep light and soil the same. Living-or-not sorting does not require a fair test. One-thing plant trials do. Do not mash them into one confused hour.

How to teach it this week

Science Hour five beats stay. This week the lookable is living or once-living material — sort tray, seed bag, houseplant, gentle worm watch, grocery produce. Soft entry: care, return organisms, wash hands. Adult present for outdoors, small parts, kitchen knives if seed coats are opened, and any animal handling.

Warm-up looking (3–8 min). Put the tray or plant down. Cover textbook captions that already say “biotic” or “organism.” Ask what they notice before any vocabulary sentence.

Predict / wonder (3–7 min). “What do you think this needs to stay alive?” At 5–7, two choices tied to visible clues (water or no water; light sill or cupboard). At 8–10, ask for a reason that points at something they can see.

Try (5–15 min). Sort; or set a sprout; or hunt plant parts on a known-safe plant; or start a one-thing water trial. Record is the child’s drawing or tally — not a generated progress photo filed as ours.

Talk and name. Now short naming: living / once-living / not; needs we can see; care and return. At 8–10, optional life-cycle card placement for one familiar organism.

Exact wording you can say

- “What does this need to stay alive — and what makes you say that?”
- “What do you see that shows a need?”
- “Is this living, once-living, or not living — how do you know?”
- “What would happen if we took away ___?”
- (8–10) “Where does this fit on a birth–growth–reproduction–death card set?”
- “You may change your mind after another look.”

First try-it for the student

- **5–7 default:** LIFE-01 Living / Not Sort (five objects; argue with reasons; adult waits before correcting).
- **8–10 default:** LIFE-01 plus a light life-cycle card sit for one familiar organism, or start LIFE-02 Seed Count Sprout with honest date records.
- Soft alternate: LIFE-05 Plant Parts Hunt on a known-safe houseplant or garden plant.

Materials prep the night before (5 minutes). Choose non-toxic plant material. Wash the tray. If worms are on the plan, confirm weather and a return spot. If grocery is on the plan, pick a low-crowd time. Print or draw a three-column sort mat: Living / Once-living / Not living. Leave the columns empty of “answer key” pictures.

First authentic question + wait. Ask the locked opening about one object only — the dried leaf is perfect — then wait. If silence stretches, point at the fresh leaf and the dried leaf together: “What’s the same? What’s different?” Do not answer yourself at the one-second mark.

Exit crumb options. (a) “___ is living because ___.” (b) Date + seed drawing. (c) Four plant-part labels. (d) Refuse sentence from LIFE-07 if needed. Pick one. Sticky triad from the assessment bank can wait for Chapter 8’s full treatment; a single crumb is enough tonight.

How to fade help. First sitting: you place two piles and think aloud about one object. Next: child sorts; you ask for one reason each. Later: child sorts five and checks one “once-living” against “not living” without your face as the answer key.

When to stop talking. After the locked opening, wait a slow three. If they are mid-sort, do not crown a pile early. Look at the tray.

Age-band moves

Band	This week’s ambition
5–7	Sort by a visible need; plant parts labels; seed sprout start; oral exit; adult scaffolds care routines
8–10	Sort with reasons; life-cycle cards light; one-thing plant trial; heartbeat-after-move as body-needs picture (not medical diagnosis); still refuse mold cultivation and cell dumps

This-week shape (hypothetical illustration)

Day	Beat	Named try
Mon	Sort by needs	LIFE-01 Living / Not Sort
Wed	Seed start	LIFE-02 Seed Count Sprout (setup)
Fri	Plant parts + optional one-thing start	LIFE-05 + begin FAIR-04
Optional OUT	Grocery or garden	OUT-03 Produce Sort or OUT-06 Insect Count Soft

Focus sentence Monday: “We sort by a need we can see.” Friday: exit crumb plus one diagnostic — rock-as-living, plants-eat-dirt, or mold-jar itch.

Practice that actually builds learning

Talk box (locked opening)

Opening question: “What does this need to stay alive — and what makes you say that?”

Follow-ups:

1. What do you *see* that shows a need? (*observe*)
2. What does your drawing or sort show? (*record*)
3. How do you know from *this* — point at the object or plant? (*reason*)
4. Is this living, once-living, or not living — what else could someone say? (*check / alternative*)
5. What would happen if we took away ____? You may change your mind after another look. (*revise*)

How to wait. Slow three after you ask. After they stop, wait again. Do not leap in with “because it needs water!” Look at the tray or sill.

Stuck silence usually means: the question was vague; materials never landed; they think “alive” means “moves”; they are hunting the textbook word; wait was zero; they fear being wrong about the rock. Fix the setup. Offer two piles to choose between. Point back at a need they can see.

In-home try-its (LIFE bank + bridges)

LIFE-01 Living / Not Sort

- **Time:** 15–25 min
- **Materials:** Picture cards or real safe objects (rock, leaf, toy bug, seed, water cup)
- **Safety:** No mold cultures; wash hands; choke-size check for small parts under 7
- **The fun:** Arguing kindly about the dried leaf
- **The skill:** Sort living / once-living / not; reasons before adult correction
- **Steps:** Sort; argue with reasons; adult waits; then short naming talk
- **Error patterns:** “Moving = alive” as only rule; calling fire alive without talk

LIFE-02 Seed Count Sprout

- **Time:** Setup 20 min + daily checks
- **Materials:** Beans; wet paper towel; bag; optional dark cupboard vs light shelf — if fair-test link, keep other factors same

- **Safety:** Adult present; wash hands; food-safe beans if any later tasting is a separate kitchen context, not “lab eating”; no fake day counts
- **The fun:** A date when green shows up
- **The skill:** Observe dry vs sprouting; refuse invented day counts and generated progress photos
- **Steps:** Observe dry seeds; set sprouting; record days to sprout; draw
- **Error patterns:** Generated progress photos; eating sprouts from an unclean setup

LIFE-03 Worm Watch Soft

- **Time:** 20–30 min
- **Materials:** Compost worm or earthworm from garden with permission; damp paper; tray
- **Safety:** Adult present; gentle handling; wash hands; return outdoors; skip if allergies or strong phobia — use a labelled video as observation of media with honesty (“this is a screen, not our tray”)
- **The fun:** Watching a living thing respond gently
- **The skill:** Observe; name a need; return; no harm
- **Steps:** Observe movement / response to light gently; return to soil; draw
- **Error patterns:** Harming the worm; salt “experiments”; keeping without a care plan

LIFE-04 Bird Feeder Tally

- **Time:** 15 min × several days
- **Materials:** Window feeder or park bench view; tally sheet
- **Safety:** Adult present; squirrel/rodent awareness; seed-dust allergies; local wildlife rules; no chasing wildlife; no harmful bread dumping
- **The fun:** Who visited today?
- **The skill:** Same time of day when possible; tally; note weather; pattern talk
- **Steps:** Tally visits; refuse fake tallies; exit: “Most visits when ____.”

LIFE-05 Plant Parts Hunt

- **Time:** 20–30 min
- **Materials:** Houseplant or garden plant known safe
- **Safety:** Adult names safe plant; wash hands; no mouthing; no pulling plants in parks
- **The fun:** Finding root / stem / leaf / flower if present
- **The skill:** Observe before textbook diagram; label four parts
- **Steps:** Find parts; draw and label; talk after looking
- **Error patterns:** Toxic houseplant sap ignored; park vandalism

LIFE-06 Heartbeat After Move

- **Band:** 8–10
- **Time:** 20–25 min
- **Materials:** Timer; quiet sit vs 1 min march
- **Safety:** Adult present; stop if dizzy; no children with known cardiac limits without clinician guidance; this is a PE-soft science picture — **not medical advice, not diagnosis, not a health sermon**
- **The fun:** Feeling the body change after moving
- **The skill:** Two counts; body needs picture
- **Steps:** Rest count with adult help; after-move count; talk “exercise made ____”
- **Error patterns:** Diagnosing illness; unsafe heart-rate contests

LIFE-07 Mold Bread Refuse Card (meta safety)

- **Band:** 8–10 (+ adult any band)
- **Time:** 10–15 min talk
- **Materials:** None required (or sealed store photo of mold)
- **Safety:** Do not cultivate mold at home for this book’s default; adult present for any sealed-kit exception
- **The fun:** Being the family that knows when to refuse
- **The skill:** Name why open mold grows are refused (spore exposure)
- **Steps:** Discuss; write exit: “We refuse open mold grows because ____.”
- **Error patterns:** Open mold jars on purpose; sniffing mold

Bridge tries from other banks (same chapter week)

TRIAL-03 Seed Coat Peek — dry bean vs soaked bean (adult prep); magnifier; draw; no tasting; adult present if any knife is used.

TRIAL-06 Plant Drink Dye Soft (8–10) — celery or pale flower; food-safe dye; overnight; draw colored path; no drinking dye water; refuse internet “our plant” photos.

FAIR-04 Plant Water Amount — one deliberate water change; same light/soil/pot; multi-day honesty.

Out-of-home try-its**OUT-03 Grocery Produce Sort**

- **Time:** 20–30 min in store or market
- **Materials:** Eyes; soft list
- **Safety:** Adult present; hygiene; allergy awareness; store courtesy; no opening

packages; no tasting unbought food; no food shaming

- **The fun:** Leaf / root / fruit hunts in kid words
- **The skill:** Once-living vs living vs not without a lab; observe before nutrition lecture
- **Steps:** Sort three foods by plant part guess; exit crumb

OUT-06 Garden Insect Count Soft

- **Time:** 20–25 min
- **Materials:** Tally sheet; agreed square of garden or park
- **Safety:** Adult present; sting plan; allergy awareness; no bees-nest approach; do not kill; do not collect endangered organisms
- **The fun:** Five quiet minutes of noticing
- **The skill:** Count what you see; classify roughly (ant / fly / other); release ethic
- **Steps:** Tally; exit totals; wash hands

Zoo or aquarium visit (motivation, NCES outing row exists for zoo/aquarium among homeschool outings): still **one** looking question — “What does this animal need that we can see in the habitat?” — not a binge worksheet packet.⁴⁰

Blocked / mixed

Blocked: LIFE-01 alone until reasons point at a need, not only at “it moves.”

Mixed: LIFE-02 + FAIR-04 across days; LIFE-05 + DRAW-style labels; Friday LIFE-07 if mold pride appeared.

One incorrect example to diagnose

Illustration (hypothetical). Child sorts a rock into “living” because the rock sat in the garden and got larger with mud. Parent laughs or says “that’s silly.” Child shuts down.

Diagnose: Rock-as-living via stuck mud; adult crush instead of another look.

Next move: “What does this need to stay alive — and what makes you say that?” Wait. Put a seed beside the rock. Start LIFE-02. Compare in three days. Keep the child’s first sort in the notebook. Add a new dated page when they revise. Do not overwrite.

⁴⁰Zoo or aquarium among homeschool outing cells: NCES 2024-113 Table A-5 (zoo/aquarium visit past month). Motivation outing; still one looking question.

Exit crumb

Sort five objects by a visible need with one reason each — or start a one-thing seed/plant sit with today's date and a drawing.

Dose and safety spine for living work

Three to five hours plus one OUT is enough. Multi-day plant trials need a daily 5-minute check, not a forgotten bag in a backpack.

Living-things safety, repeated because it is load-bearing:

1. Adult present for outdoor collection edges, water, small parts, and animal handling.
2. No harm; return organisms; wash hands.
3. Known-safe plants only; no toxic foraging; no mouthing.
4. Mold cultivation refused by default (LIFE-07).
5. Heartbeat sit is not medical advice.
6. No unauthorized mixing if a “plant food” chemical curiosity appears — authorized materials only.

Needs language you can keep plain

You do not need the word *organism* on day one. You need eyes on a need.

For animals the child knows (pet, bird at the feeder, worm on the tray), needs talk sounds like: water, food, air, a place that is not harming them, care from someone who notices. For green plants on the sill: water, light, a place to grow, air. You are not writing a complete physiology. You are training the child to point at evidence.

When they say “it needs love,” translate gently toward care they can see: water on a schedule, not leaving a worm in a dry tray, returning an insect outside. Love as a word can stay in the family. Science Hour asks what care looks like on the table.

Once-living is the category that unlocks grocery and wooden spoons without a fight. An apple slice is not living on the counter the way a tree is living in the yard. It was part of something living. A dried leaf is once-living. A plastic leaf is not. That distinction prevents “everything from nature is alive” and “nothing that isn't moving is alive” from becoming the only two rules.

Fossil light (optional crumb at 8–10)

If you have a fossil photo in a named book, or a museum postcard, one sentence is enough: “This was once living — what makes us say that?” Do not turn the hour into a deep-time lecture. One evidence piece. Then back to the sill plant or the sort tray.⁴¹

Sibling notes

Shared tray, parallel jobs: younger child places into two piles (living / not) while you hold once-living as a third pile you introduce; older child defends once-living with a reason. Grocery OUT-03 works well with mixed ages — one cart, three finds, no shaming talk about anyone’s lunch.

What this chapter is not

Not a Biology transcript line. Not *Little Thinkers Science I*. Not a nature-study dump that never asks for a child’s reason. Not a culture-war pamphlet about food systems. Safe science objects only: seed, leaf, plant, worm watch soft, grocery produce, feeder tally. Soft entry. Care. Return. Wash hands.

Library picture books and zoo visits motivate. They do not replace the sort tray or the seed diary. If the book shows a perfect sprout sequence and your bag is still quiet on day three, your bag is still the data. The book is a comparison after looking — OUT-07 shape from the banks, when you want it — not an overwrite.

Practice menu reminder

Primary cards this chapter: LIFE-01 through LIFE-07 (LIFE-07 is the refuse meta). Pair TRIAL-03, TRIAL-06, FAIR-04. OUT: OUT-03 grocery produce sort; OUT-06 insect count soft. Error focus: rock-as-living; plants eat dirt; mold jar pride. Exit crumb: sort by a visible need; or one-thing plant started.

⁴¹Fossil or museum postcard as one evidence crumb, not a deep-time lecture: Framework / NGSS evidence grain light.

A longer worked parent picture (hypothetical)

Monday: five objects on a tray. Child puts the rock with living things “because it got bigger in the garden.” You wait. You do not laugh. You ask the locked opening about the rock. Child shrugs. You put a dry bean beside the rock and say you will both look again Wednesday.

Wednesday: LIFE-02 setup. Wet towel, bag, date written by the child. Rock still on the tray for comparison. Talk is short.

Friday: a root tip shows. Child moves the rock to “not living” and the dried leaf to “once-living” without being told. You uptake: “You said the rock needed to stay with living things on Monday. Today you put it here — what makes you say that now?” Wait. Write both days in the notebook. That is teaching. That is not a percentile.

For the student

Living things need things to stay alive. You can look for those needs.

This week’s science looking is not “memorize a list of kingdoms.” It is not “grow the fuzziest mold.” It is: sort what is living, what was once living, and what was never living — and say what a living thing needs, with something you can point at.

Tiny worked example

On the tray: a seed, a fresh leaf, a rock, a metal spoon, a dried leaf.

- Seed: living (or will show living when it sprouts under care) — needs water and the right place to sprout.
- Fresh leaf on a plant: living — needs the plant’s water and light.
- Dried leaf: once-living — it was part of a living plant.
- Rock: not living — mud stuck on is not the same as growing like a seedling.
- Metal spoon: not living.

You may disagree at first. That is fine. Look again. Change your mind when the evidence changes.

Try 1

Sort five things. For each one, say: living, once-living, or not living — and **what makes you say that**. Draw the sort.

Try 2

Start a seed sit. Write today's date. Draw the dry seed. Check again on later days. Write the date when something changes. Do not invent a day you skipped.

Explain it back

Tell a grown-up: “___ is living because it needs . *I know because I saw* .”

A note about care

If you watch a worm or count insects outside, be gentle. Put living things back. Wash your hands. Science looking does not mean keeping everything in a jar forever.

If you feel squeamish

You do not have to hold a worm to do this chapter. You can sort pictures. You can watch a plant. You can tally birds through a window. Tell your grown-up what feels okay. Brave looking is not the same as forcing your hands.

If you want more

Keep the seed diary for two weeks. Each day: date + one word or one tiny drawing. If nothing changed, write “same.” “Same” is honest data. Skipping days and then inventing them is not.

Challenge

At the grocery (with a grown-up), find one food that is a leaf, one that is a root, and one that is a fruit or seed in your words. Say which are once-living. No tasting food you did not buy. No opening packages.

Life-cycle light (mostly 8–10)

With a grown-up, place cards for one familiar animal or plant: beginning, growing, making new ones, ending. Use a bean you grew or a picture set you looked at after real looking. You do not have to know every creature on Earth this week.

If it isn't clicking

Diagnostic 1 — Prior idea loud (rock-as-living; plants eat dirt).

Next move: Another look, not a lecture. Seed beside rock. Celery dye path or wilt-then-water sit. Keep old words; add new dated page.

Diagnostic 2 — Safety fear or disgust blocking the hour.

Next move: Choose a softer card — LIFE-01 with pictures; LIFE-05 on a familiar houseplant; skip worm watch; use labelled media with honesty. Never force handling.

Diagnostic 3 — Coverage itch / mold jar pride / cell dump.

Next move: LIFE-07 Refuse Card. One phenomenon this week. Needs first. Put the diagram book away until after the sort.

Slow down when sorting reasons never point at a need — stay on LIFE-01 and grocery sort.

Go ahead when reasons are visible-need-based and a seed diary is honest — add FAIR-04 and life-cycle cards.

Human help when allergies, phobias, or outdoor safety feel beyond what you want to hold alone — pick picture sorts and sill plants; ask a trusted adult or local nature program for handling norms. This book is not medical advice.

Traits-like-parents (soft 8–10 edge)

If you have two generations visible — a parent plant and a cutting, or a family pet and a photo of the parent animal from a named album — ask one light question: “What is the same about the young and the parent — what do you see?” Stop there. Do not open Mendel. Do not open identity fights. One looking crumb is enough for this book's edge.⁴²

⁴²Traits-like-parents soft 8–10 edge: one looking question only; no Mendel; no identity fights.

Records title reminder

If you need a subject name on the notebook, use **Science** or, when the work is actually that, **Life Science**. Never a curriculum brand. Never Little Thinkers Science I. The sort, the seed diary, and the care notes live inside Science.

Tools, including AI

Point back to the Science Hour AI box. Short version for this chapter:

You may ask a tool *on your account* to explain a named needs idea to you, or to draft a SCRIPT you will say after the child's sort. You may not generate a "microscope photo" or mold close-up and file it as the student's result. Generated image $\neq$ data. You may not invent a seed diary the child did not keep. Parent holds the account; 5–7 no live open chat as the child's partner; 8–10 still parent in the room; unaided first on looking and trying.⁴³

What "done enough" looks like

- ☐ Child sorts at least five objects living / once-living / not with a reason that points at a need or at once-living evidence.
- ☐ Child can answer the locked opening for one living thing on the table or sill.
- ☐ At least one care-ethic sit happened (return worm / no-harm insect count / wash hands named) *or* LIFE-07 refuse spoken if mold pride arose.
- ☐ One seed or plant sit started with real dates — or a plant-parts labelled drawing completed.
- ☐ Parent refused open mold cultivation as the default.
- ☐ No generated growth photos filed as the child's data.
- ☐ At 8–10 soft edge: one life-cycle card placement for a familiar organism *or* one honest pulse-before/after move sit with safety limits held.

Showable two ways: sort the tray *and* say one need sentence; or draw the seed *and* retell what changed since the start date.

Not done-enough: a kingdom chart copied with no tray; an open mold jar as portfolio pride; a multi-variable plant "proof" that ignores Chapter 3.

⁴³Science Hour AI box / COPPA under-13: parent holds account; generated image $\neq$ data; unaided first on looking and trying.

When this holds, weather-window work can use the same eyes on sky patterns — and the one-thing muscle from Chapter 3 still guards any plant water trial you keep running on the sill. Revisit the sort tray next Tuesday if rock-as-living returns; mix last week's seed check; stop while a look remains.

Chapter 5

Weather window and sky patterns



Figure 7: A kitchen-table still-life near a window: a clothesline or row of three weather cards (sun, cloud, rain symbols drawn simply), a small thermometer in shade, a mini cloud booklet, a dated notebook, a ribbon for wind. No people. No logos. No readable package branding. No phone screen as the hero object.

Why this matters

What is the same about today and yesterday — and what is different?

That question is the weather window. It is not a disaster documentary. It is not a phone-radar competition. It is not an adult climate-policy fight smuggled into Science Hour. It is looking out the window, hanging a card, and — after a few days — saying a pattern in ordinary words.⁴⁴

Children this age can notice sky, wind evidence, wet or dry ground, and warmer or cooler air. They can hang three to five weather cards. At 8–10 they can keep a simple table or pictograph of rainy versus dry days, or morning versus afternoon temperature in the same shady spot. Severe weather belongs as preparedness questions — what do people do to get ready? — not as a binge of frightening footage.⁴⁵

What this idea unlocks is Earth-science looking at home scale. Local weather over days the child recorded is honest data. A forecast app can check *after* the window look; it cannot replace eyes. Puddles that shrink, ribbons that lift in wind, clouds drawn at a real time stamp — those are the lookables. Library weather picture books motivate. They do not overwrite a week the child did not actually see.

Two refusals sit beside the idea, or the week becomes either a screen hour or a culture-war hour.

Radar-first is refused as the opening move. The phone may come out after notices are on paper. Inventing a week of data the child did not record is refused the same way generated observations are refused in every chapter — it is not the child's looking.⁴⁶

Climate-as-policy sermon is refused as the hour's purpose. Climates in different regions from a *named* atlas page can be an 8–10 edge. Grade-band maps mention seasonal patterns in tables and, later, climates from combined information — use that as a trail edge, not as a 5–7 dump and not as a pamphlet from the left or the right.⁴⁷ Weather is patterns. Preparedness is practical questions. Live controversy

⁴⁴Weather window as looking and pattern in ordinary words — not disaster documentary or phone-radar competition: NGSS / Framework weather and sky pattern grain for early grades.

⁴⁵Severe weather as preparedness questions, not frightening footage binge: Framework / NGSS severe-weather preparedness language light.

⁴⁶Invented week of data refused the same way generated observations are refused: Science Hour honesty spine; parent-side record integrity.

⁴⁷Climate-as-policy sermon refused as the hour's purpose; regional climate from a named atlas

is not the default try box.

You do not need to be a meteorologist. You do need to hang the card, ask same/different, wait, and keep the phone radar as a check after the window look — not instead.

This week you can learn to hear radar-first and invented week-data, and to answer with another look out the real window. Today the child can fill one weather card and compare it to yesterday — or mark a simple sky pattern (clouds, wind, temperature) with a time stamp.

For the parent: understand it yourself

Many adults feel rusty on weather beyond the lock screen. That is ordinary. Five minutes here, then the warm-up, is enough for tomorrow. The child still does the looking.

Everyday picture. Deciding on a jacket by looking out the window. Cold glass, moving trees, wet sidewalk — you gather same/different from yesterday without naming a front. That hallway moment is weather-window science in ordinary clothes. The precise work is to slow it down: notice before the app, record before the claim, pattern after several cards hang together.

Precise picture. Weather at 5–10 is **local and recorded**. Kid variables you can hang without jargon:

- Sky look (clear, cloudy, mixed) — draw before forcing type names.
- Wet or dry (rain, snow, puddles, dry ground).
- Wind evidence (calm, breezy, strong — ribbon or leaves).
- Warmer or cooler (hand on the glass is a start; shady thermometer is better when you have one).

At 8–10, add a simple table: date / sky / wet-dry / wind word / optional temperature. Seasonal talk is “what our weeks have usually looked like,” not a causal climate model. Rain gauge (straight-sided jar, off the ground, adult present) turns one storm into a measured row — still not a national weather-minutes study.⁴⁸

page is an 8–10 edge: NGSS climate/season trail as map, not pamphlet.

⁴⁸Simple tables, rain gauge, morning/afternoon temperature with place held same: NGSS 3–5 weather-table / measurement grain light — not a national weather-minutes study.

Severe weather preparedness questions are allowed and useful: What do we do if thunder is close? Where is our flashlight? Who do we tell if we leave the park early? Disaster documentary binges are not the hour. Fear without agency is not Science Hour.

Wrong answers you should be able to hear

1. *Weather = climate policy fight.* Hear adult controversy arriving as content. Redirect: “What is the same about today and yesterday — and what is different?” Hang the card. Park policy for adult time.
2. *Only phone radar counts as looking.* Hear screen-first. Cover the phone. Two minutes of window silence. Child lists five notices. Then — optional — compare to the app.
3. *Severe weather binge without preparedness questions.* Hear fright-as-curriculum. Stop the footage. Ask what people do to get ready. Practice one family prep crumb if storms are in your season.
4. *Inventing a week of data.* Hear empty cards filled from memory or from “what the week should look like.” Trash the fake rows. Go to the window now. One real card beats seven invented ones.
5. *Cloud type quiz before several days of drawings.* Hear name-first. DRAW-06 and WEATH-01 want drawings and words first; names after patterns exist.

A sixth you will also hear: standing in the street for “better wind,” climbing on sills for “better sky,” or sun-staring during cloud look. Hear safety failure. Adult present outdoors; no sun stare; no sill climbing; weather extremes — stay in.

Jacket test as warm-up transfer. Before you wake the child, decide whether *you* need a jacket from the window alone. Note your reasons. After the child’s card, compare. The point is not to match the app. The point is to notice that looking gathers evidence before a label.

Five-minute parent warm-up

Do this alone at the real window before the lesson.

Minute 1. Look outside two minutes in silence. List five notices on a scrap (cloud, wet/dry, wind evidence, light, something else).

Minute 2. Fill today’s weather card yourself. Do not check the phone yet.

Minute 3. Practice the locked opening. Count a slow three. Look at the glass, not at an imaginary child.

Minute 4. Check yesterday's card if it exists. Say one same and one different aloud.

Minute 5. Optional: open the app and notice agreement or difference — then put the phone away before the child arrives.

Earth-science honesty. Elementary earth-science time shows up in national parent surveys as a real sub-subject cell alongside inquiry.⁴⁹ This chapter can honestly own a weather window without pretending every home will log the same minutes. Dose is still three to five Science Hours plus one OUT when storms and schedules allow — short daily cards count.

How to teach it this week

Science Hour five beats stay. Shorten the try beat on a daily card day (10 minutes can be enough). Lengthen on rain-gauge setup, puddle map after rain, or Friday week look-back.

Warm-up looking (3–8 min). Stand at the window — or on a porch with adult present. Cover forecast captions. OBS-01 shape: silence first, then “What do you notice?”

Predict / wonder (2–5 min). “What do you think the card will say this afternoon?” or “Will the puddle be bigger or smaller when we come back?” Two choices at 5–7; reason at 8–10.

Try / record (5–15 min). Hang or draw today's card; ribbon wind test; shady thermometer morning/afternoon; puddle outline; cloud booklet page. Child's record is the data.

Talk / pattern. Same/different versus yesterday. Friday: week look-back in ordinary words. Names for cloud types only after several drawings exist.

Exact wording you can say

⁴⁹Elementary earth-science / geology among homeschool science sub-subjects: NCES Parent and Family Involvement in Education 2016 (NCES 2020-001) Table 9 cells — quote as survey rows, not minutes-per-week promises.

- “What is the same about today and yesterday — and what is different?”
- “What do you see in the sky?”
- “Warmer or cooler — how do you know?”
- “Wet or dry? Windy or still?”
- (8–10) “What does our week’s table show?”
- “What would change your mind about the pattern?”

First try-it for the student

- **5–7 default:** WEATH-01 Cloud Week Book start + today’s card compared to yesterday (OBS-01 first sitting if cards are new).
- **8–10 default:** WEATH-01 or WEATH-04 Temperature Morning/Afternoon with same shady spot; begin a simple week table.
- Soft alternate: WEATH-03 Wind Flag Test; after rain, WEATH-05 Puddle Map.

Fade example across five cards.

Card 1: You ask notices; you write their words on the card.

Card 2: They draw the sky; you add the wet/dry word they choose.

Card 3: They draw and choose wind word; you only ask same/different.

Card 4: They run same/different first; you wait before adding anything.

Card 5: Friday clothesline — they teach a sibling or stuffed animal the week pattern while you listen for radar-first language to redirect.

When to stop talking (again, because screens pull hard). If you hear yourself narrating the whole sky — “That’s a cumulus, and the wind is from the west, and the app says...” — stop. Put one finger on the glass. Ask one notice question. Wait. The hour is their looking, not your forecast career.

How to fade help. First days: you scribe the card from their words. Next: they draw sky and dictate one word. Later: they fill the card and tell same/different without your summary first.

When to stop talking. After same/different, wait. If they stare at your phone pocket, put the phone in another room for the looking minutes.

Age-band moves

Band	This week's ambition
5–7	Daily card; cloud drawings; wind words; oral same/different; adult handles thermometer; short oral exit
8–10	Week table or pictograph; rain gauge row; morning/afternoon temps same place; preparedness questions; still refuse radar-first and policy sermons

This-week shape (hypothetical illustration)

Day	Beat	Named try
Daily short	Card hang	WEATH-01 page ($\approx$ 10 min)
One longer day	Temp twice	WEATH-04 Morning/Afternoon
After rain	Puddle	WEATH-05 Puddle Map
Friday	Week look-back	Cards on a clothesline + talk box
Optional OUT	Sky or water edge	OUT-05 Night Sky Sit soft; OUT-02 Puddle/Creek Edge

Focus sentence Monday: “We look out the window before the phone.” Friday: pattern sentence plus one diagnostic — radar-first, invented data, or fear-binge without prep questions.

Materials prep. Cardstock weather cards or a mini booklet; ribbon; shady outdoor thermometer (no mercury toys; adult present); straight-sided jar for rain (8–10); chalk for puddles; blanket for night sky. Cool LED flashlight for prep talks — never lasers; never sun-staring through “filters.”

Practice that actually builds learning

Talk box (locked opening)

Opening question: “What is the same about today and yesterday — and what is different?”

Follow-ups:

1. What do you *see* in the sky? (*observe*)
2. What does your card or drawing show? (*record*)
3. How do you know from *this* window look — not from the app alone? (*reason*)
4. Warmer or cooler / wet or dry / windy or still — what else could someone notice? (*check*)
5. (8–10) What does our week’s table show? What would change your mind about the pattern? (*revise*)

How to wait. Slow three. If they are mid-draw of a cloud, do not correct the shape into a textbook cumulus. Look at the sky with them.

Stuck silence usually means: they think the answer is on your phone; the question was vague; yesterday’s card never existed so same/different has no anchor; wait was zero; they fear “wrong weather words.” Hang yesterday’s card beside today’s. Point at the glass. Ask for one notice only.

In-home try-its (WEATH + OBS/DRAW bridges)

WEATH-01 Cloud Week Book

- **Time:** 10 min daily × 5
- **Materials:** Mini booklet; pencil; gray crayon
- **Safety:** No sun stare; adult for outdoor
- **The fun:** A tiny book that is only yours
- **The skill:** Daily cloud drawing + word; Friday pattern talk; names after patterns exist
- **Steps:** Draw; note time; Friday: “This week clouds were mostly ____.”
- **Error patterns:** Copying app icons; skipping real looking

WEATH-02 Rain Gauge Simple

- **Band:** 8–10

- **Time:** Setup 20 min + storm checks
- **Materials:** Straight-sided jar; ruler; open place off the ground
- **Safety:** Adult present; no rooftop; lightning — stay inside
- **The fun:** Measuring a real storm's gift
- **The skill:** Record date/amount; empty after measure; refuse invented mm; refuse roof-drip as "rain"
- **Steps:** Mark cm; place; measure; table row for one storm

WEATH-03 Wind Flag Test

- **Band:** 5–7 (both welcome)
- **Time:** 15–25 min
- **Materials:** Ribbon or light scarf; compass optional
- **Safety:** Adult present; no street standing for "better wind"; no climbing; extremes — stay in
- **The fun:** Ribbon as a wind detector
- **The skill:** Calm / breezy / strong words; same flag different days
- **Steps:** Observe; record wind word + direction guess

WEATH-04 Temperature Morning/Afternoon

- **Band:** both
- **Time:** 10 min × 2 same day
- **Materials:** Outdoor thermometer; shade placement
- **Safety:** Adult present; glass thermometer care; no mercury toys; no licking thermometer
- **The fun:** Two numbers, one place
- **The skill:** Same shady spot; morning vs afternoon; sun/heat picture after evidence
- **Steps:** Record two numbers + place kept same
- **Error patterns:** Thermometer in direct sun then comparing unfairly

WEATH-05 Puddle Map

- **Band:** both
- **Time:** 20–30 min after rain
- **Materials:** Chalk; sidewalk puddles
- **Safety:** Adult present; traffic awareness; no contaminated flood-water play
- **The fun:** Chalk outline that lies about the future
- **The skill:** Outline; return later; notice shrink; wonder where water went, then name
- **Steps:** Chalk; photo or draw; exit: "Puddle got ____."

WEATH-06 Snow / Ice Crystal Soft

- **Band:** both (climate permitting; else skip)
- **Time:** 15–25 min
- **Materials:** Black cloth; magnifier; cold day
- **Safety:** Adult present; time limits in cold; dry clothes; frostbite awareness
- **The fun:** Catching a flake that will not wait
- **The skill:** Look fast; draw real flake; refuse generated snowflake printouts as “ours”; refuse hairdryer erase-before-look
- **Steps:** Catch; look; draw; exit: “Real flake looked ____.”

OBS-01 Window Weather First — silence; five notices; name weather words only after.

DRAW-06 Cloud Shape Record — time stamp; draw today’s look; no forced type names yet.

Out-of-home try-its**OUT-02 Puddle or Creek Edge Soft**

- **Time:** 20–30 min
- **Safety:** Adult present; no flood/fast current; no drinking; waterproof plan; local rules
- **The fun / skill:** One focused water-edge look; return scoops; record “I saw ____.”
- Bridge to weather: clarity and level after rain versus a dry week — same/different across visits.

OUT-05 Night Sky Sit

- **Time:** 20–30 min
- **Materials:** Blanket; optional star app **after** 5 min naked-eye looking
- **Safety:** Adult present; traffic; cold; **no lasers**; local light/safety
- **The fun / skill:** Notice bright vs faint; moon shape if present; “I noticed ____ before any app.”
- Sky patterns cousin to weather window — still observe before the screen.

Library weather picture books: motivation after a real card week; compare one book picture to your Friday clothesline — book does not overwrite your cards.

Blocked / mixed

Blocked: OBS-01 + one card until same/different uses yesterday's real card.

Mixed: Daily WEATH-01 with Friday table at 8–10; WEATH-04 on a clear day; WEATH-05 after rain; night sky on a clear evening as cousin sit.

One incorrect example to diagnose

Illustration (hypothetical). Parent opens a radar app at the start of Science Hour. Child copies green blobs onto the weather card without looking outside. Friday's "week" is seven app icons. Parent feels productive.

Diagnose: Radar-first; invented looking; screen as the lookable.

Next move: Trash none of the child's pride in coloring — but mark those cards "app copy" in the margin. Start a new clothesline from today's real window. OBS-01 silence. Locked opening tomorrow with a real yesterday. Phone only after notices. Do not generate a sky photo and file it as the student's result.

Severe weather preparedness (short, practical)

When storms are in your season, add three questions to one hour — not a binge:

1. What do we do if we hear thunder while outside?
2. Where are our flashlights and shoes?
3. Who decides when we leave the park or stop the puddle sit?

Practice one crumb (flashlight location). Stop. That is enough for Science Hour. Deeper family emergency plans belong to family preparedness time, not to a fear documentary.

Exit crumb

One card versus yesterday (same + different) — or, at 8–10, one week-table sentence: "This week was mostly ____."

Dose and safety spine for weather work

Daily cards can be ten minutes. One longer sit plus Friday look-back completes the week. Lightning — stay inside. No rooftop gauges. No sill climbing. No sun

stare. Adult present for outdoor thermometer, night sky, creek edges, and cold-weather flake sits. Prefer cool LED if you need indoor light talk; refuse lasers. Stop when weather extremes make looking unsafe; the notebook waits.

How weather connects to fair-test light (Chapter 3)

Morning/afternoon temperature is a natural fair-test cousin: change only the time of day; keep the shady spot the same. If the thermometer sits in sun in the morning and shade in the afternoon, you changed two things — place *and* time — and you cannot tell for sure what the sun did. WEATH-04's "same shady spot" line is Chapter 3 in outdoor clothes.

Rain gauge honesty is the same muscle: roof drip is a second water source. Put the jar in the open, off the ground, adult present, and empty after you measure. Lightning ends the try. Investigation continues another day.

How weather connects to living needs (Chapter 4)

Bird feeder tallies (LIFE-04) get more interesting when weather cards hang beside them: "Most visits when ____." Do not force a causal sermon. Notice same/different. Plants on the sill still need their one-thing water trial kept fair — a storm week outside does not excuse moving one seedling into better light while claiming you only studied rain.

Cloud names — when and how

After several days of drawings, you may offer one or two names that fit what is on the page — not a quiz of ten Latin labels. If the child's drawing shows puffy heaps, you may say some people call those cumulus. If the drawing shows a gray sheet, you may say stratus. If they disagree because their sky looked mixed, keep their word and add "mixed." Name-after-pattern is observe-before-explain applied to vocabulary.

Seasonal talk without a dump

At 8–10, Friday look-backs across a month can sound like: "Our September cards were often ____." That is seasonal weather in ordinary language — the trail that

standards maps mark with tables and pictographs.⁵⁰ Comparing your town to a *named* atlas page about another region’s climate is an edge sit for a curious older child, once, with a book on the table — not a 5–7 spine and not a policy fight.

A longer hypothetical week (illustration only)

Monday: OBS-01 at the kitchen window. Five notices. First card hung. Phone still in the other room.

Tuesday: WEATH-03 ribbon on the porch with adult. Wind word: breezy. Card two.

Wednesday: WEATH-04 morning number in shade; afternoon number in the same shade. Child writes both. Same/different talk uses Tuesday’s card too.

Thursday: Rain. Stay inside for thunder. Measure the gauge after the storm passes and the adult says it is safe — or skip measure and only do a window card: wet, cloudy, calm. Honesty beats a rooftop risk.

Friday: Clothesline of five cards (one short indoor-only day still counts). Talk box. Pattern sentence. Optional library picture book *after* the clothesline.

Diagnostic the parent jots privately: Tuesday the child asked for the phone at second one. Plan Wednesday: phone stays in the bedroom until after the card.

Apartment, winter, and “nothing happening” weeks

If the sky looks “the same” for five days, the pattern *is* the sameness. Write it. “Mostly clear and dry” is a real Friday sentence. If winter cancels puddle maps, use window frost noticing (no tongue on metal) or skip to night sky when clear and cold gear is enough. If you have no outdoor thermometer, warmer/cooler by jacket-needed and glass-feel is acceptable at 5–7 while you borrow or buy a simple shady thermometer later. Do not invent numbers.

What this chapter is not

Not a meteorology course. Not a climate-culture-war pamphlet. Not a 180-day weather worksheet dump. Not activity mania (ten instruments, no card). Not kitchen-as-Chemistry anything. Not generated radar art as student data. Title: **Science** or **Earth Science** when it is actually that.

⁵⁰Seasonal patterns in tables and pictographs on standards maps: NGSS Lead States (2013); California 2016 *Science Framework* — trail language for Friday look-backs.

For the student

Weather is what you notice outside — today, and compared with yesterday.

This week’s science looking is not “win at the forecast app.” It is not “watch scary storm videos.” It is: look out the window, make a card, and say what is the same and what is different.

Tiny worked example

Yesterday’s card: cloudy, dry, calm.

Today’s card: cloudy, wet, breezy.

Same: cloudy.

Different: wet instead of dry; breezy instead of calm.

You know because you looked — and because yesterday’s card is still hanging where you can see it.

Try 1

Fill today’s weather card. Compare it to yesterday. Say one same and one different. Draw the sky if you can see it.

Try 2

Make a ribbon wind test. Is it calm, breezy, or strong? Write the word. Do the same tomorrow. Did the wind change?

Explain it back

Tell a grown-up: “Today is ____ . Yesterday was ____ . The same is ____ . The different is ____ . I know because I saw ____.”

A note about phones

Phones can check after you look. They are not the first look. If the app and your eyes disagree, write both — and trust that your eyes saw *your* street.

If the weather is wild

If thunder is close, looking from a safe inside window is still looking. Do not go out for a “better” puddle. Tell your grown-up what you see from inside. Science Hour can be short on storm days. Being safe is part of being a good looker.

If you miss a day

Leave a blank. Write “missed” and the date if you want. Do not fill in fake weather to make the week look pretty. Pretty fake weeks are not science looking.

Challenge

Keep cards for five days. On Friday, hang them in a row. Say a pattern in ordinary words: “This week was mostly ____.” If you are ready, make a simple table with a grown-up: rainy days vs dry days.

Night sky cousin

On a clear night, sit with a grown-up on a blanket. Notice for five minutes before any app. What is bright? What is faint? Is the moon out? No laser pointers. Come back inside when you are cold.

If it isn’t clicking

Diagnostic 1 — Radar-first / screen hunger.

Next move: Phone in another room for looking minutes. OBS-01 silence. Five notices before any forecast word.

Diagnostic 2 — No yesterday anchor.

Next move: You cannot do same/different without a prior card. Make today’s card. Tomorrow the talk box works. Do not invent yesterday.

Diagnostic 3 — Fear binge or policy fight arriving as content.

Next move: Stop the footage or the argument. Preparedness three questions or return to the window card. Live controversy is not the try box.

Slow down when cards are empty or copied from icons — stay on OBS-01 and one real card a day.

Go ahead when same/different is easy — add WEATH-04, rain gauge, week table. **Human help** when severe weather anxiety is big: keep Science Hour on calm window looking; use family preparedness resources you trust; this book is not trauma therapy and not emergency management certification.

Instruments: enough and not too many

A ribbon, a shady thermometer, a straight-sided jar, and a pencil beat a pile of unused kit parts. If a vendor weather station arrives as a gift, use **one** sensor in one fair way — same place, dated log — and still begin with eyes out the window. Instruments extend looking. They do not replace it. If setup eats the whole hour, you bought a project, not a Science Hour; simplify tomorrow.

Talk-box stuck scripts (copy-ready)

If silence hardens after the locked opening, try one of these — then wait again:

- “Let’s look at yesterday’s card together. Point to one thing.”
- “Is the ground wet or dry — just that one word?”
- “Show me with your hands: big wind or little wind?”
- “You may look for ten more seconds before you answer.”

If they answer only with app language (“80% chance”), say: “That might be the phone. What do *your* eyes see right now?” Point at the glass.

Records crumb

Date the cards. A stranger flipping the notebook should see **Science** (or Earth Science if you use that honestly) and a clothesline or booklet of looking — not a brand name and not a generated collage.

Revisit rhythm

Next Tuesday: pull Friday’s clothesline back out. Mix one new card. Ask same/different against last week’s “mostly ____.” Stop while a look remains. Life-of-the-habit for weather is the cards you keep, not a diploma and not a ninth teaching chapter.

Cross-check with the book's focus lock

Observe before explain: window before app, drawing before cloud-type quiz. Fair test light: same shady spot for temperatures; open placement for rain; name the second change when wind and rain arrive together and you cannot tell for sure which soaked the sidewalk first. Both locks still hold in the sky.

Tools, including AI

Point back to the Science Hour AI box.

You may ask a tool *on your account* to help *you* understand a named weather idea, or to draft a SCRIPT for after the child's card talk. You may not generate a sky photo, radar collage, or "what I saw" paragraph and file it as the student's observation. Generated image $\neq$ data. Invented week tables $\neq$ data. Parent holds the account; unaided first on looking; 5–7 no child-facing open chat; 8–10 still parent in the room.⁵¹

Parent wrong-answer drill (two minutes)

Read these aloud to yourself once this week:

1. Child opens with radar language → phone away; window silence; five notices.
2. Child invents Tuesday because they forgot → blank "missed"; today's real card only.
3. Adult relative starts a climate argument at the table → "We're doing today/yesterday cards now; adult talk later."
4. Child wants storm videos → preparedness three questions; then a calm window card.

You are not managing a deficient parent. You are practicing hearing. Many adults feel rusty. That is normal. The move is always: lookable first, one good question, wait.

⁵¹Science Hour AI box / COPPA: parent holds account; generated sky or radar collage $\neq$ student observation; unaided first.

What “done enough” looks like

- ☐ Child can fill a weather card from a real window look and compare to a real yesterday card (same + different).
- ☐ At least three cards (ideally three to five) hang or sit in a booklet with dates or order.
- ☐ Parent kept phone radar as after-check, not as the opening lookable, for at least one solid run of days.
- ☐ One OUT or after-rain sit happened when safe (puddle, night sky soft, or library compare-after-looking).
- ☐ No invented week of data; no generated sky filed as student result.
- ☐ Severe weather, if discussed, included at least one preparedness question — not only fright.
- ☐ At 8–10 soft edge: a simple week table, rain-gauge row, or morning/afternoon temp pair with place kept the same.

Showable two ways: hang the cards *and* say the pattern aloud; or draw today’s sky *and* retell same/different.

Not done-enough: seven app icons pasted as a “week”; a climate pamphlet with no window; a storm-video binge with no card and no prep question.

When this holds, Chapter 6 can borrow sky and shadow cousins (shadow stick, light path) while this chapter’s clothesline stays as a habit you can revisit any Tuesday. Mix last week’s card with a new puddle sit; stop while a look remains. Title the work **Science** or **Earth Science** only when it is actually that — never a brand, never Little Thinkers Science I.

Practice menu reminder

Primary: WEATH-01 through WEATH-06; OBS-01; DRAW-06. OUT: OUT-02 puddle; OUT-05 night sky soft. Error focus: radar-first; climate culture-war as the hour; invented week of data. Exit crumb: one card vs yesterday; or week table at 8–10.

Sibling and apartment notes

One window is enough. A balcony with adult present counts. If the “window” is a shared hallway view, take a short supervised step outside when safe. Siblings can

alternate scribe and notice roles. Disagree soft: two honest sky notices can both stay until a second look.

What “pattern” means in kid words

A pattern here is not a secret code. It is “mostly cloudy this week,” “rainy after three dry cards,” “afternoon warmer than morning in our shady spot.” If the week is all over the place, “no clear pattern yet — we need more cards” is an excellent Friday sentence. That honesty is science looking.

Done-enough is a path through weather looking, not a percentile and not a diploma. If the clothesline shows real days and the child can say same/different without grabbing your phone, you can move on — and still hang a card any morning the sky asks for eyes.

Access date for the teaching moves and safety spine behind this chapter: **15 September 2026**. Limits belong in one plain sentence when you cite a study elsewhere in the notes: this is useful guidance for a home Science Hour, not a promise that every home will see the same sky or the same result.

Chapter 6 — Light, path, push and pull



Figure 8: A flashlight, a card, a clear cup, and a small board ramp with a toy car on a wooden table — still-life, no people, no logos.

Why this matters

Light on a wall and a car on a ramp are physical-science looking. They are not a laminated Newton poster. They are not $F = ma$. They are not a laser pointer as a “better flashlight.”

At ages 5–10, a child can block a beam and say what happened. They can push a toy car harder and say what changed. They can notice a shadow move across a stick without being asked to recite Earth-turn vocabulary first. That is enough for this chapter.

This is worth the struggle because two wrong habits love this week. One habit dumps equations before anyone has looked. The other habit treats force as fuel — the push “runs out,” the thrown coin still has an upward force, the book sits on a table that “does nothing.” Prior ideas are the mind at rest. Hear them. Repair with another look. Do not scold.

The move that unlocks the week is small. Put a flashlight and a card on the table, or a board and a car. Ask what happened when the light hit this — or when we pushed or pulled. Wait. Let them draw the path or say the motion. Then, if it still fits, give one short name in ordinary words: block, pass, bounce, push, pull, farther, slower.

Kindergarten and early elementary science maps already treat push harder / pull harder, objects seen when illuminated, and materials in a light path as lookable jobs. Later maps add balanced and unbalanced in qualitative language, magnets as cause and effect, and a light energy-transfer edge. Those maps are texture, not a mandate for your kitchen. You do not need to live in a standards state. You do not need equations. You need one light path try and one ramp try this week.

You do not need to be a physicist. You do need a lookable before the explanation, and one question you do not already have the vocabulary sentence for.

Sister chapters already built the spine: notice before name, draw what happened, one deliberate change with “can we tell for sure?” This chapter puts those moves on light and motion. Weather had a window. Living things had needs you can see. Here the lookable is a beam path and a push or pull you can feel with a toy car or a drawer. High-school Physics stays in the older book when the child is older. One pointer is enough. Then teach the flashlight and the ramp.

For the parent: understand it yourself

Here is the everyday picture.

A flashlight under a blanket fort. A child blocks the beam with a hand, then with a card, then with a clear cup. They laugh when the wall goes dark and bright again. Nobody has said “opaque” yet. Tomorrow they can still point to the card and say what the light did.

Here is the precise picture.

Objects are seen when light reaches them. Materials in a light path can block, pass, or bounce light enough that a spot moves. A harder push or a taller ramp can change how a car moves — farther, faster, or both — when other things stay the same. Magnets get their own sitting so “invisible push” does not swallow the ramp language in one confused hour. Energy language, if it appears at all, stays at a labelled grade-four edge and still is not an equation dump.

Force-as-fuel is the prior idea to hear. A child may say the push got used up, or that something keeps pushing upward after a throw. You do not crush the idea with a lecture. You set another try: push gently, push harder, keep the car the same, keep the ramp the same, ask what they saw. The repair is another look.

Wrong answers you should be able to hear

1. Force-as-fuel as the final answer.

“The push ran out, so the car stopped.” Full stop. No second look.

What it usually means: a common prior idea is doing the talking.

What you say: “Let’s try again. Same car. Same ramp. Push harder this time. What do you notice?”

2. Lasers as better flashlights.

A pointer appears because it looks “more science.” Eyes get risk for free.

What it usually means: the kit won over the bank.

What you say: “We use a cool LED flashlight. Lasers stay off this table.”

3. An equation dump.

$F = ma$ or a force diagram lands before anyone has blocked a beam or run a car.

What it usually means: secondary school grain arrived early.

What you say: “First we look. What happened when the light hit the card — or when we pushed?”

4. The table does nothing.

“The book just sits. The table is not pushing.” Said as settled fact without another look.

What it usually means: contact forces are invisible, so they feel unreal.

What you say: “Interesting. What happens if we take the table away?” Soft book-drop with adult hands — then talk. Still no equation.

5. Glow-stick as a chemistry path.

A commercial glow stick becomes a mixing license or a “chem lab.”

What it usually means: a change picture got promoted to a credit.

What you say: “We observe. Adult bends the stick. We follow the package. This is not a chemistry credit.”

Five-minute parent warm-up

Tomorrow’s flashlight and a card sit alone with you. Turn on the light. Block the beam. Say out loud, without saying “opaque”: “The wall went dark where the card sat. The light stopped at the card.”

Then set a short board as a ramp and a toy car. Push gently. Push harder. Say what changed without saying “acceleration.”

Practice the locked opening to the empty chair:

“What happened when the light hit this — or when we pushed or pulled?”

Count a slow three. Do not answer yourself. Put lasers away. Put the Newton poster away. Five minutes. Lookable first.

How to teach it this week

The Science Hour shape stays the same. Light and force share this chapter so the week stays one physical-science looking week, not two thin units.

Session shape (~20–40 minutes)

1. **Lookable down (3–8 min).** Flashlight and card, or board and car, or shadow stick. Materials visible before vocabulary.
2. **Notice / predict before naming (3–7 min).** “What do you notice?” “What do you think will happen if we block the beam / raise the ramp?”

3. **Try (5–15 min).** One light path try *or* one push-pull try per hour. Magnets as a separate sitting at 8–10.
4. **Record (3–8 min).** Draw the beam path or the ramp run at 5–7. Picture-plus-sentence or a two-row result at 8–10.
5. **Exit crumb / talk box (1–5 min).** Locked opening. One follow-up. Stop while a look remains.

Twenty-minute day: block the beam, draw the path, locked question. That is a full hour.

Exact wording you can say

Opening, locked:

“What happened when the light hit this — or when we pushed or pulled?”

Then wait. A slow three is a convention, not a sacrament. Look at the wall spot or the car track, not at the child, if the silence is hard.

If they freeze: “Point first. Where is the dark patch? Where did the car stop?”

If they say the push ran out: “Show me with another try. Same car. Same ramp. What changes when we push harder?”

If they ask for the science word: “We can name it after. What do you *see*?”

If they aim the light at eyes: stop the hour. Safety first. Investigation continues another day.

Age-band moves

Ages 5–7. Block / pass with flashlight and toys. Shadow stick marks on a sunny day. Push-pull sort with doors, drawers, soft rope, toy car. Wind-push sail with a low fan and adult. Draw the path or the run. Oral exit. No variable vocabulary tests. Prefer weak fridge magnets if magnets appear at all; keep that sitting short.

Ages 8–10. Mirror bounce soft indoors with adult aim rules. Friction shoe rub — change surface only. Balance beam book — notice distance. Magnet strength distance as its own sitting. Fair-test light on ramp height: change height only; same car. Sketch the light path. Name change and same. Still no $F = ma$. Still no lasers. If looking habits are weak, stay on block-the-beam longer. Place by skill, not birthday.

First try-it for the student

Put a flashlight and a card on the table. Predict the shadow. Block the beam. Ask the locked opening. Draw the path. That is LIGHT-02, Block the Beam. It is the week's home base for light.

Or: same car, board ramp, change height only once. Ask what changed when we pushed or when the ramp got taller. That is FORCE-02, Car Ramp Height.

This week, honestly

Focus move: one light path try and one ramp or push-pull try; hear force-as-fuel; ask what they saw.

- **Monday:** LIGHT-02 Block the Beam (5–7) or LIGHT-03 Mirror Bounce Soft (8–10).
- **Tuesday:** FORCE-01 Push Pull Sort, then a short ramp notice.
- **Wednesday:** FORCE-02 Car Ramp Height with fair-test light language.
- **Thursday:** LIGHT-01 Shadow Stick Hour (yard) *or* FORCE-04 Magnet Strength Distance as a separate sitting (8–10).
- **Friday:** one diagnostic wrong (force-as-fuel as final answer; or laser request) and a done-enough check — draw beam path *or* say what changed when pushed harder.
- **Once this week, out of the home:** OUT-05 Night Sky Sit (light/dark noticing, app after five minutes naked-eye) *or* playground ramp/slide noticing with adult — push-pull language, not daredevil.

Three to five Science Hours. One out-of-home try. Friday a crumb. Not a physics semester. Not a circuit dump with household-unsafe setups.

Practice that actually builds learning

Named try-its below carry time, materials, safety, the fun, and the skill. At least one lives in the home. At least one lives outside it. The yard shadow stick, the playground ramp, and the night-sky sit *are* the practice. They do not replace the looking move.

Prefer cool LED flashlights. Refuse lasers in the default banks. Adult present for water, outdoors, fans near water, night work, and magnets with small parts.

Kitchen chemistry is a picture, not a credit — a glow stick stays observe-only.

Composite homes below are illustrations, not reported families.

Talk box (use every light or force hour)

Opening (locked): “What happened when the light hit this — or when we pushed or pulled?”

Follow-ups:

1. What do you *see* in the path of the beam — or on the track?
2. What does the drawing / notebook show?
3. What changed when we pushed harder — or when we put the card in the way?
4. (8–10) Was anything balanced or unbalanced — what makes you say that? Can we tell for sure?
5. You may change your mind after one more try.

How to wait: After you ask, count a slow three. After they stop, wait again. If they are mid-look or mid-draw, do not cut them off. Look at the wall spot or the car, not at their face, if the silence is hard.

Stuck silence usually means: the question was vague; wait-time was zero; they are hunting “opaque” or “force” because you asked name-first; they think science means repeating your explanation; the materials never landed; they offered a book-only guess. Pose, wait, point back at the lookable. Do not fill the silence with the vocabulary.

One incorrect example to diagnose: Parent shines a flashlight, says “This is opaque,” and asks the child to repeat the word. The card never gets a child’s drawing. The child can recite “opaque” and cannot show what the beam did. Repair: cover the vocabulary. Block the beam. Ask the locked opening. Draw the path.

LIGHT-01 Shadow Stick Hour (in-home / yard)

Band: both.

Time: 15 minutes × 2–3 same day.

Materials: Stick in soil or pot; chalk or tape marks; sunny day.

Safety: Adult present; sun safety; no sun-staring.

The fun: The shadow “walks” while the stick stays still.

The skill: Observe shadow path before Earth-turn vocabulary.

Steps

1. Plant the stick. Mark the tip of the shadow morning.
2. Mark again near noon. Mark again afternoon. Do not move the stick between marks.
3. Ask what they notice before any turning-Earth picture.
4. Exit: three marks photo or drawing + “shadow moved ____.”

Ages 5–7: oral notice; you scribe.

Ages 8–10: simple sketch with time labels.

Error patterns: Moving the stick; sun-staring; explaining Earth turn before the third mark.

Exit crumb: “Shadow moved ____.”

LIGHT-02 Block the Beam (in-home)

Band: 5–7 (8–10 may join and sketch).

Time: 20–25 minutes.

Materials: Flashlight (cool LED); small toys; wall or ceiling spot; card; clear cup.

Safety: No eye shine; adult present; no lasers.

The fun: Making the wall go dark and bright on purpose.

The skill: Predict shadow; notice block / pass; draw path without requiring “opaque.”

Steps

1. Predict: what will the shadow look like?
2. Block with a toy. Block with a card. Try a clear cup.
3. Ask the blocked opening.
4. Draw the beam path and the dark patch.
5. Exit: “Shadows need ____ and ____.”

Error patterns: Eyes in beam; fear talk without trying; vocabulary first.

Exit crumb: Drawing of path + oral sentence.

Illustration. A parent sets a flashlight on books so the beam hits the wall. The child predicts a circle. The toy makes a bumpy dark patch. The child changes the prediction for the card. That is looking. “Opaque” can wait.

LIGHT-03 Mirror Bounce Soft (in-home)

Band: 8–10.

Time: 20–30 minutes.

Materials: Flashlight; small mirror; paper target; adult aim rules.

Safety: Adult present; **no lasers**; no sun reflected into eyes; indoor LED preferred.

The fun: Making the spot land on a target feels like a quiet game.

The skill: Light direction picture — not a full optics course.

Steps

1. Predict where the spot will appear.
2. Bounce to the target. Adjust once.
3. Sketch the path: flashlight → mirror → target.
4. Exit: path sketch.

Error patterns: Mirror toward sun outdoors aggressively; laser pointers; equation talk.

LIGHT-04 Color Filter Peek (in-home)

Band: both.

Time: 15–25 minutes.

Materials: Transparent color report covers or safe filters; magazine page or colored blocks.

Safety: Adult present; never sun-viewing; filters are not eclipse glasses.

The fun: Familiar colors look “wrong” through a filter.

The skill: Notice which colors change before “wavelength” words.

Steps

1. Look through one filter. Say what changed.

2. Try a second filter. Record before naming.
3. Exit: “Through red, ____ looked ____.”

Error patterns: Staring at sun through filters; believing filters make “night.”

LIGHT-05 Glow Stick Observe (commercial; in-home)

Band: both.

Time: 15–20 minutes.

Materials: One commercial glow stick; dim space.

Safety: Adult present; adult bends; follow package; no chewing; wash if leak; dispose properly; no cutting open.

The fun: Light that does not feel hot.

The skill: Observe before/after; light-without-heat picture — not a chemistry path.

Steps

1. Observe before bend.
2. Adult activates. Observe after.
3. Exit: “Light without feeling hot: ____.”

Error patterns: Cutting open; fluid on skin; calling it fire; “now we do chemistry.”

LIGHT-06 See in Dim Room (in-home)

Band: 5–7.

Time: 15–20 minutes.

Materials: Dimable room; familiar objects; nightlight OK for comfort.

Safety: Adult present; no locking children in dark; no pitch-black panic.

The fun: Eyes “wake up” after a minute.

The skill: Needing light to see — careful, not scary.

Steps

1. Dim the room. What can you see right away?

2. Wait about one minute. What can you see now?
 3. If someone turns on a phone light, note it as a change.
 4. Exit: “I needed ____ to see ____.”
-

FORCE-01 Push Pull Sort (in-home)

Band: 5–7.

Time: 15–20 minutes.

Materials: Door, drawer, toy car, soft rope.

Safety: Adult present; no slamming doors on fingers; no hitting siblings as “force examples”; no heavy lifting contests.

The fun: Acting push vs pull with real house jobs.

The skill: Force as push/pull in kid words after acting.

Steps

1. Act a push. Act a pull.
 2. Sort everyday actions into two piles or two lists.
 3. Ask the locked opening for one action.
 4. Exit: two pushes + two pulls listed.
-

FORCE-02 Car Ramp Height (in-home)

Band: both.

Time: 25–35 minutes.

Materials: Toy car; board ramp; height blocks; same car each try; clear track.

Safety: Adult present; clear track; no cars at faces; stairs-as-ramp only with adult judgment.

The fun: Higher start, longer run — if you keep the try fair.

The skill: Fair-test light: change height only; talk farther/faster without formulas.

Steps

1. Predict: taller ramp → what happens?
2. Change height only. Same car. Same push style (or no push — release from rest).

3. Mark runout with tape.
4. Ask: what did we change — what did we keep the same — can we tell for sure?
5. Exit: fair sentence + result.

Ages 5–7: oral fair sentence; you help the “same” list.

Ages 8–10: child names change / same; optional two-row table.

Error patterns: Pushing differently each time; confounding surface and height; force-as-fuel as the only explanation with no redesign.

Illustration. A child insists the push ran out. You keep the car and the low ramp. One gentle release, one from a taller block. The child says the taller start went farther. Force-as-fuel can wait for another look next week. Today’s repair was the second run.

FORCE-03 Friction Shoe Rub (in-home)

Band: 8–10.

Time: 20–30 minutes.

Materials: Same shoe block or eraser; tile vs carpet; optional gentle spring scale if owned.

Safety: Adult present; no sandpaper on skin.

The fun: Feeling how “sticky” a floor is without a lecture.

The skill: Change surface only; name friction after.

Steps

1. Same object. Tile push. Carpet push.
2. Feel or measure effort.
3. Exit: “Rougher surface → ____.”

Error patterns: Different pushers with different strength unnoticed; skin abrasion tests.

FORCE-04 Magnet Strength Distance (in-home; separate sitting)

Band: both (prefer weak fridge magnets for 5–7).

Time: 20–30 minutes.

Materials: Magnet; paperclip chain; ruler; optional cards between.

Safety: Pinch caution with strong magnets; keep magnets away from devices you care about; adult present; small parts.

The fun: The clip “lets go” at a distance you can mark.

The skill: How far until the clip drops; keep orientation notes; do not mash with ramp language in the same confused hour.

Steps

1. Make a short clip chain. Measure distance until drop.
2. Change one thing only — distance cards, or magnet, not both unnoticed.
3. Exit: distance number + setup sketch.

Error patterns: Confounding magnet type and distance; device damage; treating magnets as the whole “force unit.”

FORCE-05 Balance Beam Book (in-home)

Band: 8–10.

Time: 20–30 minutes.

Materials: Ruler; fulcrum block; coins or cubes.

Safety: Adult present; no standing on chairs to “be the beam.”

The fun: Finding the quiet middle.

The skill: Distance matters; refuse “magic balance.”

Steps

1. Predict balance.
2. Try positions. Notice distance.
3. Exit: “To balance, I moved ____.”

Error patterns: Changing coin number and distance together without noticing.

FORCE-06 Wind Push Sail (in-home)**Band:** 5–7.**Time:** 20–25 minutes.**Materials:** Toy car or boat in tub; paper sail; fan on low with adult.**Safety:** Adult present; dry hands near electric fan; GFCI awareness; no hair in fan; no high fan near water.**The fun:** Air that you cannot see still moves the toy.**The skill:** Push of air picture with/without sail; same fan setting.**Steps**

1. Predict with sail / without sail.
 2. Same fan setting. Observe.
 3. Exit: “Air can ____.”
-

Cousin tries (optional; keep light)**TRIAL-07 Flashlight Through Stuff** — predict pass/block for paper, foil, clear cup; record.**OBS-05 Shadow Notice First** — notice before name on any sunny wall.**TRIAL-04 Balloon Static Soft** — soft static notice; adult present; not a shock dare.**OUT-05 Night Sky Sit (out-of-home)****Band:** both.**Time:** 20–30 minutes.**Materials:** Blanket; optional star app **after** five minutes naked-eye looking.**Safety:** Adult present; traffic; cold; **no lasers**; local light/safety.**The fun:** Dark that is not a scare if an adult is there.**The skill:** Light/dark noticing; bright vs faint; moon shape if present.**Steps**

1. Sit. Notice for five minutes with eyes only.
2. Then, if you want, open an app.

3. Exit: “I noticed ____ before any app.”

Error patterns: App-first; laser pointers at sky; road viewing without adult.

Playground ramp / slide noticing (out-of-home)

Band: both.

Time: 15–25 minutes on site.

Materials: Eyes; notebook crumb afterward.

Safety: Adult present; playground rules; push-pull *language*, not daredevil; no forcing scary heights.

The fun: The slide they already love becomes a lookable.

The skill: Push, pull, start high / start low — noticed, not performed as a stunt.

Steps

1. Watch one safe go (child or sibling with consent).
2. Ask: what happened when they pushed off — or when they started higher?
3. At home: one sentence or drawing.

Error patterns: Turning science hour into a bravery contest; ignoring posted rules.

Blocked vs mixed practice

Blocked (new move): one hour is only Block the Beam, or only Push Pull Sort, until the child can point without hunting vocabulary.

Mixed (when the move is warm): beam path Monday, ramp Wednesday, night-sky Friday — still one focus question per hour.

Library science shelf after a try (not instead): compare one book picture of a shadow or ramp to *your* drawing. Book does not overwrite home data.

Mixed week illustration (labelled illustration, not a reported family)

Monday: Block the Beam. The child draws a dark patch and says the wall went dark. Tuesday: Push Pull Sort — door push, drawer pull, car push, rope pull. Wednesday: Car Ramp Height — one block, then two, same car, tape marks on the floor. Thursday: Shadow Stick in the yard — three chalk marks; no Earth-turn speech yet. Friday: locked question; parent hears “the push ran out”; second gentle ramp try; notebook line: “taller start → farther.” Saturday evening optional: Night Sky Sit, five minutes eyes-only. That week is physical-science looking. It is not a physics credit.

Incorrect example bank (diagnose one on Friday)

Name-first light hour. Parent says “transparent, translucent, opaque” and quizzes the list before the flashlight is on. Repair: materials down; locked opening; name after.

Confounded ramp “proof.” Child changes ramp height *and* gives a harder shove, then claims height is the reason. Repair: “Can we tell for sure? Let’s keep the push the same.”

Demo with no record. Adult does a perfect mirror bounce while the child watches. No drawing. Repair: child aims (with adult rules) once; child sketches.

Generated beam diagram as data. A fluent machine image of “light rays” is taped into the notebook as today’s result. Repair: real flashlight, real card, child’s messy path drawing. Generated image $\neq$ data.

Magnet-and-ramp mash. One hour tries to teach gravity, friction, and magnetic attraction with three demos and no child’s sentence. Repair: magnets get their own sitting.

How to fade help

First you hold the flashlight steady and ask what they notice. Then they hold the light (beam away from eyes) while you ask. Then they predict, try, and draw with you only naming the “same” list when asked. Stop talking when they are mid-draw. Struggle before rescue: ask, wait, hint (“look at the wall”), then model one path

arrow if needed. Do not grab the answer. Do not fill the silence with “because opaque materials...”

When to stop talking

When the car is rolling. When the chalk is marking. When the child is mid-sentence about the dark patch. When a sibling is restating a claim before agreeing. Your job is one good question and a lookable that stays in view.

For the student

You are allowed to look before anyone gives you the science word.

Light needs a path. When something sits in the way, the wall can go dark. When something lets light through, the wall can stay bright. When light bounces, a bright spot can move.

A push and a pull are both ways to make something move — or to try to. Push a car gently. Push harder. Keep the car the same. Keep the ramp the same. Say what you saw. You may change your mind after one more try.

Tiny worked example

A flashlight points at a wall. You put a card in the way. The wall has a dark patch shaped a little like the card. You draw a line for the light and a blob for the dark. You say: “The light hit the card. The wall went dark there.” You did not need the word “opaque” yet.

Two tries

1. Block a beam with a card. Draw the path. Say what happened when the light hit the card.
2. Push a toy car on a ramp harder, then gentler — or raise the ramp once. Say what changed. Keep one thing the same on purpose.

Explain it back

Point at your drawing. Tell someone: “This is where the light went. This is what changed when we pushed.”

Challenge

Without looking at a book first, make a shadow on purpose. Then make it bigger or smaller by moving the toy — not by guessing a vocabulary word.

One more challenge (optional)

On a sunny day, mark a stick’s shadow three times. Do not move the stick. Tell someone what the shadow did. You may still wonder why. Wondering after looking is allowed.

If it isn’t clicking

1. Force-as-fuel will not budge.

They keep saying the push ran out, and they refuse a second try.

Next move: smaller materials. Same car. Two releases only. You write what stayed the same. Ask only “what do you see?” Do not lecture Newton. If fear of being wrong is loud, celebrate the redesign: “Scientists try again.”

2. Safety fear freezes the hour.

Dark room, fan, or magnets feel scary.

Next move: choose a softer bank card — Block the Beam in a bright room; Push Pull Sort with a drawer; Shadow Stick outdoors at a comfortable hour. Night-sky sit can wait. Comfort is not a failure.

3. Coverage itch.

A parent (or a curriculum map) wants circuits, simple machines posters, and energy formulas this week.

Next move: one phenomenon. Beam or ramp. Put the map away until the child can draw a path or name one change. Depth is the hour. Coverage is a map.

4. “I’m bad at drawing paths.”

The notebook stays blank because art fear arrived.

Next move: three lines and an arrow count. You may scribe the oral sentence under

their scribble. Science drawing is data, not art class. Point back to Chapter 2 if needed.

5. Magnets swallowed the week.

Every hour became clip chains; ramps and beams never returned.

Next move: park magnets for ten days. Return to Block the Beam and Car Ramp Height. Magnets were a sitting, not the spine.

When to slow down: if they cannot yet point to a dark patch or a car stop, stay on LIGHT-02 / FORCE-01.

When to go ahead: if they name change and same on a ramp and sketch a bounce path, offer Mirror Bounce Soft or Friction Shoe Rub.

When to get a human tutor: if motor planning or anxiety makes every try unsafe even with soft banks — a local teacher, co-op adult, or occupational support can help the body side while you keep looking gentle. No shame.

Correct a wrong reason without crushing the look. “Never correct the child” is not this book’s rule. Crushing the look is not teaching either.

Prior ideas are resource and barrier. A child who says the table does nothing is thinking. Your job is another look — soft book, no table, adult hands ready — then talk. A child who wants a laser is curious about bright spots. Your job is a firm boundary and a cool LED that still makes a path worth drawing.

Tools, including AI

Point back to the Science Hour AI box for the full allow-and-ban list. Parent holds the account. Ages 5–7: no live open chat as the child’s partner. Ages 8–10: still parent in the room. Unaided first on looking and trying. SCRIPT after the try.

Optional analog tools this week: cool LED flashlight, small mirror, board-as-ramp, weak fridge magnets, chalk for shadow marks, notebook.

Do not generate a sky photo or a “perfect beam path” image and file it as tonight’s result. Generated image is not data. Do not invent a runout table the child never measured. Do not ask a tool to write what you should have seen. A labelled schematic you look at together, then put away before the try, is fine.

What “done enough” looks like

Before you move on from this chapter’s focus:

- ☐ Child can block or pass a beam and say what happened without needing the word “opaque.”
- ☐ Child can push or pull (or run a ramp) and say what changed when we pushed harder or raised the height.
- ☐ One drawing shows a beam path *or* a ramp result a stranger could roughly guess.
- ☐ You heard force-as-fuel at least once and answered with another look, not a scold.
- ☐ Lasers stayed off the default table; adult was present for night, fan, and magnet work.
- ☐ One out-of-home light or motion notice happened (yard stick, night sky, or playground language).

Placement by skill, not birthday. A seven-year-old who fair-tests ramp height may use 8–10 cards. A nine-year-old who still freezes at dim-room noticing may stay on bright Block the Beam. A publisher’s “grade 2 science” scope is not a legal grade and not a transcript line. The work lives inside **Science** — or **Physical Science** only when the week’s work is actually that. Never a brand. Never Little Thinkers Science I.

Done enough is a path through elementary looking — not a diploma, not a percentile, not a culture-war win.

If Friday’s beam drawing still makes sense after a night — they can point to the dark patch again Monday — the look held. If it does not, you have next Tuesday’s job: same flashlight, same card, shorter hour. Revisit is not failure. Mixing last week’s ramp with this week’s mirror bounce is allowed once each move works alone.

Chapter 7 — Talking and writing science



Figure 9: A dated notebook page with a trial drawing and one sentence caption beside sticky talk prompts — still-life, no people, no logos.

Why this matters

A child can watch ice melt, run a ramp, and still leave the hour with only your words in their mouth. The plot of the demo sits in the ear. The drawing sits blank. Tomorrow the plot is thinner. The looking never became theirs.

Talking science at 5–10 is one authentic question, a wait, and a finger on the record. Writing science at 5–10 is a ladder: label, then caption, then a short exit sentence, then — when the child is ready — a claim that points at *this* trial’s evidence. It is not a lab-report genre. It is not a vocabulary quiz. It is not a blank page meant to shame a six-year-old who still draws the sentence.

This is worth the struggle because talk is how looking becomes theirs, and a caption is how looking lasts past the hour. A fluent summary from a grown-up, or from a machine, is performance. A sentence that could only come from *this* drawing or *this* notebook line is science.

Classroom research on wait-time started in elementary *science* lessons. Teachers often waited about a second after a question and less than a second after a child stopped talking. When they waited longer — a few seconds — the talk changed. A later digest called the pause *think-time* and named kinds of silence. Use the pause at your table. This is a useful study, not a promise that every home will see the same result, and it is not a kitchen experiment you are failing if Wednesday is noisy.

Materials alone rarely make discussion. Ice on a plate is not yet a conversation. You still pose, wait, and point. The child answers you, not a tab.

The locked question for this chapter is the whole craft in one line:

“What do you know from what you saw — not from the book alone?”

If they can answer that, with a finger or a caption, the week did its job.

Chapters 1–6 put lookables on the table: notice before name, draw what happened, one change at a time, living-by-needs, weather window, light path and push-pull. This chapter teaches the mouth and the pencil to stay honest to those looks. Sister English manuals care about oral before heavy print; borrow the kindness, not the phonics scope. Sister thinking manuals care about wait and change-your-mind; borrow the craft, not a playground debate without a lookable. Then teach science talk. One authentic question beats ten quiz words. One caption that needs the drawing beats a fluent page that never looked.

For the parent: understand it yourself

Here is the everyday picture.

You tell a short story about why the car went farther. You ask, “So what happened?” The child repeats your story, brightly. You feel the hour worked. You put the ramp away. On Friday you ask what they know from what they saw. They blink. They remember your explanation. They cannot point to the tape mark on the floor.

Here is the precise picture.

The trial is done. The drawing or the two-row table is on the table. You ask what they know from what they saw, not from the book alone. You wait a slow three. They point at the farther tape mark. You uptake: “You said the car went farther; our marks show the taller start.” For a five-year-old you scribe a three-word caption. For an eight-year-old they write one sentence that ends on the data. The story can fade. The mark stays.

An authentic question is one you do not already have the exact sentence for. “What is friction?” with the definition facing you is recitation. “What do you know from what you saw — not from the book alone?” is closer to authentic. You may still know more than they do. You have not prespecified their sentence.

Oral comes before heavy print. Spelling help is allowed. You may scribe. Science looking does not wait for fluent writing. Place by skill, not birthday.

Claim-because-evidence at 8–10 stays light: “I think ____ because I saw ____.” It is not a five-paragraph essay. It is not Wineburg sourcing transplanted onto a ramp. It is a finger on *this* trial.

Rowe’s wait-time work is elementary science-class tape work. Stahl’s think-time digest labels kinds of silence and a roughly three-second convention. Both are classroom-origin. Do not mash them into a kitchen-table science RCT. Do not invent a wait coefficient for your home. Do count three when your habit is to rescue at one.

Wrong answers you should be able to hear

1. Rapid-fire quiz.

“What is the vocabulary word? What is the vocabulary word?” No wait. No lookable.

What it usually means: talk time became a test.

What you say: put the materials back. Ask the locked opening. Count three.

2. “Guess what I’m thinking.”

Only one sentence will count, and it is yours.

What it usually means: the question was not authentic.

What you say: “There may be more than one true notice. Point at the record.”

3. Overwriting mid-sentence.

They start a caption; you erase and write the textbook aim instead.

What it usually means: adult voice replaced child looking.

What you say: stop. Keep their words. Add a second line for your note if you must — labelled as yours.

4. Public shaming.

A wrong prediction becomes a joke at the child’s expense.

What it usually means: wrong predictions were treated as character.

What you say: “Wrong predictions are useful. Keep the circle. We looked.”

5. Generated paragraph filed as the child’s caption.

A fluent machine sentence sits under a blank drawing as “what I observed.”

What it usually means: performance replaced looking.

What you say: refuse it. Real two-minute look. Child’s words. Generated observation is not data. Point back to the Science Hour AI box.

Five-minute parent warm-up

Find the child’s last caption — or the last sentence you scribed. Cover it with a sticky.

Ask yourself: what follow-up will I use tomorrow? Practice uptake aloud: “You said ; *our drawing shows* .”

Practice the locked opening to the empty chair. Count three. Do not answer.

If you like a SCRIPT for later, write four lines for yourself now — questions, not their observation. The child never chats a tool. You hold the account. Point back to the Science Hour AI box when you want the full allow-and-ban list.

How to teach it this week

The Science Hour shape stays. Talk and write are the try and the crumb, not a second subject named Little Thinkers Science Writing.

Session shape (~20–40 minutes)

1. **Lookable / materials (3–8 min).** Something already tried this month — ice, ramp, plant, weather card, beam path — or today’s new try.
2. **Notice / predict (3–7 min).** Before naming. Before the book paragraph.
3. **Try (5–15 min).** The science try itself, *or* a talk-box sit on yesterday’s record.
4. **Record (3–8 min).** Label, caption, exit strip, mini table, or claim-because — one print load, not a packet.
5. **Exit crumb / talk box (1–5 min).** Locked opening. One follow-up. Uptake one phrase accurately.

Twenty-minute day: uncover yesterday’s drawing, locked question, one caption strip. That is a full hour.

Exact wording you can say

Opening, locked:

“What do you know from what you saw — not from the book alone?”

Then wait.

If they retell your lecture: “Put your finger on the drawing. What is there that you did not need me to say?”

If they freeze: “Point first. Then give the point a word.”

If they say “because science”: “Because you *saw* what?”

If siblings talk over each other: “I heard you say ; ***the table shows*** . Do you agree — point at the evidence.”

When you fade help, stop supplying the because-clause. When you stop talking, look at the notebook.

Age-band moves

Ages 5–7. Talk stays concrete: notice, predict soft, because the drawing. Writing is three labels on parts actually drawn, or a three-word caption, drawn or dictated. Exit sentence strip with your scribe: “Today I learned ____ because I saw ____.” Wrong predictions stay circled without a score. Dates optional.

Ages 8–10. Add claim-because-evidence. Add a full caption under a trial. Add mini table two rows when they measured. Fair box chorus: What changed? What stayed same? Can we tell for sure? Disagree soft redesign when two results conflict. Refuse generated obs as a meta sit once. If decoding or handwriting is weak, they speak; you write. If looking habits are weak, stay on labels longer. Skill, not birthday.

First try-it for the student

Ask the locked question once over yesterday’s drawing. Sit on your hands for three to five seconds. Accept the first real try. Follow up once: “How do you know from *this*?” That is TALK-01 nested with TALK-02. Then WRITE-02 Caption Under Trial or WRITE-03 Exit Sentence Strip.

This week, honestly

Focus move: because-the-drawing / because-the-notebook; one caption that could only come from looking.

- **Monday:** TALK-01 Three-Second Notice on whatever phenomenon is out + TALK-02 Because the Data.
- **Tuesday:** WRITE-01 Object Label Three (5–7) or WRITE-02 Caption Under Trial (both).
- **Wednesday:** TALK-05 Fair Box Chorus (8–10) or TALK-04 Predict Soft (5–7) before a short try.
- **Thursday:** WRITE-05 Claim Because Evidence (8–10) or WRITE-03 Exit Sentence Strip; optional TALK-06 Child Asks — park one wonder.
- **Friday:** WRITE-03 strip or sticky caption. One diagnostic: fluent summary with no finger on the record. Optional WRITE-06 Refuse Generated Obs (8–10).
- **Once this week, out of the home:** OUT-04 Museum One Exhibit — one oral retell pointing at the exhibit; or OUT-07 Library Science Shelf —

choose one book *after* a try, not instead.

Siblings: TALK-03 once. The older child needs their own looking move, not only the tutor role. TALK-07 when two honest results disagree — redesign, not a winner crown.

Three to five hours. One outing. Friday a crumb. Not a writing program. Not a debate club. Not a detector-graded caption week.

Practice that actually builds learning

Named try-its carry time, materials, safety, the fun, and the skill. Talk try-its and write try-its share a table. The museum and the library are practice. They do not replace the because-what-you-saw move.

Composite homes are illustrations, not reported families.

Talk box (use every talk/write hour)

Opening (locked): “What do you know from what you saw — not from the book alone?”

Follow-ups:

1. What do you *see*?
2. What does the drawing / notebook show?
3. Say it because of the data — finger on the record.
4. What would change your mind?
5. (Sibling) Do you agree — point at the evidence. You may change your mind after another look.

How to wait: After you ask, count a slow three. After they stop, wait again. If they are mid-sentence or mid-label, do not overwrite. Look at the drawing, not at their face, if the silence is hard.

Stuck silence usually means: the question was vague; wait-time was zero; they are hunting a vocabulary word; they are guessing what you want; the materials never landed; they offered a book-only guess; they think science means repeating the parent’s explanation; they think looking means saying what the screen already

told them. Pose, wait, point back at the lookable. Do not fill the silence with the right vocabulary. Do not turn the table into a fight.

One incorrect example to diagnose: Parent asks five quiz words in a row while the ice plate sits ignored. Child answers two. Parent finishes the definitions. Notebook empty. Repair: cover the quiz. Locked opening. Finger on what happened. One caption.

TALK-01 Three-Second Notice (in-home; nests in any hour)

Band: both.

Time: 10–15 minutes.

Materials: Whatever phenomenon is already out.

Safety: Matches the paired try.

The fun: The hush after your question feels like a dare. They almost always fill it if you do not.

The skill: Wait after the question; uptake one phrase accurately.

Steps

1. Ask “What do you notice?” or the locked opening.
2. Silently count three. Do not leap in.
3. Uptake one phrase: “You said ____.”
4. Exit: child’s notice sentence repeated back accurately.

Error patterns: Answering your own question at one second; praising only “right” science words; timer as game show.

TALK-02 Because the Data (in-home)

Band: both.

Time: 10–15 minutes.

Materials: Child’s drawing or table from today.

Safety: Ordinary table.

The fun: The sentence is legal only if it ends on the record.

The skill: Claim + because + point.

Steps

1. “What happened?”
2. “How do you know?”
3. Finger on the record.
4. Exit: claim + because + point.

Error patterns: Accepting “because science”; ignoring the notebook; turning the clause into a grammar scold.

TALK-03 Sibling Uptake (in-home)

Band: both (2+ children).

Time: 10–15 minutes.

Materials: Shared trial record.

Safety: Kindness rule; adult present for conflict; no mocking.

The fun: The second child gets to restate before agreeing — a fair turn.

The skill: Listen; restate; agree or disagree with evidence.

Steps

1. Child A claims.
2. Child B restates A’s claim before agreeing or disagreeing.
3. Both point at data.
4. Exit: “I heard you say ; *the table shows* .”

Error patterns: Mocking; adult crowning a winner without redesign; older child only tutoring.

TALK-04 Predict Soft (in-home)

Band: 5–7.

Time: 10 minutes.

Materials: Upcoming trial materials visible.

Safety: Matches the try.

The fun: Choosing before the reveal.

The skill: Two-choice predict; wrong predictions stay useful.

Steps

1. Show materials. Two-choice predict.
2. Wait. Circle the prediction.
3. After the try, keep the circle even if wrong.
4. Exit: circled prediction kept.

Error patterns: Scoring predictions; hiding materials so predict is pure guess-work without clues.

TALK-05 Fair Box Chorus (in-home)

Band: 8–10.

Time: 10–15 minutes.

Materials: Today's fair test.

Safety: Ordinary.

The fun: Saying the three questions aloud together feels like a team huddle.

The skill: What changed? What stayed same? Can we tell for sure?

Steps

1. Chorus the three questions.
2. Answer while looking at the setup — not as empty ritual.
3. Exit: three answers written or oral.

Error patterns: Rushing past “can we tell?”; ritual words without looking at setup.

TALK-06 Child Asks (in-home)

Band: both.

Time: 10–15 minutes.

Materials: Wonder box paper.

Safety: None beyond kindness.

The fun: Their question gets dated and kept.

The skill: Child asks; adult writes; hour may not answer; park for later.

Steps

1. Child asks one question.
2. Adult writes it and dates it.
3. Decide: answer next week, or design a try, or leave as wonder.
4. Exit: wonder card dated.

Error patterns: Converting every wonder into an immediate lecture; discarding child questions.

TALK-07 Disagree Soft Redesign (in-home)

Band: 8–10.

Time: 15–20 minutes.

Materials: Two conflicting results or two sibling claims; redesign materials.

Safety: Matches redesign.

The fun: Being allowed to disagree without a loser.

The skill: Name disagreement without blame; plan second trial.

Steps

1. Name the disagreement.
2. Ask what second trial would help.
3. Plan redesign one sentence.
4. Exit: redesign plan.

Error patterns: Declaring one child “the scientist”; erasing minority data.

WRITE-01 Object Label Three (in-home)

Band: 5–7.

Time: 10–15 minutes.

Materials: Drawing from today; pencil.

Safety: None.

The fun: Naming parts they actually drew.

The skill: Three labels; invented spelling OK.

Steps

1. Point to three parts on the drawing.

2. Child labels (or dictates).
3. Exit: three labels.

Error patterns: Adult rewriting in “nice” handwriting over child’s; labels for unseen parts.

WRITE-02 Caption Under Trial (in-home)

Band: both.

Time: 10–15 minutes.

Materials: Photo or drawing of setup.

Safety: None.

The fun: One sentence that “belongs” to the picture.

The skill: What we did + what we saw.

Steps

1. Look at the picture.
2. One caption sentence.
3. Exit: caption.

Error patterns: Caption that only restates the textbook aim without seeing; adult overwrite.

Illustration. Drawing shows ice smaller in a cup. Child writes: “We left ice on the counter. It got smaller.” Parent wanted “matter changed state.” Parent’s sentence can wait for a second line labelled parent note. The child’s caption is the record.

WRITE-03 Exit Sentence Strip (in-home)

Band: both.

Time: 5–10 minutes.

Materials: Strip of paper.

Safety: None.

The fun: A finishable strip — not a page.

The skill: “Today I learned ____ because I saw ____.”

Steps

1. Say the frame.
2. Fill it. Scribe if needed.
3. Exit: the strip itself.

Error patterns: “I learned science”; no because; strip becomes a lecture.

WRITE-04 Mini Table Two Rows (in-home)

Band: 8–10.

Time: 15–20 minutes.

Materials: Notebook; ruler optional.

Safety: None.

The fun: A tiny table that still looks “official.”

The skill: Condition / result; two rows minimum; numbers or words the child measured or saw.

Steps

1. Make columns: condition / result.
2. Two rows from today’s try.
3. Exit: completed two-row table.

Error patterns: Table decorated but empty; adult filling numbers child did not measure; inventing a data table for the portfolio.

WRITE-05 Claim Because Evidence (in-home)

Band: 8–10.

Time: 15–20 minutes.

Materials: Today’s data.

Safety: None.

The fun: Sounding like a careful claim without a huge essay.

The skill: I claim ____ because ____ (point to data).

Steps

1. Sentence frame aloud.

2. Child fills and points.
3. Optional one reasoning word — still light.
4. Exit: one CER-light sentence.

Error patterns: Claim with no data pointer; CER as a five-paragraph essay; detector grading the sentence.

WRITE-06 Refuse Generated Obs (meta; in-home)

Band: 8–10.

Time: 10–15 minutes.

Materials: Adult-made sample clearly marked FAKE — a fluent filled observation worksheet that was never lived; child’s real window or table look.

Safety: Emotional honesty; no shame spiral.

The fun: Spotting a fake feels like detective work.

The skill: Real data means a real look; generated or invented obs are not student data.

Steps

1. Put FAKE beside a blank page.
2. Ask which one could be *today’s* looking — and why not the FAKE.
3. Do a real two-minute look. Write from that.
4. Exit: “Real data means ____.”

Error patterns: Keeping FAKE as portfolio; AI homework as science; using the sit to scare rather than train honesty.

OUT-04 Museum One Exhibit (out-of-home)

Band: both.

Time: 30–40 minutes in museum (fits Science Hour+).

Materials: Notebook; pencil.

Safety: Adult present; museum rules; lost-child plan; hands off forbidden objects.

The fun: Choosing *one* thing in a huge building.

The skill: Observe 10 min; draw; one question; oral retell pointing at the exhibit — refuse whole-museum worksheet dump.

Steps

1. Choose one exhibit.
2. Look before reading the whole plaque.
3. Draw. One question card.
4. Exit: drawing + one oral retell that points.

Error patterns: Racing galleries; copying plaque before looking; public cold quiz.

OUT-07 Library Science Shelf Peek (out-of-home)

Band: 8–10 (5–7 may do a shorter picture compare with scribe).

Time: 25–35 minutes.

Materials: Library card; notebook.

Safety: Adult present as needed; street safety.

The fun: Finding a book that matches this week's try.

The skill: Compare one picture to your own trial drawing; note agreement/difference — book after try, not instead.

Steps

1. After a home try, find one trade book on the idea.
2. Compare one picture to your drawing.
3. Exit: "Book showed ; *my trial showed* ."

Error patterns: Replacing own data with book picture; unchecked internet print-outs as "library."

Blocked vs mixed

Blocked: a week that only practices wait + uptake on familiar drawings until silence is survivable.

Mixed: caption Monday, fair chorus Wednesday, museum retell Saturday — once wait works.

Shared looking, kitchen talk, and outings motivate. They do not replace observe-before-explain.

A library animal book and a retell can be wonderful. A museum fossil and one pointed sentence can be wonderful. Neither replaces the because-what-you-saw move on a home trial. Book after try. Plaque after look. Caption after drawing. That order is the week's honesty.

When Friday comes, pick one diagnostic wrong from the bank above. Plan Monday's isomorphic retest: same shape, fresh lookable. The child who gave a book-only answer Monday gets a covered caption and a finger rule — not a personality label.

Mixed week illustration (labelled illustration, not a reported family)

Monday: Three-Second Notice on a leftover beam drawing; parent counts three; child says the wall went dark; parent uptakes. Tuesday: three labels on a leaf drawing from an earlier week — invented spelling kept. Wednesday: Fair Box Chorus on a ramp height try. Thursday: Claim Because Evidence — “I think the taller ramp made a longer run because the tape is farther.” Friday: Exit Sentence Strip; parent catches a book-only sentence and asks for a finger. Saturday: museum one exhibit; oral retell pointing at a fossil, plaque read *after*. That week is talking and writing science. It is not a lab-report course.

Incorrect example bank (diagnose one on Friday)

Rapid-fire quiz. Five vocabulary words; lookable ignored. Repair: locked opening; wait; finger.

Guess what I'm thinking. Only the parent's sentence counts. Repair: “Point at anything true on the record.”

Overwrite mid-sentence. Parent erases child's caption for “better words.” Repair: second line labelled parent note.

Public shame on a wrong predict. Laughter at the circle. Repair: “Keep the circle. We looked. Predictions can change.”

Generated caption. Fluent AI paragraph under empty drawing. Repair: WRITE-06; real look; child's words.

Talk time as adult lecture. Ten minutes of explanation; one yes/no at the end. Repair: materials first; child speaks before the name.

How to fade help

First you model uptake: “You said the ice got smaller.” Then they point and you scribe. Then they write one caption word; you finish the scribe. Then they write the strip alone. Stop talking when they are mid-caption. Struggle before rescue: ask, wait, hint (“finger on the mark”), then model one because-clause if needed.

When to stop talking

When they are pointing. When they are mid-label. When a sibling is restating before agreeing. When the wonder card is being dated. Your job is one good question and a record that stays in view.

Uptake scripts you can steal

- “You said . *Our drawing shows* .”
 - “You said _____. Tell me where that lives on the page.”
 - “I heard farther. Show me farther.”
 - “You may change your mind. What would we look at next?”
-

For the student

Science talk is not a quiz show. It is you saying what you know from what you saw.

You may point before you speak. You may change your mind when the drawing shows something new. Wrong predictions are allowed. They help the next try.

Writing can be three labels. Writing can be one sentence under a picture. Writing can be “I think ____ because I saw ____.” You do not need a long report. You need words that match the look.

Tiny worked example

You drew a car on a ramp and a tape mark far down the floor. Someone asks what you know. You say: “The car went farther from the tall ramp because I see the tape mark is farther.” Your finger is on the mark. That is enough.

Two tries

1. Cover any book caption. Look at your drawing. Say one thing you know from what you saw. Then write or dictate a caption.
2. Say “I think ____ because I saw ____.” Point at the notebook when you say *saw*.

Explain it back

Teach a stuffed animal or a sibling: “Ask me what I know from what I saw. Wait. Do not answer for me.”

Challenge

Find a sentence that only repeats a book. Cross it out gently. Replace it with a sentence that needs your drawing to make sense.

One more challenge (optional)

Ask one wonder of your own. Have a grown-up date it. You do not have to answer it today.

If someone talks for you

If a grown-up or a sibling or a screen answers before you finish, you may say: “I was still looking.” Then point at the drawing and try your sentence again. Looking first is your job. Waiting is their job.

Print ladder at a glance

Step	What it looks like	Band first home
Label	Three words on parts drawn	5–7
Caption	One sentence: did + saw	both
Exit strip	Learned ____ because I saw ____	both
Mini table	Two rows condition/result	8–10
Claim-because	I think ____ because ____ (point)	8–10
Refuse generated	Spot FAKE; rewrite from real look	8–10 meta

Climb one rung when the lower rung is easy. Jumping to claim essays without a finger on data trains performance.

Sibling and mixed-age note

A five-year-old and an eight-year-old can share one trial. The younger child's notebook holds labels and an oral strip you scribe. The older child's notebook holds a caption or claim-because. Both answer the locked question. Neither is Little Thinkers Science I. Both are Science looking with talk.

If it isn't clicking

1. Wait was zero.

You ask and answer yourself. They learn to wait for the rescue.

Next move: sit on your hands. Count three out loud once if you must. Look at the drawing. Practice TALK-01 alone before the child arrives.

2. Print load too heavy.

Captions become tears; the hour dies.

Next move: oral exit. You scribe. Labels only. Picture-plus-phrase. Return to writing when the talk is warm. Sister English craft helps: oral before heavy print — without turning this chapter into a phonics scope.

3. Book-only answers.

Every sentence could have been said without the trial.

Next move: cover captions. Put the book away until after the try. Locked opening. Finger required before words count.

4. Overwrite habit.

You keep “fixing” their caption into textbook language mid-line.

Next move: second line labelled *parent note*. Their line stays. Apologize once if needed. Trust rebuilds.

5. Generated caption temptation.

A tool offers a perfect observation paragraph.

Next move: refuse. WRITE-06 once. Point back to the Science Hour AI box. The model may prepare *you*; it may not do their looking or their caption when those are the lesson.

When to slow down: if they cannot yet point to the record, stay on TALK-01/02 and WRITE-01.

When to go ahead: if captions point at data, offer WRITE-04/05 and TALK-07.

When to get a human tutor: if language delay or anxiety makes every talk sit unsafe — a speech-language or teaching adult can help while you keep science looking gentle. No shame.

Correct a wrong reason without crushing the look. Hearing “because science” is information. Ask for the finger. Then teach.

Tools, including AI

Point back to the Science Hour AI box. Parent holds the account. Ages 5–7: no live open chat as the child’s partner. Ages 8–10: still parent in the room. Unaided first. SCRIPT after the try — questions you will say, not a paragraph for them to copy as their observation.

You may ask a tool to explain a named page *to you*, to draft discussion questions you will vet, or to help you design a blank caption strip with PARENT KEY held by you. You may not write-my-lab-report. You may not invent an observation. You may not detector-grade “the ice got smaller.” Detectors fail. A child’s caption is not a fraud case.

Optional analog tools: sticky notes for talk prompts, sentence strips, a “wonder box” envelope, a sticky to cover book captions, notebook.

If you use a voice tool to dictate a caption the child already said, label it as their spoken words you typed. That is scribe help. It is not invent-observation. If the

tool invents details they never said — stop. Look again. Write only what they pointed to.

What “done enough” looks like

Before you move on from this chapter’s focus:

- ☐ Child can answer the locked question with a finger on the drawing or notebook at least once.
- ☐ One caption or scribed sentence exists that could only come from *this* try.
- ☐ You waited a slow three at least once without rescuing.
- ☐ You uptake at least one child phrase accurately (“You said ____”).
- ☐ No generated paragraph is filed as their observation.
- ☐ One out-of-home retell or library compare happened, or is scheduled.

Life of the habit (not a ninth chapter). Revisit next Tuesday: uncover last week’s caption; ask the locked question again. Mix last week’s try with a new talk follow-up. Stop while a look remains. A diploma is not in the room. A path through elementary science looking is.

Placement by skill, not birthday. A six-year-old with strong oral because-clauses may use 8–10 talk cards while you still scribe. A nine-year-old who freezes at print may stay oral. Publisher workbooks are scopes, not legal grades, not transcript lines. The work lives inside **Science**. Never a brand. Never Little Thinkers Science I.

Done enough is showable talk and a small written crumb — not a lab-report program, not a percentile, not a culture-war win. The stranger who someday opens the notebook should find a child’s sentence that points at a look — not a brand, and not a machine paragraph pretending to be a child’s eyes.

If Friday’s caption still makes sense after a night — they can point to the same mark Monday and say the because-clause again — the talk held. If it does not, next Tuesday is shorter: one drawing, locked question, scribe only. Revisit is not failure. Mixing last week’s beam path with this week’s claim sentence is allowed once each piece works alone.

Chapter 8 — Short assessments and progress checks

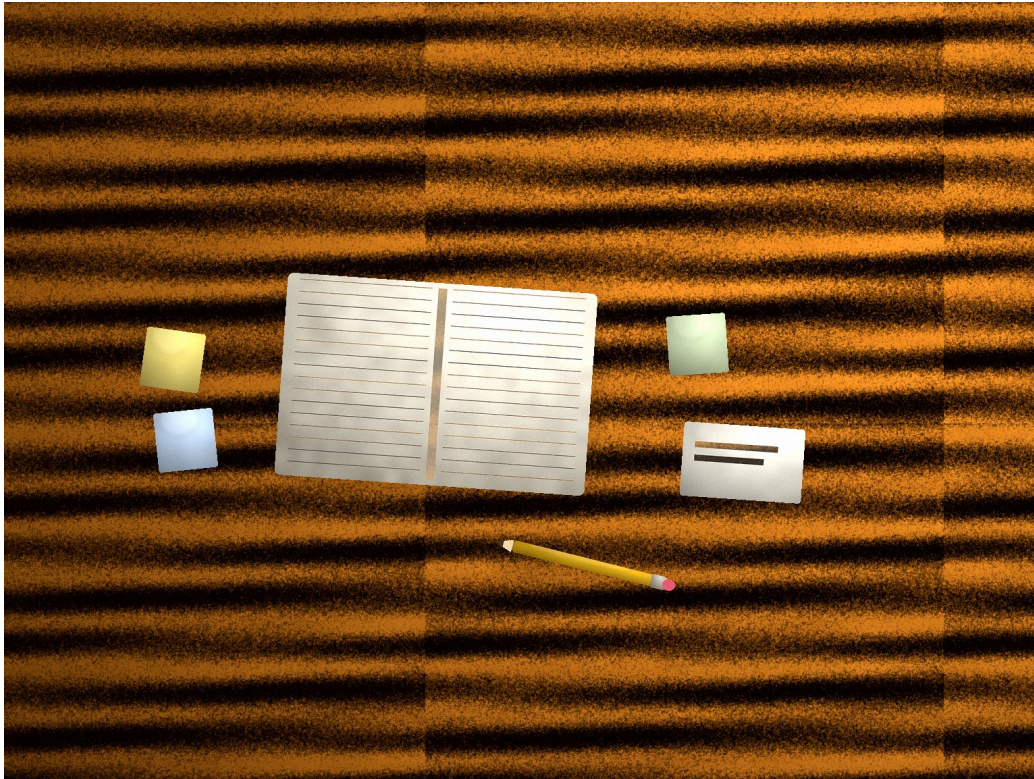


Figure 10: Three sticky exit notes and an honesty card beside a dated notebook titled Science — still-life, no people, no logos.

Why this matters

A check at ages 5–10 is a skill object, not a score that follows a child around the house.

Can they draw what they saw? Can they retell what happened? Can they name what we changed and what we kept the same? Can they sort living from not by a need they can see? If Friday’s look still holds after a night, the week taught something. If it does not, you have a next move. That is enough.

This is worth the struggle because the wrong kind of check teaches the wrong kind of science. A vocabulary list trains name-first. A percentile sticker trains identity: “science kid” or not. A public quiz trains performance. An app that awards points for minutes trains sitting, not looking. A generated observation filed in a portfolio trains dishonesty. None of those is this book’s job.

There is no nationally published homeschool NAEP Science score for this age band that this book can turn into a kitchen cut score. Some states ask for tests in reading and mathematics, or for a narrative, or for a parent-written list of subjects. Those sittings stay in their lane. They are not this chapter. This chapter does not give legal advice. You read your own statute when you need to. The stranger who later opens a notebook wants **Science** work they can see — a dated page, a draw/retell, a one-thing-changed note, a living-or-not sort — not a brand, and never a course called Little Thinkers Science I.

Screening a skill is not an identity. A child who cannot yet name what changed today is not “not scientific.” They need a shorter try, a clearer “same” list, or one more night. You already know how to reteach a shoe pair. Use that kindness here.

Kitchen skill checks are load-bearing in this book, not garnish. That is why this chapter is dedicated — the same grain as a sister assessment chapter in another subject, adapted for science looking. Exit crumbs still live in every teaching chapter. This chapter gathers the objects, the diagnostic wrongs, the progress charts without percentiles, and the done-enough language in one place.

The locked question for this chapter is how a check should feel:

“What can you show me two ways today — and what still needs practice?”

Two ways means the look is theirs, not a lucky nod. What still needs practice means the next hour already has a job. A diploma is not in the room. A path through elementary looking is — and that path is enough for this week.

For the parent: understand it yourself

Here is the everyday picture.

Friday night becomes a quiz. Ten vocabulary words. Five “what is a scientist” sentences. A parent and a child both get tight. The child guesses. The parent hears “not ready.” The notebook gets a sad face. Next week the same list, louder.

Here is the precise picture.

Friday is short. Draw what you saw from a try already on the table this week. Retell what happened — oral first. Name one thing changed and one thing kept the same, if a fair try happened. Sort five objects living / not by a need you can see, if that was the week’s work. If they can show a skill two ways — draw *and* say; sort *and* caption; point *and* name change/same — you mark the dated notebook. If a miss has a reason you can use tomorrow, you keep the reason. You do not keep a label.

Done enough is local. A five-year-old who draws the ice smaller and says it got watery is done enough for that check. An eight-year-old who names height as the one change on a ramp and writes a because-clause is done enough for that check. Holding both children to the same lab-report rubric is a different product.

When a program you already use includes chapter review questions, you may sit them as *listening and recall* inside that scope. They check whether a story was heard. They do not license a national rank. They do not replace the kitchen objects below. Label them as the program’s tools, not as a research trial of your kitchen.

Depth beats a coverage tour on tests as well as on lessons. A long vocabulary list leaves few traces. A small pattern — draw, retell, one change, one need — built slowly, is what the mind will keep. This is a useful study shape borrowed carefully, not a promise that every home will see the same result.

No CLEP for elementary. No AP. No athletic-eligibility lab language for an eight-year-old. When the child is older and the family needs those grains, open *Science for Homeschooling*. Then return here for the kitchen objects.

Wrong answers you should be able to hear

1. Percentile identity.

“This sort means you are in the top group.” Or: “This miss means you are behind

the nation.”

What it usually means: a hope, or a reading/math compliance battery, got copied onto a kitchen science skill.

What you say: “This is a skill we can practice. It is not a rank.”

2. “Failed science.”

One blank drawing becomes a personality.

What it usually means: screening turned into identity.

What you say: “You mixed the change and the same. Tomorrow we try one ramp only.”

3. Birthday lock.

“You’re 8, so you must fair-test.” Or: “You’re 5, so you cannot do science yet.”

What it usually means: birthday placement replaced skill placement.

What you say: “We place by what you can show. Looking first if looking is shaky.”

4. Invented portfolio table.

A neat data table appears for a try that never measured. Or a machine filled the blanks.

What it usually means: the portfolio wanted proof more than the child looked.

What you say: refuse. Real look. Real numbers only if measured. Invented table is not student data.

5. Detector score as honesty.

A window grades “the ice got smaller” as machine-written.

What it usually means: a fraud tool landed on a child’s sentence.

What you say: close the detector. Read the sentence yourself. Detectors fail. Looking is not a crime.

Five-minute parent warm-up

Pick one diagnostic wrong from this week — name-first, force-as-fuel, confound pride, book-only caption, generated obs, demo with no record. Write it in six words.

Plan Monday’s isomorphic retest: same skill shape, fresh lookable (new ice cube, new ramp height, new leaf).

Practice the locked opening aloud. Practice the second half kindly: “What still needs practice?” You are naming tomorrow, not a verdict.

Lay out tomorrow's check pieces: sticky triad *or* cover-the-caption *or* fair ladder *or* living sort cards. Do the check yourself once as if you were the child.

How to teach it this week

A check week is still a Science Hour week. The try *is* the check. You do not add a second, colder hour titled Assessment.

Session shape (~20–40 minutes)

1. **Lookable (3–8 min).** A familiar try or a close cousin — not a surprise stack of new content.
2. **Notice / predict (3–7 min).** Light; the check needs looking, not a lecture.
3. **Try / check object (5–15 min).** One kitchen object below — not a battery.
4. **Record (3–8 min).** Two-ways evidence in the dated notebook.
5. **Exit crumb (1–5 min).** Locked opening. One “still needs practice” line. Adult error-pattern note private.

Twenty-minute Friday: one draw-retell *or* one one-thing-changed *or* one living sort. Two ways on that one object. Stop.

Exact wording you can say

Opening, locked:

“What can you show me two ways today — and what still needs practice?”

Then wait.

If they freeze: “Show me with the drawing first. Then show me with words. We can do the second way tomorrow.”

If they perform a memorized definition: “I already know the book sentence. Show me from what you saw.”

If they ask whether they are good at science: “You can draw what happened. You are still practicing one-thing-changed. That is a map, not a name for you.”

When you fade help, do not supply the second way. When you stop talking, the notebook gets one line, not a speech.

Age-band moves

Ages 5–7. Oral retell of one or two true pieces. Draw what you saw. Living-or-not by a visible need. Exit spoken; you scribe. Two ways might be: draw *and* say; sort *and* point to a need. Fair-test language light — with adult “same” list if used at all.

Ages 8–10. Oral or short written retell. One-thing-changed named. Mini table if they measured. Fair ladder check. Claim-because pointing at the record. Two ways might be: sort *and* caption; ladder oral *and* notebook line. If looking habits are weak, return to draw-retell before fair ladder. Skill, not birthday.

First try-it for the student

Ask them to show the same idea two ways — draw the ice (or ramp, or plant) *and* say what happened. Ask what still needs practice. Write their words. That is the spirit of AS-01 plus a kitchen object.

This week, honestly

Focus move: kitchen skill objects; one diagnostic wrong; done-enough that holds after a night.

- **Monday:** Sticky Exit Triad on this week’s focus try (AS-01).
- **Tuesday:** Cover the Caption on a book page or your own aim sentence (AS-02) — then compare to their notice.
- **Wednesday:** Kitchen object — draw/retell *or* one-thing-changed *or* living-or-not (match the month’s work).
- **Thursday:** Fair Ladder Check if 8–10 and a fair try exists (AS-03); else mix last week’s lookable with a new cousin.
- **Friday:** Locked question. AS-04 Error Pattern Card (adult, private). Done-enough boxes. Sleep. Saturday or Monday, ask one piece again. If it holds, it held.
- **Once this week, out of the home (optional transfer):** museum one-exhibit retell or park notice-before-name — adult present; not a cold public quiz.
- **Once this month:** AS-05 Band Placement Soft on the plan page; AS-06 Honesty Card read aloud (8–10 + adult).

A progress chart, if you like one, has skill boxes: draw what you saw; retell; one-

thing-changed; living-or-not by needs; light path or push-pull notice. Dates of tries. No percentiles. No smiley ranks that become a personality.

Three to five hours. One optional outing. Friday a crumb that can survive a night. Not NAEP. Not a homemade high-stakes battery. Not CLEP/AP/NCAA for elementary — those belong to older grain when the child is older. One pointer to the older book is enough.

Practice that actually builds learning

Checks are named try-its. They still need time, materials, safety, the fun, and the skill. Fun here is the relief of a finishable task. The skill is showing a look two ways.

Composite homes are illustrations, not reported families.

Talk box (use every check hour)

Opening (locked): “What can you show me two ways today — and what still needs practice?”

Follow-ups:

1. Show me with the drawing.
2. Show me with words.
3. What still feels shaky?
4. Which band job fits what you just did — looking, or one-thing-changed?
5. You may try again tomorrow.

How to wait: After you ask, count a slow three. After they stop, wait again. If they are mid-draw or mid-retell, do not cut them off. Look at the notebook, not at their face, if the silence is hard.

Stuck silence usually means: the question felt like a test; wait was zero; they think two ways means two vocabulary words; they fear a rank; the materials never landed; they offered a book-only performance. Pose, wait, point back at the lookable. Do not fill the silence with a score. Do not turn the table into a verdict.

One incorrect example to diagnose: Parent prints a “science quiz” with twenty vocabulary blanks and a smiley percentile sticker at the bottom. Child cries. Note-

book has no drawing from the week. Repair: throw away the sticker. One kitchen object. Two ways. What still needs practice.

Kitchen object A — Draw what you saw (in-home)

Band: both.

Time: 10–20 minutes.

Materials: Yesterday's or this week's lookable (ice cup, leaf, ramp, beam path, plant, weather card); blank page; pencil.

Safety: Ordinary; no shame face on a “bad” drawing.

The fun: Three lines can still carry data.

The skill: Record as evidence; finger answers “what did you see?”

Steps

1. Put the lookable or the memory of it in view (materials back out if possible).
2. Child draws what happened.
3. Second way: oral sentence or scribed caption.
4. You note one missing piece kindly — not a grade.

Ages 5–7: drawing + oral; you scribe.

Ages 8–10: drawing + sentence that points at the picture.

Error patterns: “I’m bad at art” freeze; adult-filled drawing; generated image as result.

Exit crumb: Finger on the drawing answers one question overnight — ask again in the morning.

Kitchen object B — Retell what happened (in-home)

Band: both.

Time: 5–15 minutes.

Materials: A try they already did; optional stuffed audience.

Safety: Soft entry; choose a non-tender try.

The fun: Telling it to someone who only listens.

The skill: One or two true pieces; oral first.

Steps

1. “Tell what happened — from the start of the try.”
2. Second way: point to the notebook while retelling, *or* dictate one sentence.
3. Note missing pieces for tomorrow — not a personality.

Ages 5–7: one or two true pieces.

Ages 8–10: two or three pieces; optional because-clause.

Error patterns: Only repeating the parent lecture; story with no finger on record.

Kitchen object C — One-thing-changed (in-home)

Band: both (adult supports “same” list at 5–7).

Time: 10–20 minutes.

Materials: A fair try already run — ramp, plant one-thing, melt place — or a quick isomorphic retest.

Safety: Matches the try; adult present.

The fun: Being the person who catches the extra change.

The skill: Name what changed; name what stayed the same; ask can we tell for sure (8–10).

Steps

1. Look at the setup.
2. “What did we change? What did we keep the same?”
3. Second way: mark change/same on a sticky *or* redesign once if confounded.
4. Exit: fair sentence.

Error patterns: Confound pride; birthday lock into full experimental design; vinegar-soda as Chemistry credit talk.

Illustration (labelled illustration, not a study). “15 Sept 2026. Ramp: book under board. Car on carpet vs tile. Drew both. Said tile went farther. Named surface as the one thing changed.” That paragraph is a record shape. It is not a normed score.

Kitchen object D — Living-or-not by needs (in-home)

Band: both.

Time: 10–20 minutes.

Materials: Five kitchen or yard objects — seed, rock, dried leaf, beetle photo or careful live insect plan, plastic toy.

Safety: Soft living things; no harm; wash hands; return organisms; mold cultivation refused.

The fun: Arguing about the dried leaf is allowed.

The skill: Sort by a need you can see — food, water, air, place to live/grow — not a kingdom chart.

Steps

1. Sort living / once-living / not (kid words OK).
2. For one card, name the need that decided it.
3. Second way: teach *you* how to sort a leftover object.
4. Exit: one need sentence.

Error patterns: Memorized list with no lookable; harm; coverage itch into cells-as-tiny-mouths.

AS-01 Sticky Exit Triad (in-home)

Band: both.

Time: 5 minutes.

Materials: Three stickies.

Safety: None.

The fun: Three finishable squares.

The skill: Notice / wonder / claim crumb — child voice.

Steps

1. Sticky 1: notice.
2. Sticky 2: wonder.
3. Sticky 3: claim (light).
4. Keep for Friday review.
5. Second way later: uncover and retell from stickies alone.

Error patterns: All three identical adult sentences.

AS-02 Cover the Caption (in-home)

Band: both.

Time: Built into an observe sit.

Materials: Sticky over book caption or aim sentence.

Safety: None.

The fun: Reveal drum roll after their notices.

The skill: Observe before read.

Steps

1. Cover caption.
2. Child notices.
3. Uncover. Compare.
4. Exit: “I said ____ before the caption.”

Error patterns: Peeking early as a joke that becomes habit.

AS-03 Fair Ladder Check (in-home)

Band: 8–10.

Time: 10 minutes.

Materials: Today’s setup.

Safety: Ordinary.

The fun: Climbing five short rungs aloud.

The skill: Question → changed → same → result → can we tell?

Steps

1. Adult asks the ladder.
2. Child answers while looking at setup.
3. Soft pass/fail — reteach tomorrow if “can we tell?” was skipped on a confound.
4. Second way: child asks *you* the ladder once.

Error patterns: Skipping “can we tell?” when confounded; ritual without eyes on materials.

AS-04 Error Pattern Card (adult tool)

Band: adult (after hour).

Time: 5 minutes.

Materials: Notebook margin (private).

Safety: Emotional safety — private adult notes; child does not read shame.

The fun: Clarity for Tuesday.

The skill: One diagnostic error → one fix move.

Steps

1. Jot one pattern: name-first; force-as-fuel; confound pride; adult-filled obs; generated obs; demo with no record; book-only caption; birthday lock.
2. Plan one fix (isomorphic retest; cover caption; same-list sticky; refuse FAKE).
3. Exit: adult’s one-line plan — not shown as a score.

Error patterns: Grading personality; shame notes the child finds.

AS-05 Band Placement Soft (adult planning)

Band: adult.

Time: Planning.

Materials: This book’s 5–7 / 8–10 notes.

Safety: None.

The fun: Permission to meet the child where the skill is.

The skill: Place by skill, not birthday.

Steps

1. If observe-before-explain fails, stay on 5–7 looking longer even if birthday says 9.
2. If fair-test masters early, offer fair ladder and redesign.
3. Write this week’s band choice on the plan.

4. Exit: band choice dated.

Error patterns: Birthday lock; “gifted” skipping looking habits; shame “going back.”

AS-06 Honesty Card — Three Refuses (in-home)

Band: 8–10 + adult.

Time: 10–15 minutes.

Materials: Card posted on fridge or notebook inside cover.

Safety: Culture card — reinforces physical safety too.

The fun: Pointing at a refuse that *almost* happened feels brave.

The skill: Honesty habits for records.

Steps

1. Read aloud and keep: (1) No worksheet dump as Science Hour. (2) Activity mania $\neq$ investigation. (3) Generated observation is not student data.
2. Add: adult present; kitchen chemistry picture $\neq$ chemistry credit; no unauthorized mixing.
3. Child points to one refuse that almost happened this week.
4. Exit: pointed refuse + one better move.

Error patterns: Card ignored while packet continues; using the card to scold rather than reset.

Optional OUT transfer check (out-of-home)

Band: both.

Time: 20–40 minutes on site.

Materials: Notebook crumb.

Safety: Adult present; trail/museum/street rules; not a cold public quiz.

The fun: Showing the skill where the world is bigger.

The skill: Transfer — notice before name at a park tree; or one-exhibit retell; same two-ways spirit.

Steps

1. One focused look.
2. Two ways: draw/point *and* say.
3. Ask what still needs practice.
4. Home: one notebook line.

Error patterns: Twenty-item scavenger mania; grading in front of strangers.

Progress chart without percentile identity

If you want a chart, use skill rows and date columns:

Skill	Date tried	Showed two ways?	Still needs
Draw what you saw			
Retell what happened			
One-thing-changed			
Living-or-not by needs			
Light path / push-pull notice			
Caption / claim-because			

No ranks. No sibling comparison fridge chart. Self-to-self only. Screening $\neq$ identity.

Blocked vs mixed checks

Blocked: a week that only practices draw-retell until two ways are easy.

Mixed: sticky triad Monday, living sort Wednesday, fair ladder Friday — once looking is warm.

Vendor review questions: optional, labelled as the program's. They do not replace rows above.

What a stranger-readable Friday line can look like

Date. Materials. What we did. Child drew . **Child said** . Two ways: yes/partial. Still needs: _____. Adult note (private): one error pattern → Monday move.

That paragraph is enough for many homes. Thick jurisdictions may want more samples in a portfolio; put real looking samples there. Thin jurisdictions may want nothing on paper; the looking still happened. Neither fact is a reason to invent a table. Both facts are a reason to keep the Honesty Card.

Weather, light, and living crumbs as check cousins

You do not need new toys for assessment. A weather card pair (today/yesterday) checks same/different. A beam-path drawing checks light noticing. A plant one-thing photo checks fair-test light. Recycle prior banks. The AS cards shape the *ask*, not a shopping list.

Incorrect Friday you should refuse

A stopwatch race through twenty cold vocabulary items. A sibling rank chart on the fridge. An app streak celebrated as mastery. A detector score on a caption. A generated “lab report” printed in color for grandparents. Replace any of those with one kitchen object and the locked question.

Mixed month illustration (labelled illustration, not a reported family)

Week 1 Friday: Sticky Exit Triad on ice melt; draw+say two ways; still needs practice = “what we kept the same.” Week 2: Cover the Caption on a leaf page; isomorphic leaf look Monday. Week 3: One-thing-changed on ramp height; Fair Ladder once. Week 4: Living-or-not sort; Honesty Card point (“we almost filed a generated caption”); overnight hold check on the sort. Band placement note: stay 5–7 oral on captions even though birthday is 8. That month is assessment as teaching. It is not NAEP.

Diagnostic wrong codebook (keep in the notebook cover)

Hear this	Usually means	Monday move
Name-first rush	Vocabulary before lookable	Cover caption; OBS sit
Force-as-fuel final	Prior idea untested	Second ramp try; same list
Confound pride	Two changes claimed as one	Fair ladder; redesign
Book-only caption	Story swallowed evidence	Finger rule; WRITE-02
Generated obs	Performance as data	Honesty Card; real look
Demo, no record	Watching $\neq$ looking	Child draws; child says
Birthday lock	Age replaced skill	AS-05 rewrite plan
Percentile talk	Rank as identity	Skill chart self-to-self

Pick one row per Friday. Do not pick all eight.

How to fade help on a check

First you model two ways: you draw a quick path and say one sentence. Then they draw; you scribe the sentence. Then they do both. Stop supplying the second way. Struggle before rescue: ask, wait, hint (“show me with the stickies”), then model if needed.

When to stop talking

When they are mid-draw. When they are naming what still needs practice. When overnight hold is being tested in the morning. Your job is one locked question and a notebook line — not a speech about their future.

Three columns that stay separate

	Screening / compliance sitting	Formal evaluation (if sought)
Kitchen check		
What to teach tomorrow	Risk flag or state reading/math object	Qualified professional; written map

	Screening / compliance sitting	Formal evaluation (if sought)
Kitchen check		
Draw, retell, one-thing, living-or-not	Not a science percentile for this book	Not a DIY from this chapter
Parent-run; skill-based	Stay in its lane	Not replaced by stickies

Say it until it sticks: screening $\neq$ identity $\neq$ kitchen check.

For the student

A check is a chance to show what you can do — two ways — and to name what still needs practice. It is not a name for who you are.

You might draw and then say. You might sort and then point. You might name what changed and then mark it on a sticky. If something is still shaky, that is tomorrow's job. Shaky is not “failed.”

Tiny worked example

You draw the ice smaller. You say: “It got watery on the counter.” Two ways. You say what still needs practice: “I forget what we kept the same.” Tomorrow you will say the “same” list before the try. That is a good check.

Two tries

1. Show one idea two ways today (draw + say, or sort + caption).
2. Tell a grown-up one thing that still needs practice — without calling yourself a name.

Explain it back

“Checks find the next try. They do not decide if I am a science kid.”

Challenge

Cover a caption. Say what you notice. Uncover. What matches? What is new?

One more challenge (optional)

Point to the Honesty Card. Name one refuse that almost happened. Choose the better move.

After a night

Tomorrow, before any new try, open yesterday's page. Show the same idea two ways again — shorter. If you still can, the look held. If you cannot, that is useful. Ask for a smaller practice, not a bigger quiz.

If it isn't clicking

1. Identity spiral after a miss.

They say "I'm bad at science" after one blank retell.

Next move: screening $\neq$ identity. Shrink the pile. Two cards only. Isomorphic retest Monday. Celebrate the second way when it appears.

2. Wait was zero; the check felt like a trap.

You fired questions; they froze.

Next move: count three. One object only. Let them choose draw-first or talk-first.

3. Print load too heavy.

Written battery kills the Friday.

Next move: oral exit. You scribe. Stickies only. Return to print when talk is warm.

4. Portfolio pressure.

Someone wants a thick binder of perfect tables.

Next move: dated notebook with real looks beats a fake table. Honesty Card. Refuse invented data.

5. Birthday lock from a guide.

The guide says Grade 3 fair tests; the child still cannot retell.

Next move: AS-05. Stay on looking. Skill, not birthday. The guide is a map, not a verdict.

When to slow down: if two ways fail on draw-retell, stay there before fair ladder.

When to go ahead: if two ways hold after a night, mix skills and add redesign.

When to get a human tutor: if anxiety or a learning difference makes every check unsafe — a local teacher or specialist can help while you keep kitchen objects kind. Seeking help is not failure. Formal evaluation, when sought, is a different column from kitchen checks. This book does not diagnose.

Correct a wrong reason without crushing the look. A confounded “proof” is information. Ask can we tell for sure. Then redesign. Silence in the face of a fake portfolio table is not kindness.

Tools, including AI

Point back to the Science Hour AI box. Parent holds the account. Checks are unaided first on the looking and trying that is the lesson. You may use a tool to help *you* design blank sticky templates or isomorphic practice with PARENT KEY. You may diagnose work already produced (crop to the paper; no child’s face). You may not invent a data table for the portfolio. You may not generate a result image. You may not detector-grade a child’s sentence. You may not print a certificate of science.

Optional analog tools: stickies, honesty card, skill chart without ranks, covered-caption sticky, kitchen sort objects, dated notebook titled **Science**.

A timer is optional and dangerous if it becomes the point. Speed is not the skill. Two ways and a kind “still needs practice” are the skill. If a co-op or tutor runs a check, the same locked question travels — and the title of the subject, if written down, is still **Science**.

What “done enough” looks like

Before you treat this chapter’s rhythm as installed:

- ☐ Child can show at least one science idea two ways (draw+say, sort+caption, or change/same+point).
- ☐ Child can name one thing that still needs practice without taking a personality hit.
- ☐ One kitchen object from A–D ran this month on real materials.

- ☐ One AS card (01–06) shaped a Friday or a plan.
- ☐ A look held after a night at least once — or you know Tuesday’s reteach.
- ☐ No percentile sticker, no invented table, no generated observation in the file.
- ☐ Transcript-minded title, if any, is **Science** (or Life/Physical/Earth Science only when the work is actually that) — never a brand, never Little Thinkers Science I.

Life of the habit (not a ninth chapter). Revisit next Tuesday: same skill, fresh lookable. Mix last week’s draw-retell with this week’s one-thing-changed once each is warm. Stop while a look remains. Progress is self-to-self dates on a skill chart — not a diploma, not a national rank, not a culture-war win.

Placement by skill, not birthday. A seven-year-old who fair-ladders may use 8–10 checks. A nine-year-old who still needs scribed retells may stay oral. Publisher “grade 2 science” scopes are not legal grades. State reading/math sittings did not measure the plant trial. Optional vendor review questions stay optional.

Done enough is showable two ways and a next practice named kindly — a path through elementary science looking, held in a dated notebook a stranger can read as **Science**.

If Friday’s two ways still work after a night — same drawing recognized, same change named — the check held. If it does not, next Tuesday is smaller, kinder, and isomorphic. Revisit is not failure. Mixing last month’s living sort with this week’s ramp ladder is allowed once each skill works alone. Life of the habit lives here, in Chapter 7’s talk crumbs, and in the records chapter — not in a ninth teaching chapter with that title. Next week you will open the records chapter for the file you keep; this week you only need two ways, one practice named, and a notebook date a stranger could trust.

Chapter 9

Records

The science *work* can be as ordinary as ice in a cup and as serious as any classroom hour. The *credit* is a family claim until a stranger can use it — a receiving third-grade teacher, a Pennsylvania evaluator, a New York quarterly reader, a later high-school transcript the family will someday write. That stranger will not have sat at your table. They will have a page.

This chapter is how you make that page at ages 5–10. Lawful, honest, and readable to a stranger are three jobs. Keep them in view.

This week you can date one drawing and one one-thing-changed note in a notebook so a stranger could almost read them. Today the student tries the page. You write the line.



Figure 11: An open notebook titled Science, a dated line with a draw/retell note and a one-thing-changed check, a pencil, a leaf, a library postcard. No diploma seal. No grade. No brand. No child.

What you are making

You are making a **dated notebook of Science**, with the task named, plus a **draw what you saw** or **retell what happened**, plus a **one-thing-changed** check, plus a **living-or-not** sort by needs you can see, plus optional publisher review questions when you use a purchased program. You are not making a national diploma. You are not making Little Thinkers Science I.

There is often **no** elementary transcript. When a stranger asks, the title is **Science**. Life Science, Physical Science, or Earth Science only when the work is actually that. Never a brand. Never Apologia Zoology 1. Never Elemental Biology for the Grammar Stage. Never RS4K Book 1. Never Science Unlocked Wonder Year

1. Never BJU Science 2. Never Berean Science in the Beginning. Never **Little Thinkers Science I**. The dated notebook, the draw/retell, the one-thing-changed check, and the living-or-not sort live *inside* Science.

There is no federal homeschool diploma and no national science credit at this age.⁵² Texas, North Carolina, and New York do not issue one. Pennsylvania’s supervisor or approved-organization diploma after listed graduation courses is a high-school object, not a five-year-old’s file. Virginia’s list of subjects is a list, not a transcript. A parent-issued diploma can be a real piece of paper later. It is not a registrar’s national science credit now.

Two readers, even at 5–10, if anyone outside the kitchen asks. The **state**, in a thick jurisdiction: an IHIP, an affidavit, a portfolio — with science samples where the statute named science. A **later stranger**: a public school on transfer; a private school; a future high-school transcript. State compliance is not a science diploma. A thick Pennsylvania portfolio can be in statutory compliance with science samples in the log and a grade 3 *reading/math* sitting. A silent Texas year can be lawful with no file at all, provided bona fide written reading, spelling, grammar, math, and good citizenship are happening. Neither fact is a reason to skip looking and trying. Both facts are a reason not to fake a credit.

A family that invents “Little Thinkers Science I, 1.0, A” for a seven-year-old has created a slogan. A receiving teacher cannot map it. Athletic eligibility and college exams belong to an older year and to *Science for Homeschooling*.

The national survey of homeschool subjects did **not** offer one overall “science” checkbox. It offered science *sub-subjects*. In 2016, scientific inquiry or experiments was taught that year to 51 percent of K–2-equivalent homeschoolers and 46 percent of the 3–5 band (41 percent of all K–12-equivalent homeschoolers); earth sciences or geology to 43 and 47 percent (44 percent total); biology to 19 and 18 percent (25 percent total); chemistry or physics to 9 and 12 percent with NCES caution flags (15 percent total).⁵³ Informal looking does not arrive as “Science,

⁵²No national homeschool diploma: Texas Education Agency Home Schooling and Alternative Schooling pages (state does not award a homeschool diploma); North Carolina DNPE (the home school issues any later diploma; the State does not); Pennsylvania 24 P.S. § 13-1327.1(d.1) diploma is a high-school object; New York 8 NYCRR 100.10 (home instruction does not yield a local or Regents diploma); Virginia list of subjects is a list, not a transcript. Access date for URLs in these notes: 15 September 2026.

⁵³Jiashan Cui and Rachel Hanson, *Homeschooling in the United States: Results from the 2012 and 2016 Parent and Family Involvement Survey*, NCES 2020-001, Table 9: subjects taught that

1.0, B+.” If a thick state later wants a log, the translation is: name the task, name the materials or place, date it, keep a sample. History’s social-studies percents are different rows — leave them there.

Titles a stranger can read

Print these and only these as elementary names, if a name is needed at all:

Science. Life Science or Physical Science or Earth Science (only when the work is actually that). Reading / Language Arts / English. Mathematics / Arithmetic. History / Social Studies. Art, Music, Physical Education.

Those names travel. They are the names public elementary reports already use. They are the names homeschool elementary files, when they exist, already follow.

Never a brand. Never Little Thinkers Science I.

The brand, if used at all, sits in a parenthetical or in a description: “Science (library animal books; plant one-thing trial; weather window cards; museum visit notes).” Combining objects is allowed. Combining does not mint a new title. Library book plus one purchased lesson plus a park bark rub is still Science.

A publisher’s “grade 2 science workbook” is a scope, not a legal grade, and not a transcript line. Placement is by skill, not birthday.

Here is an illustration, labelled as such, not a reported family:

15 September 2026. Lookable: ice in a cup on the table. Task: draw what happened; say one thing that changed. Child drew smaller ice and a puddle. Retell: “It got smaller and water came.” One-thing-changed: we left one cup on the table and one in the fridge — child named “warm room” after one reteach. Living-or-not: leaf vs rock — sorted by “needs water / does not.”

year — scientific inquiry or experiments 51 percent in K–2, 46 percent in 3–5, 41 percent of all K–12-equivalent homeschoolers; earth sciences or geology 43 / 47 / 44; biology 19 / 18 / 25; chemistry or physics 9! / 12! / 15 (NCES caution flags on the elementary chem/physics cells). Quote those science *sub-subject* rows. Do not paste social science/history/social studies 70/78/78, arithmetic, computer science, basic reading, or foreign-language percents from the same table into a science sentence. The 2023 First Look (NCES 2024-113) did not republish subject-taught tables.

That is a record a parent, a Pennsylvania evaluator, or a New York quarterly narrative can actually use. It is not a credit. It is not NAEP. It is not a diploma. It is not a publisher placement test, though a publisher review sheet may sit in the same folder as a starting document.

The law is a pattern, not one form

U.S. homeschooling is state law. There is no federal science office. Dates and subject lists change. Read your current department page and the statute it cites. A color-coded chart on a membership site is not your statute. Five official-page examples show the range. **None of them requires “Little Thinkers Science I.”** Science is sometimes named, sometimes left to the parent’s list, and sometimes neither named nor tested.⁵⁴ This chapter is not legal advice. The parent reads their own statute.

Texas. The Agency does not regulate home-school programs. Parents follow a bona fide written curriculum: reading, spelling, grammar, math, and a course in **good citizenship**. *Leeper*, restated on TEA’s Alternative Schooling page, does **not** name the word science. The kitchen may still look and try every day. The state does not award a homeschool diploma. At 5–10 there may be no external reader at all. The kitchen still benefits from a dated notebook if the family later transfers. Transfer is treated like an unaccredited private school. “We did Exploring Creation with Zoology 1” is not, by itself, a public-school credit. A district may test for placement. TEKS Science is the public-school map TEA points curious parents toward. It is not a homeschool mandate.

North Carolina. Notice of Intent. Nine calendar months. Each year, a nationally standardized test in **English grammar, reading, spelling, and mathematics** — not science, and not a franchise. The home school, not the State, issues any

⁵⁴Texas *Leeper*, restated on TEA Alternative Schooling: reading, spelling, grammar, math, good citizenship — science not named as a word. North Carolina DNPE: annual nationally standardized test includes English grammar, reading, spelling, and mathematics — not science. Pennsylvania 24 P.S. § 13-1327.1: elementary shall include science; portfolio tests in grades 3, 5, and 8 in reading/language arts and mathematics. New York 8 NYCRR 100.10(e)(2)(i): grades 1–6 require science among other subjects; a unit is 6,480 minutes; quarterly reports; annual assessment named tests or narrative. Virginia § 22.1-254.1: parent writes a list of subjects; evidence of progress; children under six as of 30 September exempt from evidence-of-progress. None of the five requires a named “Little Thinkers Science I.” This chapter is not legal advice. The parent reads their own statute.

later diploma. A five-hour day is a recommendation, not the statute, and not a science-minutes table. DNPE tells the family not to name the homeschool after a curriculum brand. A looking-and-trying notebook is extra. It is not the annual test. The annual battery is compliance. It is not this book's kitchen check, not NAEP, and not a percentile promise.

Pennsylvania. Elementary **shall include** English (spelling, reading, writing); arithmetic; **science**; geography; history of the United States and Pennsylvania; civics; safety; health; physical education; music; and art. No “Little Thinkers Science.” Portfolio tests in grades 3, 5, and 8 are in **reading/language arts and mathematics**, not a science sitting. Drop science samples into the log of materials. Skip inventing a parallel “Little Thinkers” portfolio. 180 days or 900 hours elementary are whole-program hours, not a science-minutes table. A supervisor diploma is Pennsylvania's, and it is high school. The evaluator's experience list may include science among other subjects. The annual *test* object is still not science.

Virginia. Notice, a parent-written **list of subjects**, and evidence of progress: a composite score at or above the fourth stanine, or an equivalent SAT, ACT, or PSAT score, or an evaluation letter. Science is not named as required because no subject is named as required; the parent writes the list. **Science** is the ordinary word to write. Those sittings are not this book's short check. Children under six as of 30 September are exempt from the evidence-of-progress subsection. A publisher science quiz does not, by itself, satisfy Virginia's fourth-stanine object.

New York. Letter of intent, an Individualized Home Instruction Plan, quarterly reports, and an annual assessment. Grades 1–6 required subjects include reading, spelling, writing, English language, arithmetic, geography, United States history, **science**, health, music, visual arts, physical education. Put the work under those titles in the quarterlies: material covered in each IHIP subject. A unit is 6,480 minutes; a library animal book and a plant trial can be named under science. Apologia Zoology 1 is a **material**, not a required unit. Home instruction does not yield a local or Regents diploma. Annual assessment: named commercially published tests or, in some grades, a written narrative. Composite above the 33rd percentile or one year of growth is a probation threshold in that regulation. It is not a goal this parent manual advertises as success. Regents lab-evidence language is a later object — not this age band.

If you live in none of those states, you still have a department page. Open it.

What a short check is

Five objects, not a battery.

1. **A dated notebook.** Date, task named, what the child looked at or tried, one sentence of what happened. Under **Science** if a title is needed. A five-year-old's check is a drawing and an oral retell, not a silent twenty-item page.
2. **Draw what you saw / retell what happened.** The child shows the ice, the leaf, the ramp run, the weather card. Oral first. Written caption optional at 8–10. You may scribe their words. Compare the drawing or retell to the lookable on the table.
3. **One-thing-changed.** Fair-test light: what did we change, and what did we keep the same? Can we tell for sure? One deliberate change is enough. A confounded “proof” is a reteach, not a character flaw.
4. **Living-or-not by needs.** Cards or objects — plant, rock, toy, bug picture — sort by a need you can see: food, water, air, a place to grow or live. Reteach mix-ups kindly. Four to six pictures is enough. Not a memorized kingdom chart.
5. **Optional publisher review** (the organization sells the questions). Apologia narration prompts; Elemental notebooking checks; RS4K Show What You Know; BJU chapter review; Science Unlocked workbook pages — when you use that program. Use to check looking and recall inside a scope. They do not license a national percentile. They do not satisfy North Carolina unless the chosen test is a nationally standardized achievement battery covering English grammar, reading, spelling, *and* mathematics — a publisher science chapter quiz generally is not that battery.

A parent who feels rusty can turn the check into a performance with a tight jaw. The useful move is still: date the work, keep one wrong turn you can reteach, and leave the looking in the child's hands while they try.

Screening is not an identity label. A short check that finds a child cannot yet name what changed, or invents an observation the ice never made, is a teaching signal. It is not “this child is not a scientist.” Reteach with a shorter try, a clearer lookable, or two cups only. Do not file the miss as a permanent label.

Bands: **5–7** — draw what happened; oral retell of one sentence; living-or-not with pictures; one-thing-changed as “what did we do different?” **8–10** — picture-plus-

sentence; oral or short written retell; name change and same; living-or-not with a need named; “can we tell for sure?” after a messy try. Fit across a sibling pair is allowed.

What a short check is not

NAEP. A percentile promise. A homemade high-stakes battery. Homeschooling was 3.4 percent of K–12 students in 2022–23 (~1.765 million); instruction-at-home 5.2 percent is a different bucket (homeschool *or* full-time virtual).⁵⁵ There is no nationally representative homeschool NAEP Science score for this age band. A North Carolina annual battery and a Pennsylvania grade 3/5/8 sitting are compliance objects in reading and math, not a science percentile for this book. CLEP. AP. No CLEP for elementary. No AP at this age. Those objects belong to *Science for Homeschooling*. This book does not reprint that fight.

NCAA, Common App, and high-school transcript export are the older book. Dual enrollment, Advanced Placement, and credit-by-exam are not this chapter. A fake elementary “with Lab” line is not either. Kitchen chemistry is a picture, not a Chemistry credit. Scores help a stranger believe a *high-school* transcript. They do not replace a dated drawing at seven.

If they transfer

Transfer is a local placement problem. A brand in the title column will not be read as Science. Another reason the title should already look like Science.

A receiving teacher who sees “Little Thinkers Science I, A” has been given a slogan. A teacher who sees “Science — dated notebook; draw of ice melt; retell: ice got smaller; one-thing-changed: warm room vs fridge; living-or-not: leaf needs water” has been given a year they can place.

⁵⁵NCES 2024-113: 3.4 percent homeschooled in 2022–23, about 1,765,000; K–2 4.0 percent; 3–5 3.6 percent. Instruction-at-home 5.2 percent is a different bucket (homeschool *or* full-time virtual). NCES has never published a nationally representative homeschool NAEP Science score for this age band as a lock for this book. There is no opened national study of daily homeschool science minutes.

Place by skill, not birthday

Can the child draw what they saw before they name the vocabulary word? If not, they are still in the looking-and-drawing chapter, whatever their age. Can they name one deliberate change and one thing kept the same? If not, they are still in fair-test light. Can they sort living from not by a need they can see? If not, they are still before life-cycle cards. High-sounding *titles* cannot skip those moves. A ten-year-old may still need oral retell before written captions. A six-year-old may already hang three weather cards in order. Skill first. Publishers opened for this book say the same in their own age-band language: choose by age, development, interest, and ability; levels and grade stickers are not legal grades.

What to keep in the file

1. A dated notebook: date, task named, what the child looked at or tried, one wrong turn you will reteach. Under **Science**.
2. Drawings and retells from the week — oral scribbled by you, or a short caption at 8–10 — not a crate of every worksheet.
3. One-thing-changed notes and living-or-not sorts — a few honest pages, not every card.
4. Optional publisher review, if you used one — as a looking check, not as a yearly IQ.
5. State objects that actually apply: a Pennsylvania portfolio sample under science if you are in Pennsylvania; a New York quarterly under science if you are in New York; a Virginia list that includes Science if you write the list; a North Carolina annual battery in its own lane if you are in North Carolina. Compliance stays in its lane.

Life of the habit, as the file you keep, lives here: date the work; name the skill; keep one wrong turn you can reteach. Revisit next Tuesday is a move in Chapters 7 and 8. The notebook is how Tuesday leaves a trace.

A path through elementary science looking is the promise. A diploma is not. A percentile is not. A culture-war win is not.

For grades 1–12 survey, transcript titles, and later AP/CLEP edges, see *Science for Homeschooling* — this book teaches the 5–10 Science Hour.

Chapter 10

Resources

This chapter names programs families actually meet. It does not rank them. It does not sell them. It is a fit list so you can match an object to the table you already have: how much of the teaching you will carry, whether a kitchen can use it tonight, what it costs in money and in minutes, and whether the child will still look, try, and record what happened.

A “grade 2 science workbook” is a publisher’s scope, not a legal grade. Placement is by skill. The record, as Chapter 9 said, still says **Science**. Combining two honest objects does not mint a new title. Library animal book plus one purchased lesson plus a park bark rub is still Science.

Independent trials of these programs *in homeschool* remain scarce. A chapter review quiz is not an IES-certified randomised trial. Choose by fit: parent load, style, age label, cost shape, and whether the child’s eyes stay on the lookable and the child’s hand stays on the drawing.

This week you can name *why* tonight’s object fits this child — usually a library book and a cup of ice. Today the student tries that page. They are not on a shopping task.



Figure 12: A library card, a museum postcard, a leaf, a generic book spine with the title unreadable, a lined notebook page and pencil. No logos. No portraits.

How to read a program

Four questions do the work.

Parent load. Are you reading aloud from a library book you already have? Sitting with a scripted Teacher Guide every day? Hunting supplies for a kit box? Booking a museum Sunday?

Style. Conversational Christian narrative? Classical doing-reading-writing? Atoms-first short lessons? Textbook with optional video? Kit-in-a-box? Free digital text? Nature study and living books as a method?

Cost shape. Nothing tonight with a library card. A book you already own. A print set or kit whose dollars sat on a page this access day and will stale. Fees this book

did not verify. Confirm a current price on the page before you buy. This chapter omits unverified list prices.

Age label. A five-year-old is not an upper-elementary Chemistry-and-Physics default. A ten-year-old may still need draw-what-you-saw if looking is the skill. Birthday is a weak prior.

Reach numbers on an organization's page are that organization's copy, not a home-school census and not an effect size.

Library, museum, and nature — \$0 first

A public-library card. A children's science, animal, weather, or space shelf. A story hour. A museum, science center, or natural-history hall. A zoo or aquarium day. A yard bird list, a sidewalk crack plant, a puddle after rain. That is tonight.

Fit. Parent load: presence and talk, not a script. Looking: objects and labels in a gallery, live animals at a zoo, ice or leaf at home are natural notice-before-name prompts. Placement: the child's interest and looking stamina, not a grade sticker. Cost shape: **\$0 first** with a card, a free museum day where it exists, or the yard. Fines, holds, museum admission, zoo tickets, and parking are local objects — this book does not invent a national free-museum guarantee.

In 2016, 66 percent of homeschool households used the library as a source of curriculum and books (78 percent in K–2, 70 percent in 3–5). In 2022–23, 53 percent of the Homeschool-total row had visited a library in the past month, against 29 percent of all K–12; 43 percent had visited an art gallery, museum, or historical site, against 23 percent; 28 percent had visited a zoo or aquarium, against 21 percent.⁵⁶ Those are different questions in different years. All three outing cells are a free-or-low-cost hour. Library mission language (inspire lifelong learning; a person's right to use a library should not be denied because of age) is mission, not a kit.

A story hour, a gallery walk, a zoo visit, or a creek look is not a transcript line by itself. It does not replace a purchased spine for a family that wants one. It does not replace fair-test light for a child who cannot yet name what changed. It is the

⁵⁶NCES 2020-001 Table 6: library as a source of curriculum and books, 66 percent of homeschool households in 2016 (K–2 78; 3–5 70). NCES 2024-113 Table A-5: Homeschool-total row, library visit past month 53 percent versus 29 percent of all K–12; art gallery / museum / historical site 43 percent versus 23 percent; zoo or aquarium 28 percent versus 21 percent (2022–23). Different questions, different years — do not swap 53 for 66. Access date: 15 September 2026.

honest starting inventory before a catalog purchase. Library, museum, and nature outings motivate. They do not replace the observe-before-explain move.

Transcript. Not a course. Materials under Science if a log is needed: “library read-alouds, titles listed; museum visit notes; yard bird list.” Never Little Thinkers Science I. Never “Library Hour 1.”

The teaching chapters of this book are written for this object first. A purchased program is a fit, not a requirement.

Nature study and living books — method, not a brand

Reading (usually aloud) a well-written animal, plant, weather, or sky picture book or chapter book, then asking the child to tell back what they noticed in their own words; optionally draw or caption; optionally walk outside and look again. That is the method. It needs a library card, a blank page, and often a window or a yard — \$0 beyond paper and whatever the library already holds.

Fit. Parent load: select a book, sit nearby, listen without interrupting the first retell, put a lookable next to the page when you can. Looking: strong when the book and the leaf (or cloud, or bug) share the hour; weak if the hour is only vocabulary from the page. Placement: length and difficulty track listening stamina, not a birthday. Pairs honestly with Apologia Young Explorer reading lists, Elemental encyclopedia reading, or a naked library bag.

Nature study plus living books is a method. Put the work under Science, not under a brand. Charlotte Mason-style outdoor looking and narration sit beside many Christian publishers’ elementary copy; the method can be used without buying that program.

Transcript. Materials under Science: “living books + outdoor looking, titles listed.” Never “Charlotte Mason Year 2” as a fake credit unless the family’s whole program is being described in a narrative log — and even then the stranger-readable title remains Science.

Apologia Young Explorer (Christian, conversational, K–6)

A Christian Young Explorer series for elementary grades **K–6** (the publisher also names ages ~5–11), Charlotte Mason style, usable multi-age — as the publisher describes it. Titles across the elementary line include Astronomy, Botany, Zoology (1–3), Earth Science, Anatomy and Physiology, Chemistry and Physics. A support article dated 16 March 2026: titles may be completed in **any order**; parents choose by interest; Earth Science, Anatomy & Physiology, and Chemistry & Physics are best reserved for upper elementary (4th–6th) on the publisher’s advice, with younger siblings able to follow. A recommended creation-order sequence is labelled one possible option. Hands-on work with household items is described; experiments are encouraged.⁵⁷

Fit. Parent load: moderate if the adult reads aloud and picks activities; higher if every project runs. Conversational Christian narrative, multi-age. Age: sits cleanly in **5–10** when Zoology, Botany, or Astronomy leads; upper-elementary titles lean **8–10 edge**. Not a fit if the parent needs a secular text. Looking and trying still want an adult present — do not treat any “experiments optional” FAQ language from older product lines as a safety plan for this age.

Apologia Young Explorer is a Christian multi-age narrative option with topic choice by interest. Fit, not rank.

Transcript. Science. Never “Apologia Zoology 1.” Never “Young Explorer.”

Elemental Science — Grammar Stage (classical, ages 5–10)

Grammar Stage programs for ages **5 to 10** — as the publisher describes it. Typical progression named on opened pages: Intro to Science; Biology for the Grammar Stage; Earth Science & Astronomy; Chemistry; Physics for the Grammar Stage. Teacher guides offer options for a full year (**36 weeks** named on Grammar

⁵⁷Apologia elementary science pages and support article “Should we do the elementary science books in a particular order?” (16 March 2026): K–6 / ~5–11; any order; Earth Science, Anatomy & Physiology, Chemistry & Physics best reserved for upper elementary on vendor advice. Access date: 15 September 2026. Omit unverified 2026 list prices.

Stage pages); student workbooks for notebooking; scientific demonstrations; reading from children's encyclopedias. Biology for the Grammar Stage opened detail: ages **5–8**, grades K–2, living world / animals / human body / plants.⁵⁸

Fit. Parent load: moderate-to-high if doing, reading, and writing-it-down all run; lower if the parent picks one track. Classical method, not a high-school lab course. Age: **5–10** as labelled. Still a curriculum, not a lab safety plan. Adult present for demonstrations.

Elemental Science Grammar Stage is a classical doing-reading-writing neighbor for this exact age band.

Transcript. Science (or Life / Earth / Physical Science if the year's work truly matches). Never "Elemental Biology for the Grammar Stage."

Real Science-4-Kids / Gravitas (atoms-first, secular-leaning)

An atoms-first K–8 system: **K–2 Digital Teaching Dashboards** and **Science Books 1–7** — as the publisher describes it. Curricula Map: Book 1 around grades 2–3; Books 2–4 through about grade 5 — the elementary band for this book. Lessons the publisher describes as **35–45 minutes**, often **3–4 days/week**. Open and Go materials plus Digital Teaching Dashboard are sold. Placement is ability-based on the publisher's copy. Climate and molecules language on opened pages is secular-leaning as the publisher presents it.⁵⁹

Fit. Parent load: moderate; short Open-and-Go lessons reduce script hunting. Atoms-first secular-leaning. Age: K–2 dashboards for the younger band; Books 1–4 for the middle and upper edge of **5–10**. Lesson-length claims are the publisher's, not an NCES finding. Not a high-school Biology–Chemistry–Physics replacement — and this book does not need that object.

⁵⁸Elemental Science Grammar Stage collection pages: ages 5–10; Intro to Science; Biology / Earth & Astronomy / Chemistry / Physics for the Grammar Stage; 36-week options named; Biology Grammar Stage ages 5–8 / K–2. Access date: 15 September 2026. Omit unverified 2026 list prices.

⁵⁹Real Science-4-Kids / Gravitas curricula map and product pages: K–2 Digital Teaching Dashboards; Science Books 1–4 in the elementary band; lessons vendor-described 35–45 minutes, often 3–4 days/week; ability-based placement. Vendor lesson claims are vendor, not NCES. Access date: 15 September 2026. Omit unverified 2026 list prices.

Real Science-4-Kids is an atoms-first short-lesson option for families who want secular-leaning elementary science.

Transcript. Science. Never “RS4K Book 1.”

Berean Builders — chronological elementary (Christian)

Christian science by Dr. Jay Wile. Elementary chronological titles on the opened catalog include *Science in the Beginning*, *Science in the Ancient World*, *Science in the Scientific Revolution*, *Science in the Age of Reason*, *Science in the Industrial Age* (and neighboring elementary listing language). Hands-on work is described as woven in.⁶⁰

Fit. Parent load: moderate if the adult reads and runs the woven experiments; higher if every activity is mandatory. Christian chronological stories with looking woven in. Age: elementary titles sit in **5–10** when used as read-aloud with adult-present trials. Worldview is part of the product. Label it honestly. Neither hide it nor mock it.

Berean Builders chronological elementary is a Christian timeline-shaped science option.

Transcript. Science. Never “Berean Science in the Beginning” as the credit line.

BJU Press Science 1–6 (Christian textbook)

Elementary science as a foundation for understanding and applying science, with grade-labelled **Science 1–6** textbooks — as the publisher describes it. Video Course kits and Textbook kits exist on neighboring homeschool pages. Sibling science pages include biblical worldview / young-earth framing — that framing is part of the product on opened pages, not an optional sticker.⁶¹

⁶⁰Berean Builders elementary catalog pages: *Science in the Beginning* through *Science in the Industrial Age* and neighboring elementary titles; Christian chronological; hands-on woven in. Exact lesson counts not verified from a scope PDF this stream. Access date: 15 September 2026. Omit unverified 2026 list prices.

⁶¹BJU Press elementary science pages: Science 1–6; textbook and video course kits on neighboring pages; biblical worldview / young-earth framing on opened sibling science pages. Access

Fit. Parent load: moderate; more textbook-shaped than a picture-book pack; video can carry some of the load. Christian textbook path. Age: Science 1–3 maps toward **5–8**; Science 4–6 toward the **8–10** edge and slightly beyond. Fit for a family already using BJU for other subjects. Not a fit if the parent needs a secular text or refuses young-earth framing.

BJU Press Science 1–6 is a Christian textbook/video option with grade stickers that are scopes, not legal grades.

Transcript. Science. Never “BJU Science 2.”

Home Science Tools — Science Unlocked Wonder / Accelerate (kit curriculum)

An all-in-one kit curriculum: **Wonder** for grades **K–2** (Early Elementary Year 1 / 2 / 3 bundles; six units per year); **Accelerate** for grades **3–7** (upper elementary through middle); Launch for 8+ (out of band except a fence) — as the publisher describes it. Each kit: Teacher’s Guide, Student Workbook, activity materials. FAQ: aligns with NGSS; does not dictate philosophical, religious, or worldviews; inclusive language. Home Science Tools remains, separately, a supply shop for other programs’ kits on the main store — that supply path is not a curriculum rank.⁶²

Fit. Parent load: lower on supply hunting (materials in the box); moderate on Teacher Guide pacing. Secular-leaning / worldview-non-dictating kit path. Age: Wonder for **5–7**; Accelerate for the **8–10** edge and into middle grades. Not a fit if the parent refuses NGSS-aligned framing. Adult present for kit activities.

Science Unlocked Wonder / Accelerate is a materials-in-the-box option with low cupboard hunting.

Transcript. Science. Never “Science Unlocked Wonder Year 1.”

date: 15 September 2026. Omit unverified 2026 list prices.

⁶²Home Science Tools Science Unlocked pages: Wonder K–2 (Year 1/2/3; six units/year); Accelerate 3–7; Teacher’s Guide, Student Workbook, materials; FAQ NGSS-aligned, does not dictate worldview. NAPPA award language is vendor. Access date: 15 September 2026. Omit unverified 2026 list prices.

CK-12 Science K–5 (free digital)

FlexBooks described as free, online, customizable digital textbooks. Science K–5 appears in opened initiative lists (including California Free Digital Textbooks Initiative naming). Simulations may be named as features; they are not a kitchen trial substitute by themselves.⁶³

Fit. Parent load: parent selects pages and puts a real lookable next to the screen. Cost shape: **free** digital text. Secular or mixed-worldview adaptable. Age: K–5 maps to **5–10**. COPPA caution: parent holds the account; ages 5–10 are under 13; no child companion login as the science partner. Not a packaged lab kit.

CK-12 Science K–5 is a free digital text path — pair it with looking, not with a tab alone.

Transcript. Science. Never “CK-12 Science Grade 2” as the credit line.

Placement by skill at the shelf

A five-year-old may follow a picture-book animal story and still need short sittings and a drawing. A ten-year-old may still need oral retell before a written claim. That is fit. It is not a percentile ladder.

Ages 5–7. Shorter lookables. Draw what happened. Oral retell of one sentence. Living-or-not with pictures. Prefer library picture books, nature study, Apologia Zoology/Botany/Astronomy as read-aloud, Elemental Intro or Biology Grammar Stage, RS4K K–2 dashboards, Science Unlocked Wonder, CK-12 pages next to a real object. Soft entry on upper-elementary Chemistry-and-Physics titles: wait, or listen beside an older sibling without claiming the credit.

Ages 8–10. Longer chapters. Picture-plus-sentence. Name change and same. Weather tables light. Prefer Elemental Earth/Chemistry/Physics Grammar Stage when ready, RS4K Books 1–4, Science Unlocked Accelerate, Apologia upper-elementary titles with eyes open, BJU Science 3–6 with worldview eyes open, Berean chronological titles as read-aloud with trials, CK-12 with a real try beside the page.

⁶³CK-12 overview / info pages and California Free Digital Textbooks Initiative lists naming Science Grade K to 5. Individual FlexBook internals may be unavailable; treat page-level detail cautiously. Free digital; parent holds account for under-13 users. Access date: 15 September 2026.

A family may teach both bands in one hour with different exit tasks. Share the lookable; split the try. The older child needs their own looking move — not only tutoring the younger one.

Put the kitchen checks from Chapter 8 and Chapter 9 next to whatever spine you choose: date the work; ask what they noticed before a name; ask what the drawing shows; ask what changed and what stayed the same; sort living-or-not by needs. The program supplies the story or the kit. The notebook supplies the stranger-readable line.

Combining honestly

Need: \$0 start → library + museum + nature (yard, zoo free day, creek look).

Need: outdoor looking without a brand → nature study + living books + narration.

Need: Christian multi-age narrative, topic by interest → Apologia Young Explorer.

Need: classical doing-reading-writing at ages 5–10 → Elemental Science Grammar Stage.

Need: secular-leaning atoms-first short lessons → Real Science-4-Kids.

Need: Christian chronological stories with experiments woven in → Berean Builders elementary.

Need: Christian textbook / optional video → BJU Press Science 1–6.

Need: materials-in-the-box, NGSS-aligned inclusive → Science Unlocked Wonder / Accelerate.

Need: free adaptable digital text → CK-12 Science K–5 (parent account; lookable on the table).

A family may combine. Common honest combinations: library shelf + one purchased spine; nature walk + Elemental or Apologia reading; Science Unlocked kit as the year's hands-on path; CK-12 pages beside a kitchen trial; Berean or BJU for a Christian family already in that lane. Combining does not authorize a franken-course title. The transcript still says **Science**.

A looking-heavy path that assumes the child can already design multi-variable experiments will not teach notice-before-name to a five-year-old. That is a fit warning, not a slur. A child who cannot yet draw what they saw is not “behind in RS4K”; they are still in a looking-and-drawing year.

When Klahr, NGSS, or the *Framework* are named next to a program, label class-

room or standards origin. Do not launder a classroom trial into a kitchen randomised study. A publisher quiz is not an IES RCT. We name fit: parent load, style, placement, cost shape, age label. Fit chooses. Ranking does not.

For grades 1–12 survey, transcript titles, and later AP/CLEP edges, see *Science for Homeschooling* — this book teaches the 5–10 Science Hour.

A Note on Sources

Studies named in the chapters are listed in Notes, in one series at the back. That is where the full citations live, so the teaching pages can stay a teaching voice.

Some items the research behind this book did not open stay out of the teaching voice. I did not invent a coefficient, a statute, or a product feature to fill a hole. If a study is in the chapter, it is in the notes. If we could not open it, it is not used as a finding here.

Program hours, credit claims, and product descriptions in the resources chapter are taken from the companies' own pages. Access date 15 September 2026.